

Breeding biology and the evolution of dynamic sexual dichromatism in frogs

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Abstract

Dynamic sexual dichromatism is a temporary colour change between the sexes and has evolved independently in a wide range of anurans, many of which are explosive breeders wherein males physically compete for access to females. Behavioural studies in a few species indicate that dynamic dichromatism functions as a visual signal in large breeding aggregations; however, the prevalence of this trait and the social and environmental factors underlying its expression are poorly understood. We compiled a database of 178 anurans with dynamic dichromatism that include representatives from 15 families and subfamilies. Dynamic dichromatism is common in two of the three subfamilies of hylid treefrogs. Phylogenetic comparative analyses of 355 hylid species (of which 95 display dynamic dichromatism) reveal high transition rates between dynamic dichromatism, ontogenetic (permanent) dichromatism and monochromatism reflecting the high evolutionary lability of this trait. Correlated evolution in hylids between dynamic dichromatism and forming large breeding aggregations indicates that the evolution of large breeding aggregations precedes the evolution of dynamic dichromatism. Multivariate phylogenetic logistic regression recovers the interaction between biogeographic distribution and forming breeding aggregations as a significant predictor of dynamic dichromatism in hylids. Accounting for macroecological differences between temperate and tropical regions, such as seasonality and the availability of breeding sites, may improve our understanding of ecological contexts in which dynamic dichromatism is likely to arise in tropical lineages and why it is retained in some temperate species and lost in others.

Introduction

Sexual selection is a driving force in the evolution of conspicuous coloration and displays in animals (Darwin, 1871; Andersson, 1994; Andersson & Simmons, 2006). Whereas conspicuous displays or colour may increase the reproductive success of particular individuals, they can also be costly if the signaller is conspicuous to unintended receivers such as predators (Endler,

1980, 1983; Andersson, 1994; Ciccotto & Mendelson, 2016). Thus, animals have evolved numerous strategies to mediate trade-offs between natural and sexual selection with respect to signalling traits. For example, species with conspicuous displays typically delay trait expression until reproductive maturity or when the individual can bear the energetic investment involved in displaying a sexually selected trait (Zahavi, 1975; Booth, 1990). Likewise, some animals may largely conceal signalling traits, such as colour patches, feathers and scales, and selectively expose these traits to intended receivers (Endler, 1992; Vukusic *et al.*, 2002; Whiting *et al.*, 2003; Kemp & Macedonia, 2006; White *et al.*, 2015). Consequently, these brief displays attract the attention of intended receivers while enabling an

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organism to restrict the amount of time they are conspicuous to unintended receivers. When displays involve conspicuous coloration, another solution is dynamic colour change: the ability of an organism to switch a colour signal on and off over short periods of time using neurophysiological or hormonal control.

Cephalopods and chameleons are the banner organisms for dynamic colour change because of their ability to change colour over a time span of seconds to milliseconds (Hanlon *et al.*, 1999; Stuart-Fox & Moussalli, 2008, 2009; Allen *et al.*, 2013), but dynamic colour change is also common in anuran amphibians, fishes and other reptiles where hormonally regulated colour change typically occurs over the course of hours to days (Sumner, 1940; Cooper & Greenberg, 1992; Nilsson Sköld *et al.*, 2012; Kindermann *et al.*, 2013, 2014; Olsson *et al.*, 2013; Tang *et al.*, 2014). In many species with neurophysiologically mediated dynamic colour change, such as chameleons, conspicuous display coloration has evolved for social reasons via sexual selection and dynamic colour change allows animals to respond to a social or predatory cue by adjusting their signal according to a potential receiver's visual system (Stuart-Fox *et al.*, 2007, 2008; Stuart-Fox & Moussalli, 2008). By contrast, dynamic colour change in frogs is slower because it is hormonally mediated (but see Kindermann *et al.*, 2014) and is typically studied in the context of camouflage and thermoregulation (Camargo *et al.*, 1999; Tattersall *et al.*, 2006); however, studies in species with ephemeral colour differences between the sexes (dynamic sexual dichromatism) indicate that dynamic colour change in frogs may also evolve for social purposes via sexual selection (Wells, 1980; Ries *et al.*, 2008; Doucet & Mennill, 2010; Sztatecsny *et al.*, 2010, 2012; Rehberg-Besler *et al.*, 2015; Kindermann & Hero, 2016b).

Acoustic signals are clearly the dominant mode of communication in frogs, yet increasing evidence indicates that visual signals, such as arm waving and foot-flagging displays, play an important role in anuran communication, particularly in noisy environments (Lindquist & Hetherington, 1996; Haddad & Giarretta, 1999; Hödl & Amézquita, 2001; Hartmann *et al.*, 2004, 2005; Hirschmann & Hödl, 2006; Boeckle *et al.*, 2009; Reichert, 2012; Starnberger *et al.*, 2014). Many animals augment the conspicuousness of visual signals by enhancing the contrast between signals and the luminance, colour and/or pattern of signalling backgrounds (Bradbury & Vehrencamp, 1998), and frogs are no exception. For example, in diurnal *Staurois* frogs that engage in foot-flagging displays, foot webbing becomes brighter at sexual maturity and likely increases conspicuousness of territorial displays to potential male rivals (Stangel *et al.*, 2015). Furthermore, behavioural experiments in European treefrogs (*Hyla arborea*), squirrel treefrogs (*Hyla squirella*) and strawberry poison frogs (*Oophaga pumilio*) indicate that females prefer conspicuous males (Taylor

et al., 2007; Gomez *et al.*, 2009; Maan & Cummings, 2009), and in polymorphic red-eyed treefrogs (*Agalychnis callidryas*) and strawberry poison frogs, females mate assortatively with respect to colour pattern (Summers *et al.*, 1999; Jacobs *et al.*, 2016). Thus, luminance, colour and/or pattern are likely important components of species recognition and mate choice for many anurans, even in low-light conditions.

Dynamic sexual dichromatism is currently known from several dozen anuran species, but our present understanding of the physiological basis and function of this trait in frogs is limited (Bell & Zamudio, 2012; Rojas, 2016). Several anurans with dynamic dichromatism are explosive breeders in which males physically compete for access to females, such as the Asian bullfrog *Hoplobatrachus tigerinus* (Tabassum *et al.*, 2011) and Fleay's Barred-frog *Mixophyes fleayi* (Stratford *et al.*, 2010), suggesting that this trait functions as a visual signal in large breeding aggregations. Behavioural studies in two explosive breeding species with dynamic dichromatism (Fig. 1), the European moor frog (*Rana arvalis*) and the yellow toad (*Incilius luetkenii*), indicate that temporary divergence in coloration between the sexes prevents costly mistakes in sex recognition when males seek females in aggregations with male-biased operational sex ratios and scramble competition for mates (Doucet & Mennill, 2010; Sztatecsny *et al.*, 2012; Rehberg-Besler *et al.*, 2015). Similarly, in lek-breeding stony creek frogs (*Litoria wilcoxii*), yellow coloration in males during amplexus serves as an intrasexual signal that may diminish sperm competition or displacement by other males (Kindermann & Hero, 2016b). By contrast, male Trinidad poison frogs (*Mannophryne trinitatis*) turn black while calling and this colour change appears to play a role in dominance interactions in these highly territorial frogs (Wells, 1980). Finally, in sexually dichromatic spadefoot toads (*Scaphiopus couchii*), where dichromatism is ontogenetic (fixed), females may use carotenoid-based colour as an honest indicator of male quality in large breeding aggregations where acoustic signals are more difficult to assess (Vásquez & Pfennig, 2007). Thus, sexual dichromatism in frogs, and dynamic dichromatism in particular, may function as intrasexual (male–male competition) and/or intersexual (female choice) signals, particularly in species with explosive or lek-breeding systems.

Here, we update the global review of dynamic sexual dichromatism evolution and frequency in frogs by Bell & Zamudio (2012) and test the hypothesis that species with explosive or lek-breeding systems in the family Hylidae, a diverse and globally distributed radiation, are more likely to exhibit sexual dichromatism. Hylids are an ideal group for investigating trade-offs between natural and sexual selection with respect to signalling traits because they typically lack chemical defences and, like many frogs, rely on camouflage to avoid detection by visually oriented predators (Toledo & Haddad, 2009;

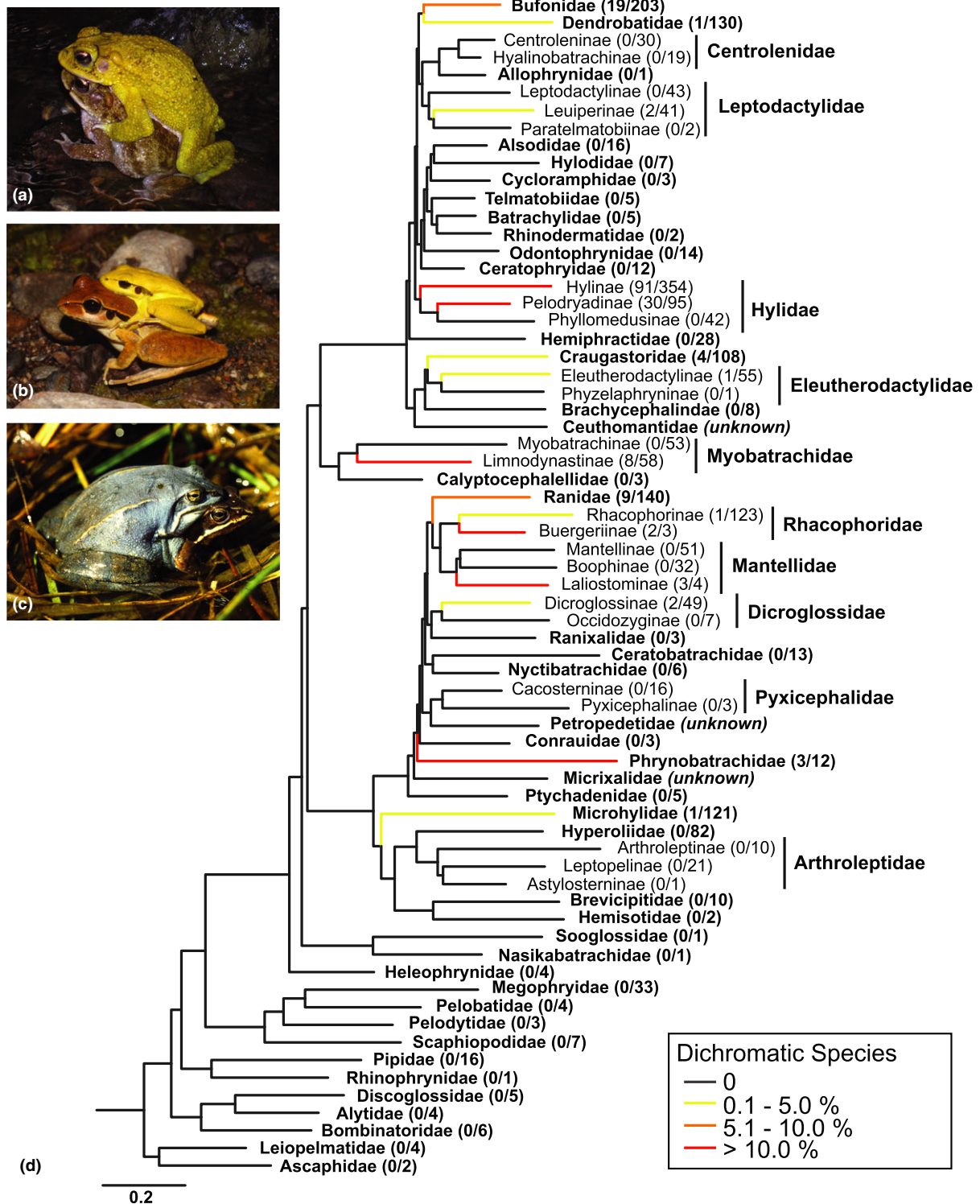


Fig. 1 Representative anurans exhibiting dynamic sexual dichromatism: (a) yellow toad *Incilius luetkenii*, photograph by Javier Sunyer; (b) stony creek frog *Litoria wilcoxii*, photograph by Matt Greenlees; (c) and European moor frog *Rana arvalis*, photograph by Lars Iversen; (d) phylogenetic distribution of dynamic sexual dichromatism. Families are shown in bold and subfamilies in regular print. Branches are coloured according to the percentage of species with dynamic sexual dichromatism. The proportion of dichromatic species over the total number of species evaluated is shown in parentheses for each tip. The phylogeny is modified from Pyron & Wiens (2011).

Wells, 2010; Rojas, 2016). Furthermore, sexual dichromatism is correlated with latitudinal variation in species richness, predation pressure and seasonality in other vertebrates (e.g. birds; Peterson, 1996; Badyaev & Hill, 2003; Friedman *et al.*, 2009; Simpson *et al.*, 2015; Dale *et al.*, 2015); therefore, we also investigated the relationship between dynamic dichromatism and species distribution (temperate or tropical) in hylids to assess whether broad macroecological factors may play a role in the prevalence and evolution of this trait.

Materials and methods

The distribution and diversity of dynamic sexual dichromatism in frogs

To establish the presence of dynamic sexual dichromatism in frogs, we searched the published scientific literature, consulted field guides, online databases (www.mphibianweb.org and www.iucnredlist.org) and expert knowledge. Due to its ephemeral nature, dynamic sexual dichromatism is rarely reported in the literature; therefore, we established a set of criteria by which to identify dynamic dichromatism using colour photographs of live frogs (Fig. S1). Briefly, we scored species as dichromatic when images of breeding condition males (vocalizing, in amplexus, or engaging in breeding behaviour) showed a distinctly different dorsal coloration than females and nonbreeding males. All photographic evidence was assessed independently by two of the authors (RCB and GNW), and species in which dynamic dichromatism is suspected but unconfirmed were not included in subsequent analyses. Therefore, our scoring system was conservative and likely underestimated the prevalence of dynamic dichromatism. The final database includes 2146 species (of ~6700 described anuran species) for which we assessed the presence/absence of dynamic dichromatism (Table S1). For each species with dynamic sexual dichromatism, we also report the baseline and display colours and associated references (Table S2).

The evolution of sexual dichromatism in the family Hylidae

We gathered data from the literature on whether species form large breeding aggregations (explosive or lek-breeding species, as defined by Wells, 1977), and biogeographic distribution (temperate/tropical) for all hylid species for which we assessed the presence/absence of dynamic dichromatism (Table S3). With some exceptions, the breeding season in many hylid frogs is not as dramatically condensed as that of classic explosive breeders, such as arid-adapted bufonid and scaphiopo-did frogs; however, following Wells (1977) we considered species in which individual populations have explosive breeding periods as 'explosive' (even if the breeding season of the species lasts for several months).

Likewise, many hylids are lek-breeding species and we included these in the large breeding aggregation category. Species that do not form large breeding aggregations included prolonged breeders and those that breed in bromeliads. Species for which minimal information on physical appearance or general ecology is available were not included in subsequent analyses.

We generated a phylogeny with nucleotide sequence data from GenBank for all hylids for which we had phenotypic data. The resulting data set included two mitochondrial and four nuclear loci for 355 hylid frogs (approximately 37% of the currently recognized species in the family Hylidae; Table S3). We aligned sequences with CLUSTAL X 2.0.10 (Larkin *et al.*, 2007), excluded any ambiguously aligned sites and used PARTITIONFINDER 1.1.10 (Lanfear *et al.*, 2012) to establish the substitution models and partitioning scheme for the analysis (Table S4). We estimated the phylogeny using Bayesian phylogenetic analyses implemented in BEAST 1.8.0 (Drummond *et al.*, 2012) with a speciation birth–death incomplete sampling tree prior and a lognormal relaxed clock that was unlinked across all markers. We enforced monophyly for the three subfamilies and 18 strongly supported monophyletic genera/lineages (bootstrap support >95 in Pyron & Wiens, 2011) and obtained posterior distributions from two independent Markov chain Monte Carlo (MCMC) simulations, each run for 350 million generations. We assessed parameter convergence using TRACER 1.5 (Rambaut *et al.*, 2013). The effective sample size (ESS) for each parameter was above 100, and simulations were repeated without sequence data to test the influence of priors on posterior distributions for all parameters. We combined tree files from replicate runs using LOGCOMBINER and discarded the first 150 million trees as burn-in prior to summarizing the posterior distribution of trees using TREEANNOTATOR and generating a maximum clade credibility (MCC) tree. To account for phylogenetic uncertainty in the subsequent comparative analyses, we randomly selected 100 post-burn-in phylogenies from the posterior distribution and conducted analyses using both the MCC tree and the randomly selected subset of trees.

Both forms of sexual dichromatism (dynamic and ontogenetic) are present in the Hylidae. To reconstruct the evolutionary history of dynamic dichromatism, ontogenetic dichromatism and monochromatism across hylid frogs, we implemented stochastic character mapping (Bollback, 2006) on the MCC tree. This method calculates the conditional likelihood of each character state at each node and then samples the posterior distribution of states to simulate transitions between character states along each branch. Character state transitions are conditioned on a rate matrix and can occur at any point along a branch (i.e. they are not restricted to speciation events). We used the package phytools (Revell, 2012) implemented in R 3.1.1 to summarize the results of 100 simulated reconstructions on the MCC tree comparing the equal-rates, symmetrical-rates and

all-rates-different models. We used a likelihood-ratio test to select the 'best' model and then reran the analysis across the set of 100 phylogenies to estimate transition rates between states.

To assess whether the evolution of dynamic dichromatism is correlated with biogeographic distribution or forming large breeding aggregations, we implemented BAYESTRAITS v3, a reversible-jump MCMC method (Pagel & Meade, 2006) that visits dependent and independent models of binary character evolution in proportion to their posterior probability. Using a randomly selected set of 100 phylogenies, we ran MCMC analyses for independent and dependent models of character evolution for dynamic dichromatism (present or absent) with respect to biogeographic distribution (temperate or tropical) and breeding system (presence/absence of large breeding aggregations). Dynamic and ontogenetic dichromatism may have important differences in terms of evolutionary lability and function; thus, we did not include the seven species with ontogenetic (permanent) sexual dichromatism in correlated trait analyses. We were unable to assess the breeding system for one dichromatic and two monochromatic species; therefore, these character states were coded as missing data in analyses. We ran two independent Markov chains of 50 million generations each for independent and dependent models, with and without the covarion model (Tuffley & Steel, 1998), which allows traits to vary their rate of evolution within and between branches. We assessed convergence and verified that ESS values for all parameters were >200 with TRACER 1.5 and compared the relative performance of the different models via Bayes factors (BF). For traits with strong support for the dependent model of character evolution (BF > 5), we discarded the first 5 million generations as burn-in and calculated mean transition rates among character states. Finally, we implemented multivariate phylogenetic logistic regression in the R package phylolm (Paradis & Claude, 2002; Ives & Garland, 2010; Tung Ho & Ane, 2014) to examine the correlation structure among dynamic dichromatism, biogeographic distribution, breeding system and the interaction between biogeographic distribution and breeding system. We used the logistic_MPLE method, which maximizes the penalized likelihood of the logistic regression, ran 10 000 bootstrap replicates on the MCC tree, and used a likelihood-ratio test to select the 'best' model. These analyses did not include the three species for which we were unable to assess the breeding system.

Results

The distribution and diversity of dynamic sexual dichromatism in frogs

Our study considerably extends the previously documented prevalence of dynamic dichromatism (31

species from nine families and subfamilies; Bell & Zamudio, 2012) to include 178 species from 15 families and subfamilies (Fig. 1, Table S2). Of these combined records, 67 are documented in the peer-reviewed literature or are based on the authors' personal observations, whereas the remaining 111 are based on the criteria described in the methods (Table S2). This is still likely an underestimate of this ephemeral trait, and we provide a list of species in which dynamic dichromatism is suspected but unconfirmed to guide future research efforts (Table S2). The vast majority of the new records are from the family Hylidae, which is a predominantly arboreal group of frogs in which dynamic dichromatism appears to be quite common and occurs in both temperate and tropical species (Figs 1 and S2). We also documented high levels of dichromatism in the family Bufonidae, a diverse lineage of chemically defended species that inhabit a variety of habitats, and in the family Ranidae and subfamily Limnodynastinae (Myobatrachidae), which include predominantly stream-dwelling, leaf litter and semi-fossorial species.

Among the 178 anuran species that exhibit dynamic sexual dichromatism, this temporary colour change typically results in yellower coloration (165 species; Table S2). In a few cases, however, male coloration changes to black/darker shades relative to baseline colour (e.g. the dendrobatids *M. trinitatis* and *Rheobates palmatus*, and the ranids *Rana graeca*, *Rana sylvatica*, *Rana temporaria* and *Clinotarsus curtipes*), brighter green coloration (e.g. the phrynobatrachid *Phrynobatrachus natalensis*, and the hylids *Hypsiboas rufitelus* and *Itapohyla langsdorffii*) or vibrant blue coloration (e.g. *R. arvalis*). Mutual dynamic colour change occurs in at least one of these 178 species, the bufonid *Xanthophryne tigerinus*, in which both males and females exhibit display coloration that is yellower relative to baseline coloration (Table S2).

The evolution of sexual dichromatism in the family Hylidae

Within the diverse and widespread family Hylidae, dynamic dichromatism is present in two of the three subfamilies and 121 species of the 491 we assessed (Fig. 1; Table S2). Over 25% of species we assessed in the Hylinae and Pelodyadinae subfamilies are sexually dichromatic, and frogs in these subfamilies are found in diverse temperate and tropical habitats throughout the Americas, Europe and Asia (Hylinae), and Australia and Papua New Guinea (Pelodyadinae). By contrast, none of the Phyllomedusinae species are sexually dichromatic. This lineage of arboreal hylids is restricted to the neotropics, and most species have specialized reproductive modes, laying their eggs on leaves overhanging water or in tree hole cavities. Species in all three subfamilies form large breeding aggregations.

To examine transitions in dynamic dichromatism, ontogenetic dichromatism and monochromatism across

hylids, we used stochastic trait mapping to reconstruct the timing and placement of changes on the hylid phylogeny. Our final data set of species for the analysis included 355 hylid species (seven with ontogenetic sexual dichromatism, 95 with dynamic sexual dichromatism and 253 sexually monochromatic species). We found the all-rates-different model was the best fit for mapping dichromatism on the hylid phylogeny (log-likelihood for equal-rates, symmetrical-rates and all-rates-different models was -285.884 , -249.797 and -233.640 , respectively; $\chi^2 = 32.3$, d.f. = 3, $P < 0.005$) and reconstructed monochromatism as the most likely ancestral state (Fig. S3). We found very high transition rates across the phylogeny (average of 205.6 changes between states across 10 000 simulations) demonstrating the high evolutionary lability of this trait. The vast majority of transitions occurred between dynamic dichromatism and monochromatism (Fig. 2), with roughly equivalent shifts from monochromatism to dynamic dichromatism (68.45) and reversals back to monochromatism (89.21). Transitions to ontogenetic dichromatism were less common and occurred more frequently from dynamic dichromatism (22.86) than from monochromatism (0.46), indicating that dynamic dichromatism may serve as an intermediate stage to ontogenetic dichromatism in this family (Fig. 2).

To test for potential selective mechanisms shaping the evolution of dynamic dichromatism in hylid frogs, we applied correlated trait evolution analyses between i) dynamic dichromatism and species distribution (temperate or tropical) and ii) dynamic dichromatism and large breeding aggregations (present/absent). We did not find evidence for concerted evolution between dynamic dichromatism and species distribution (Table 1). Although breeding aggregations in hylids are not typically considered 'explosive', more than half of

the species in our data set form large breeding aggregations and more than one-third of those species exhibit dynamic dichromatism (Fig. 3). The dependent model diagram of evolutionary rates shows that transitions to dynamic dichromatism are much more likely among species that form large breeding aggregations than those that do not (q24 vs. q13; Fig. 3, Table S5). In addition, dichromatism is retained in a few lineages that do not form large breeding aggregations (q43; Fig. 3). Thus, we see concerted evolution of dynamic dichromatism and breeding biology in hylids with the evolution of large breeding aggregations preceding the evolution of dynamic dichromatism. Finally, in our multivariate phylogenetic logistic regression analyses we found that the interaction between species distribution and breeding aggregation was a significant predictor of dynamic dichromatism in hylids (likelihood-ratio test $\chi^2 = 5.4$, d.f. = 1, $P < 0.05$; Table 2). This result indicates that the association between dynamic dichromatism and breeding aggregation differs between temperate and tropical regions such that breeding aggregation is a strong predictor of dynamic dichromatism in tropical regions and not in temperate regions.

Discussion

Widespread diversity of dynamic sexual dichromatism in anurans

We identified dynamic sexual dichromatism in 178 frog species from 15 families and subfamilies in both temperate and tropical regions. Although this is a substantial increase from the initial review in Bell & Zamudio (2012), we assessed only ~30% of described frog diversity and were conservative in our assessments based on photographs; thus, we expect that the prevalence of

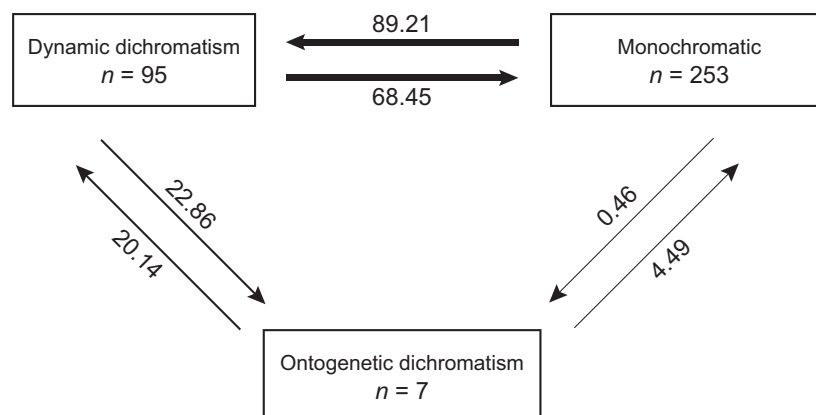


Fig. 2 Estimated transition rates between dynamic dichromatism, ontogenetic dichromatism and monochromatism in the Hylidae based on stochastic character mapping in phytools. The thickness of the arrows is proportional to the mean transition parameter estimate from 10 000 simulations across 100 phylogenies sampled from the posterior distribution of the phylogenetic analysis. Sample sizes included in the analysis are indicated for each character state.

Table 1 Transition rate models and associated marginal likelihood estimates from the stepping stone sampler for BAYESTRAITS analyses. Bayes factors for each model compared to the model with strongest support (based on Bayes factors >5) are shown, with the best model indicated in bold.

	Dichromatism vs. Species distribution		Dichromatism vs. Breeding aggregation	
	Likelihood	Bayes factor	Likelihood	Bayes factor
Independent	−318.71	43.28	−433.87	74.08
Independent cv	−297.07	–	−415.91	38.16
Dependent	−321.79	−49.44	−413.79	33.92
Dependent cv	−296.43	1.28	−396.83	–

dynamic dichromatism in frogs is much greater than what we report here. The broad phylogenetic distribution of dynamic sexual dichromatism across the anuran tree of life suggests that this trait has evolved multiple times and that the underlying mechanisms and function of dynamic dichromatism may vary among lineages. In contrast to the initial review, our updated assessment of dynamic dichromatism includes at least one species, the Amboli toad *X. tigerinus*, in which both males and females are bright yellow during the breeding season (Biju *et al.*, 2009). Mutual dynamic colour

change may also occur in a few other species including the microhylid *Cophixalus saxatilis* (Table S2). Bright coloration in female birds can reflect mutual mate choice (Hanssen *et al.*, 2006) or female territoriality and competition for limited resources (Heinsohn *et al.*, 2005; Murphy *et al.*, 2009). The Amboli toad thus presents an exciting opportunity to investigate whether similar mechanisms underlie mutual dynamic dichromatism in frogs.

Among frogs with dynamic sexual dichromatism, a temporary shift to yellow coloration is by far the most common display colour in both diurnal and nocturnal species (~95% of dichromatic species). The ability to discern colour differences is well documented in several diurnal anurans (Hailman & Jaeger, 1974; Sztatecsny *et al.*, 2012), but the extent of anuran colour discrimination in low-light conditions is unclear. Yellow coloration is typically brighter relative to background environments; therefore, in species that are active in low-light conditions, increased brightness of yellow frogs may serve as an effective achromatic signal. For example, males in the dynamically dichromatic yellow toads (*I.luetkenii*) distinguished between brown and yellow clay frog models in both daytime and night-time trials (Rehberg-Besler *et al.*, 2015), indicating that sexual differences in hue and/or luminance were

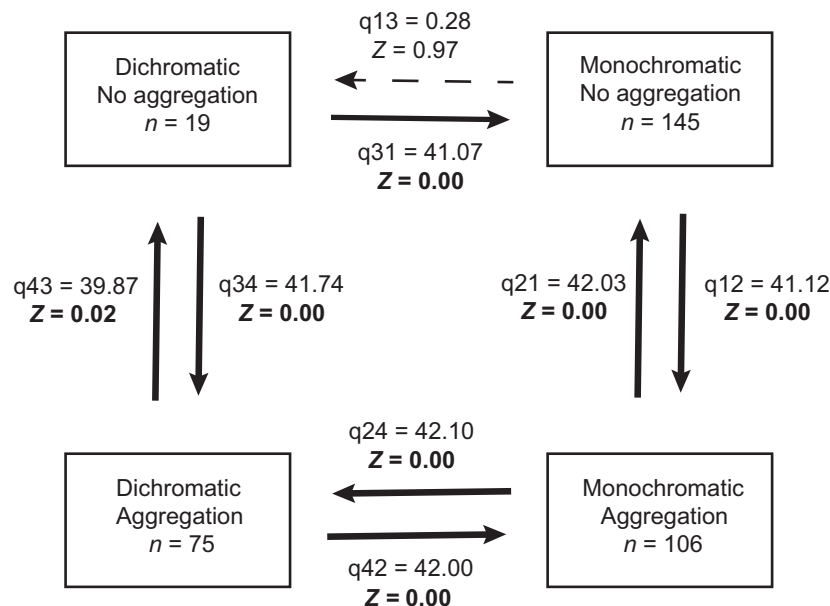


Fig. 3 Transition rates between discrete character states of dynamic dichromatism (present or absent) and large breeding aggregations (present or absent) across 348 species of hylid frogs taken from the posterior distribution of BAYESTRAITS. The thickness of the arrow is proportional to the mean transition parameter estimate (q_{ij}) from the posterior distribution of BAYESTRAITS, and Z indicates the frequency with which each was assigned to zero (Z) in the dependent model analysis. Probable transitions ($Z < 0.1$) are bolded, and improbable transitions ($Z \geq 0.20$) are represented by dashed arrows. Sample sizes (the number of species for each combination of traits) are indicated for each character state. We were unable to assess presence/absence of explosive breeding for one dichromatic and two monochromatic species; therefore, the sample sizes for these traits add up to 345 species.

Table 2 Summary of multivariate phylogenetic logistic regression analyses for the effects of large breeding aggregations (0 = absent, 1 = present) and distribution (0 = temperate, 1 = tropical) on dynamic dichromatism (0 = absent, 1 = present). Bolded predictors of dynamic dichromatism correspond to those significant at $P < 0.05$. Alpha is the phylogenetic correlation parameter estimate from phyloglm.

	Pen. Log-Likelihood	AIC	Breeding coefficient	Distribution coefficient	Breeding* Distribution coefficient	Alpha
Dichromatism ~ Breeding	−156.6	324.0	1.722	–	–	6.623
Dichromatism ~ Distribution	−192.1	395.2	–	0.641	–	11.489
Dichromatism ~ Breeding + Distribution	−150.7	315.4	1.843	0.829	–	3.799
Dichromatism ~ Breeding + Distribution + Breeding*Distribution	−148.0	313.3	0.672	−0.460	1.353	3.968

detectable in low-light conditions. The high prevalence of yellow coloration in dynamically dichromatic frogs may also reflect physiological limits as to which temporary colour changes are possible. Frog skin typically contains three pigment cell types: melanophores, which contain melanin; nonreflecting chromatophores, which contain pteridine or carotenoid pigments and are called xanthophores when they bestow yellow coloration and erythrophores when they bestow red coloration; and finally reflecting chromatophores called iridophores, which bestow bright and blue structural coloration (Bagnara, 1976). Histological studies in the dynamically dichromatic stony creek frogs (*L. wilcoxii*; Fig. 1) indicate that temporary changes to yellow coloration are accomplished by aggregating melanin within melanophores to reveal yellow xanthophores (Kindermann & Hero, 2016a), which may be a shared mechanism among frogs for displaying temporary yellow coloration. Thus, several nonmutually exclusive mechanisms may contribute to the high prevalence of yellow coloration in dynamically dichromatic anurans.

Trade-offs between natural and sexual selection with respect to signalling traits may partly explain why dynamic dichromatism is so common in hylid treefrogs. Hylids typically lack chemical defences and, like many frogs, rely on camouflage to avoid detection by visually oriented predators (Toledo & Haddad, 2009; Wells, 2010; Rojas, 2016). Thus, dynamic colour changes may serve as brief displays to attract the attention of intended receivers while enabling male frogs to restrict the amount of time they are conspicuous to unintended receivers. Anurans that are chemically defended, such as the bufonids and dendrobatids, exhibit conspicuous warning coloration year-round, indicating that increased visibility to predators is not costly in these species. Yet, several bufonids exhibit dynamic sexual dichromatism with males reverting to dull, baseline coloration after breeding (Table S1). Thus, the pervasiveness of dynamic dichromatism in both chemically defended and undefended species may reflect more universal benefits to remaining cryptic outside of the breeding season, or that maintaining bright coloration is energetically costly. Alternatively, changes in coloration may be linked to hormone fluctuation such that

these colour changes are only possible during the breeding season. Although the role of hormones in mediating dynamic dichromatism in bufonids has not yet been characterized, laboratory experiments in rhaconophorid and hylid treefrogs link increases in testosterone (*Buergeria robusta*; Tang *et al.*, 2014) or epinephrine (*L. wilcoxii*; Kindermann *et al.*, 2014) with dynamic yellow colour change. These distinct hormonal mechanisms, which are not necessarily linked with breeding activity in *L. wilcoxii*, may reflect differences in the adaptive (or possibly nonadaptive) role of dynamic colour change in different species.

The evolution of sexual dichromatism in hylid treefrogs

Within hylids, we found that transitions from dynamic dichromatism to ontogenetic dichromatism were much more frequent than those from monochromatism to ontogenetic dichromatism, suggesting that dynamic colour change may act as an evolutionarily intermediate state to ontogenetic colour change in this family. Furthermore, we found that the few hylids with ontogenetic dichromatism occur in tropical regions and do not necessarily form large breeding aggregations. Female choice is often cited as an evolutionary mechanism in sexually dichromatic vertebrates when males are the 'brighter' sex (Salthe, 1967; Andersson, 1982; Kodric-Brown & Brown, 1984; Wiens *et al.*, 1999; Cooper & Hosey, 2003; Todd & Davis, 2007; Caro, 2009) and is a more common component of courtship in frogs with prolonged breeding than in those with explosive breeding (Wells, 1977). Therefore, whereas behavioural studies and observations in dynamically dichromatic frogs indicate that temporary sexual colour differences serve as intrasexual signals (Wells, 1980; Sztatecsny *et al.*, 2012; Rehberg-Besler *et al.*, 2015), ontogenetic (permanent) colour differences, which are subject to natural and sexual selection year-round, may be more likely to serve as intersexual signals of mate quality (Vásquez & Pfennig, 2007; Maan & Cummings, 2009). Hylid genera with monochromatic, dynamically dichromatic and ontogenetically dichromatic species (*Scinax*, *Dendropsophus* and *Hypsiboas*) present an ideal opportunity to

address these hypotheses with natural history observations of courtship behaviours (e.g. foot-flagging in *D. parviceps*; Hödl & Amézquita, 2001), mating system dynamics (e.g. male–male antagonistic interactions), controlled behavioural experiments (e.g. female mate choice and male sex recognition) and assessments of predation risk (e.g. clay model attack rates; Kindermann & Hero, 2016b).

Across hylids, we found concerted evolution of dynamic sexual dichromatism and large breeding aggregations, and that the evolution of forming breeding aggregations precedes the evolution of dynamic dichromatism. Behavioural experiments in the yellow toad (family Bufonidae) and European moor frog (family Ranidae) demonstrated that dynamic dichromatism serves to improve sex recognition in large breeding aggregations (Sztatecsny *et al.*, 2012; Rehberg-Besler *et al.*, 2015). Male–male scramble competition for mates and multiple amplexus are common in hylids that form large breeding aggregations; thus, behavioural sex-recognition experiments in hylids may reveal that dynamic dichromatism serves the same function in a third anuran family. The correlated trait analysis, however, also indicates that dynamic dichromatism is retained in hylid lineages that do not form large breeding aggregations. In hylids with prolonged breeding, dynamic dichromatism may serve in territorial or dominance interactions (e.g. *M. trinitatis*; Wells, 1980), as indicators of male quality (e.g. *S. couchii*; Vásquez & Pfennig, 2007), or functions that have not yet been explored in other anurans. For example, in the stony creek frog (*L. wilcoxii*), male dynamic colour change intensifies during amplexus, which may serve as a dominance signal to other males (Kindermann & Hero, 2016b) or alternatively may simply be a by-product of hormones released during amplexus to elicit sperm release (Kindermann *et al.*, 2014). Future studies of hylid mating behaviour, colour change physiology and visual systems will highlight the social environments that drive the evolution of sexual dichromatism in frogs and will contribute to our understanding of the trade-offs between natural and sexual selection with respect to signalling traits in animals.

Although most of the hylid species that exhibit dynamic dichromatism have tropical distributions, we did not find support for concerted evolution of dichromatism with respect to biogeographic distribution across the hylid phylogeny. Dynamic dichromatism is present in both tropical and temperate hylines and pelodyadines and absent in phyllomedusines (which are entirely tropical). Nonetheless, the multivariate phylogenetic logistic regression recovers the interaction between biogeographic distribution and forming breeding aggregations as a significant predictor of dynamic dichromatism in hylids. Specifically, breeding aggregation is a strong predictor of dynamic dichromatism in tropical regions and not in temperate regions. This pattern may be driven by a few predominantly temperate genera in the

Hylinae (*Acris*, *Hyla* and *Pseudacris*), which occur in the Northern Hemisphere, and do not exhibit dynamic dichromatism even though a number of species form large breeding aggregations. Both Southern Hemisphere temperate species in the Hylinae that exhibit dynamic dichromatism belong to predominantly tropical genera, and one (*Scinax catharinae*) is an explosive breeder, whereas the other (*Hypsiboas leptolineatus*) is a pronged breeder in which males are territorial and engage in physical combat (Reinke & Deiques, 2010). By contrast, dynamic sexual dichromatism is common in both temperate and tropical pelodyadines, which occur throughout Australasia and include both explosive and prolonged breeders. Finally, phyllomedusines are an entirely tropical radiation in which many species form large breeding aggregations where visual signals, including colour pattern, are important aspects of intersexual communication (Jacobs *et al.*, 2016); yet, we did not find evidence of dynamic sexual dichromatism in this group. Amplexus and egg deposition in phyllomedusines are terrestrial (both occur on leaves overhanging water or in tree hole cavities; Zamudio *et al.*, 2016); therefore, scramble competition and mistakes in sex recognition may be less common in these species relative to frogs that deposit their eggs in shared aquatic habitats (i.e. ponds and streams). By contrast, most hylines and pelodyadines deposit their eggs in water, potentially creating more opportunities for mistakes in sex recognition or scramble competition to secure access to females when populations form large breeding aggregations (Wells, 1977). These differences in breeding biology and the biogeographic distribution of dynamic dichromatism among hylids indicate that macroecological differences between temperate and tropical regions, such as seasonality and the availability of breeding sites, may improve our understanding of ecological contexts in which dynamic dichromatism is likely to arise in tropical lineages and why it is retained in some temperate species and lost in others.

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Supporting information

Additional Supporting Information may be found online in the supporting information tab for this article:

Figure S1 Sample photographic criteria for assessing the presence of dynamic sexual dichromatism in the Australian hylid *Litoria revelata*.

Figure S2 Maximum clade credibility tree for 355 species of hylid frogs with traits (dynamic dichromatism, species distribution, and large breeding aggregation) indicated on the tips of the phylogeny.

Figure S3 Stochastic character mapping of dynamic dichromatism, ontogenetic dichromatism and monochromatism across 355 species of hylid frogs based on 100 SIMMAP simulations of character states on the MCC phylogeny implemented in phytools.

Table S1 List of anuran species assessed for the presence/absence of dynamic sexual dichromatism.

Table S2 Database of dynamic dichromatism for all families of frogs; male and female baseline colours; male and/or female display colour; evidence documenting dichromatism (D = documented in the literature, P = photographic, O = observed by authors); and associated references.

Table S3 GenBank accession numbers for two mitochondrial and four nuclear loci used to estimate the phylogeny of 355 hylid frogs in this study and trait values (dichromatism, geographic distribution, and breeding aggregation) for comparative analyses.

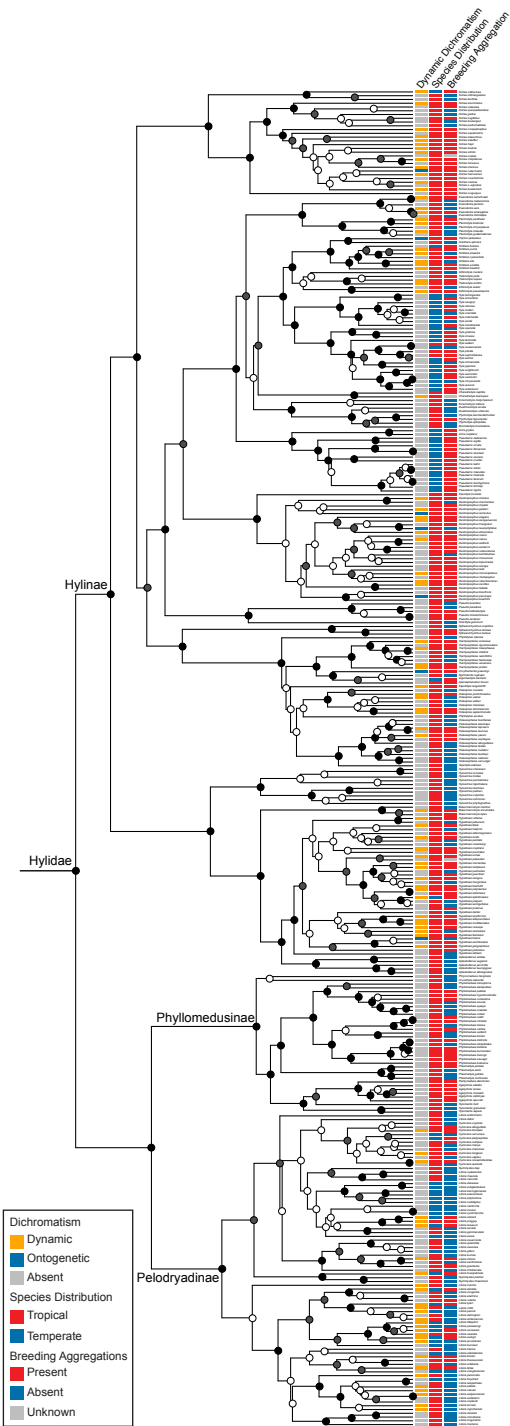
Table S4 Substitution models and partition schemes for Hylidae phylogenetic reconstruction.

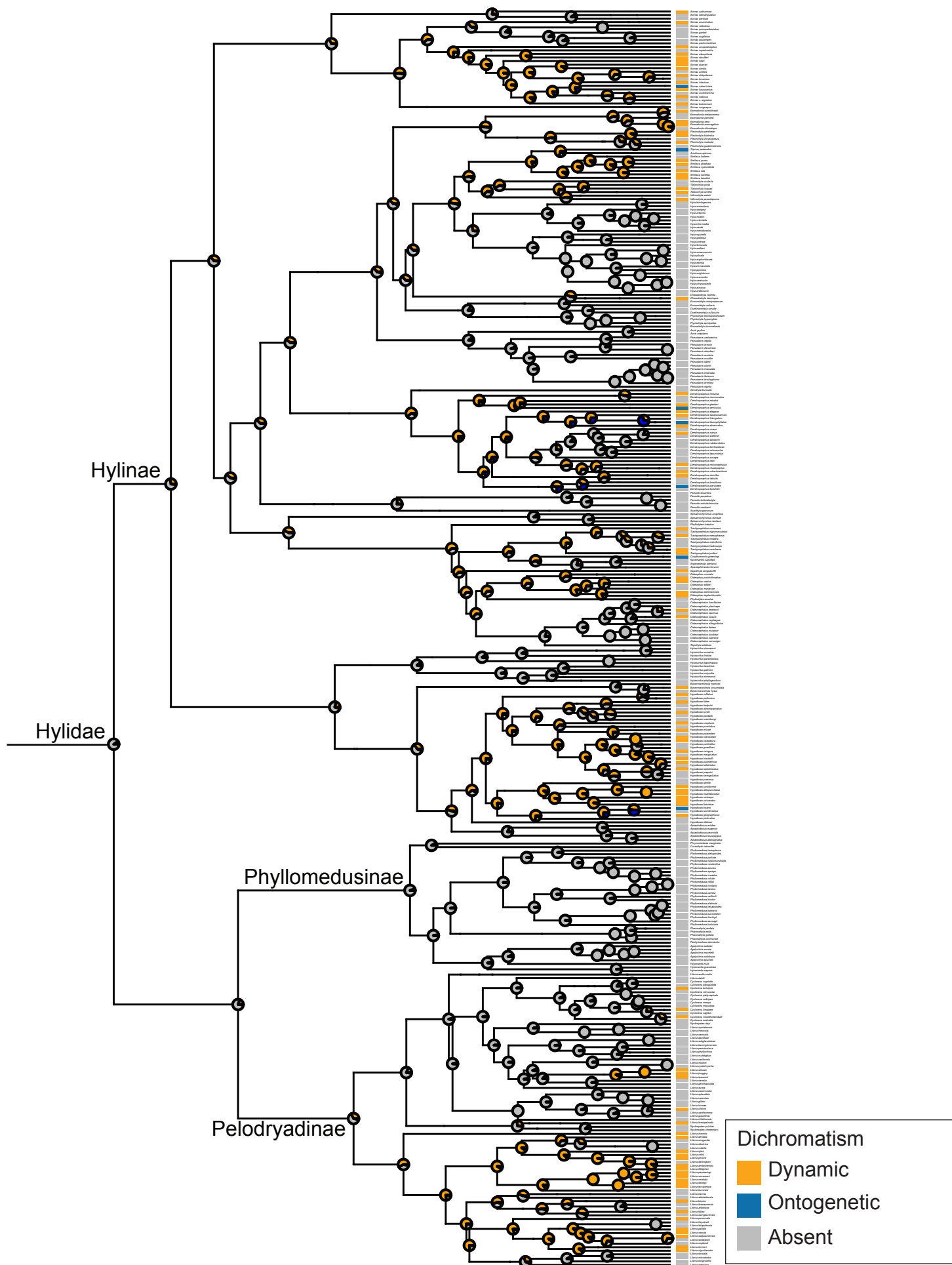
Table S5 Median and upper/lower 95% HPD for correlated dynamic dichromatism and breeding aggregation transition rates from the posterior distribution of BAYES-TRAITS analysis.

Data S1 Full citations for references in Table S2 and references used to assess breeding biology in hylids.

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Species and authority
<i>Acris blanchardi</i> Harper, 1947
<i>Acris crepitans</i> Baird, 1854
<i>Acris gryllus</i> (LeConte, 1825)
<i>Adelophryne baluensis</i> Hoogmoed, Borges, and Cascon, 1994
<i>Adelotus brevis</i> (Günther, 1853)
<i>Adelphobates castaneoticus</i> (Caldwell and Myers, 1990)
<i>Adelphobates galactonotus</i> (Steindachner, 1864)
<i>Adelphobates quinquevittatus</i> (Steindachner, 1864)
<i>Adenomus dasi</i> Manamendra-Arachchi and Pethiyagoda, 1998
<i>Adenomus vandianus</i> (Günther, 1872)
<i>Adenomus weileri</i> (Günther, 1858)
<i>Afraxalus "quadrivittatus"</i> Pickersgill, 2007
<i>Afraxalus delicatus</i> Pickersgill, 1984
<i>Afraxalus dorsalis</i> (Peters, 1876)
<i>Afraxalus enselcolae</i> Largen, 1974
<i>Afraxalus fomasini</i> (Blancoi, 1849)
<i>Afraxalus fulvivittatus</i> (Cope, 1861)
<i>Afraxalus laevis</i> (Ah!, 1930)
<i>Afraxalus caprio</i> (Ferreira, 1906)
<i>Afraxalus pendorata</i> Perret, 1960
<i>Afraxalus quadrivittatus</i> (Wehrer, 1908)
<i>Afraxalus spinifrons</i> (Cope, 1862)
<i>Afraxalus stuhlmanni</i> (Pfeiffer, 1863)
<i>Afraxalus ulugurensis</i> (Barbour and Loveridge, 1928)
<i>Agalychnis annae</i> (Duellman, 1963)
<i>Agalychnis aspera</i> (Peters, 1873)
<i>Agalychnis calliope</i> (Cope, 1862)
<i>Agalychnis dactylops</i> (Cope, 1864)
<i>Agalychnis granulosa</i> (Cruz, 1989)
<i>Agalychnis hui</i> (Duellman and Mendelson, 1965)
<i>Agalychnis lemur</i> (Boulenger, 1882)
<i>Agalychnis moreletii</i> (Duméril, 1853)
<i>Agalychnis saltator</i> Taylor, 1955
<i>Agalychnis spurrelli</i> Boulenger, 1913
<i>Aglyptodactylus laticeps</i> Glaw, Venegas, and Böhme, 1998
<i>Aglyptodactylus madagascariensis</i> (Duméril, 1853)
<i>Aglyptodactylus securifer</i> Glaw, Venegas, and Böhme, 1998
<i>Aibericus tuberculatus</i> (Richards, Johnston, and Burton, 1992)
<i>Allobates alexandri</i> (Grant and Rodríguez, 2001)
<i>Allobates bromelicola</i> (Test, 1956)
<i>Allobates brunneus</i> (Cope, 1887)
<i>Allobates caeruleodactylus</i> (Lima and Caldwell, 2001)
<i>Allobates crombiei</i> (Morales, 2002)
<i>Allobates femoralis</i> (Boulenger, 1884)
<i>Allobates gesconii</i> (Morales, 2002)
<i>Allobates granti</i> (Kok, MacCulloch, Gaucher, Poelman, Bourne, Lathrop, and Lenglet, 2006)
<i>Allobates hodii</i> Simões, Lima, and Farias, 2010
<i>Allobates inaperatus</i> (Morales, 2002)
<i>Allobates kingburyi</i> (Boulenger, 1918)
<i>Allobates marchesianus</i> (Meln, 1941)
<i>Allobates masingeri</i> (Morales, 2002)
<i>Allobates nivicola</i> (Caldwell and Lima, 2003)
<i>Allobates spumiponens</i> Kok and Ernst, 2007
<i>Allobates subulobonifrons</i> (Lima, Sanchez, and Souza, 2007)
<i>Allobates telamancus</i> (Cope, 1875)
<i>Allobates wayuu</i> (Acosta-Galvis, Cuentas, and Coloma, 1999)
<i>Allobates zaparo</i> (Silverstone, 1976)
<i>Allophryne rufiventris</i> Gaige, 1928
<i>Alzodes australis</i> Formas, Ubieda, Cuevas, and Nuñez, 1997
<i>Alzodes barreri</i> Veloso, Diaz, Ilarra-Constant, and Penna, 1981
<i>Alzodes hugo</i> Cuevas and Formas, 2001
<i>Alzodes igneus</i> Cuevas and Formas, 2005
<i>Alzodes montanus</i> (Lataste, 1902)
<i>Alzodes nodosus</i> (Duméril and Bibron, 1841)
<i>Alzodes norae</i> Cuevas, 2008
<i>Alzodes varzeolii</i> (Donoso-Barríos, 1974)
<i>Alytes ciliimanus</i> Bocak, 1979
<i>Alytes dickhilleni</i> Arntzen and García-París, 1995
<i>Alytes muletensis</i> (Sánchez and Adrover, 1979)
<i>Alytes ostetricans</i> (Laurent, 1768)
<i>Ameerega altamazonica</i> Twomey and Brown, 2008
<i>Ameerega bassleri</i> (Meln, 1941)
<i>Ameerega bilinguis</i> (Lunger, 1969)
<i>Ameerega boliviana</i> (Boulenger, 1902)
<i>Ameerega braccata</i> (Steindachner, 1864)
<i>Ameerega caranach</i> (Schulte, 1989)
<i>Ameerega erythronota</i> (Vigil and Miyata, 1980)
<i>Ameerega flavopicta</i> (Lutz, 1925)
<i>Ameerega hahnii</i> (Boulenger, 1884)
<i>Ameerega griseoides</i> Brown and Tiedney, 2009
<i>Ameerega maceo</i> (Rodríguez and Myers, 1993)
<i>Ameerega parvula</i> (Boulenger, 1882)
<i>Ameerega pascetti</i> Brown and Twomey, 2009
<i>Ameerega petersi</i> (Silverstone, 1976)
<i>Ameerega picta</i> (Bibron, 1836)
<i>Ameerega pampaleae</i> (Morales and Velasco, 1998)
<i>Ameerega pongoensis</i> (Schulte, 1999)
<i>Ameerega rubriventris</i> (Löters, Debold, Herle, Glaw, and Kreller, 1997)
<i>Ameerega silverstonei</i> (Myers and Daly, 1975)
<i>Ameerega simulans</i> (Myers, Rodríguez, and Icochea, 1998)
<i>Ameerega trivittata</i> (Spix, 1824)
<i>Amietia angolensis</i> (Bocage, 1869)
<i>Amietia fuscigula</i> (Duméril and Bibron, 1841)
<i>Amietia vandijvi</i> (Visser and Channing, 1997)
<i>Amietophrynus gemini</i> (Mook, 1897)
<i>Amietophrynus gracilipes</i> (Boulenger, 1899)
<i>Amietophrynus gutturalis</i> (Power, 1927)
<i>Amietophrynus koskenveni</i> (Loveridge, 1932)
<i>Amietophrynus emauri</i> (Boulenger, 1911)
<i>Amietophrynus maculatus</i> (Halliwell, 1854)
<i>Amietophrynus mauritanicus</i> (Schlegel, 1841)
<i>Amietophrynus partheuensis</i> (Smith, 1938)
<i>Amietophrynus rangeri</i> (Hewitt, 1935)
<i>Amietophrynus regularis</i> (Reuss, 1833)
<i>Amietophrynus lewisi</i> (Riedel and Ernst, 2000)
<i>Amietophrynus togoensis</i> (Ah!, 1924)
<i>Amietophrynus xeros</i> (Tandy, Tandy, Keith, and Duff-Mackay, 1976)
<i>Amolops chungensis</i> (Pope, 1925)
<i>Amolops granulatus</i> (Liu and Hu, 1961)
<i>Amolops hongkongensis</i> (Pope and Romer, 1951)
<i>Amolops lewisi</i> (Boulenger, 1889)
<i>Amolops marmoratus</i> (Blyth, 1855)
<i>Amolops panhai</i> Matsui and Nishitabata, 2006
<i>Amolops richetti</i> (Boulenger, 1899)
<i>Anaxyrus americanus</i> (Holbrook, 1836)
<i>Anaxyrus baxteri</i> (Porter, 1968)
<i>Anaxyrus boreas</i> (Baird and Girard, 1852)
<i>Anaxyrus californicus</i> (Camp, 1915)
<i>Anaxyrus canorus</i> (Camp, 1916)
<i>Anaxyrus cognatus</i> (Say, 1824)
<i>Anaxyrus compactilis</i> (Wiegmann, 1833)
<i>Anaxyrus debilis</i> (Girard, 1854)
<i>Anaxyrus eximius</i> (Myers, 1942)
<i>Anaxyrus fowleri</i> (Hindley, 1882)
<i>Anaxyrus hemiphysa</i> (Cope, 1886)
<i>Anaxyrus houstonensis</i> (Sanders, 1953)
<i>Anaxyrus kellyi</i> (Taylor, 1938)
<i>Anaxyrus microcephalus</i> (Cope, 1867)
<i>Anaxyrus nelsoni</i> (Sternberger, 1983)
<i>Anaxyrus punctatus</i> (Baird and Girard, 1852)
<i>Anaxyrus quercicus</i> (Holbrook, 1840)
<i>Anaxyrus reiformis</i> (Sanders and Smith, 1951)
<i>Anaxyrus speciosus</i> (Girard, 1854)
<i>Anaxyrus terrestris</i> (Bonnatere, 1769)
<i>Anaxyrus woodhousei</i> (Girard, 1854)
<i>Anodonthys boulengeri</i> Müller, 1952
<i>Anomaloglossus baobabtrachus</i> (Bostel and Massary, 1999)
<i>Anomaloglossus beckeri</i> (Noble, 1923)
<i>Anomaloglossus degranvillei</i> (Lescure, 1975)
<i>Anomaloglossus stapheni</i> (Martins, 1989)
<i>Anopheps spinosa</i> (Steindachner, 1864)
<i>Ansonia hantzi</i> Inger, 1960
<i>Ansonia latiffi</i> Wood, Grismer, Ahmad, and Senawi, 2008
<i>Ansonia leptopus</i> (Günther, 1872)
<i>Ansonia longidactyla</i> Inger, 1960
<i>Ansonia malayana</i> Inger, 1960
<i>Ansonia spinifer</i> (Mocquard, 1890)
<i>Apapapendodon brunus</i> Miranda-Ribeiro, 1920
<i>Apistodiscus albopagulus</i> (Lutz and Lutz, 1938)
<i>Apistodiscus aridae</i> (Cruz and Peixoto, 1987)
<i>Apistodiscus ehrhardti</i> (Müller, 1924)
<i>Apistodiscus eugenioi</i> (Carvalho e Silva and Carvalho e Silva, 2005)
<i>Apistodiscus buraplangae</i> (Cruz, Pimenta, and Silvino, 2003)
<i>Apistodiscus leucopygus</i> (Cruz and Peixoto, 1985)
<i>Apistodiscus pervillei</i> Lutz, 1960
<i>Arenophryne rotunda</i> Tyler, 1976
<i>Argenteohyla siemensi</i> (Mertens, 1937)

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<i>Ariequius krebsi</i> (Mertens, 1938)
<i>Arrivoletia adelphus</i> Perret, 1956
<i>Arrivoletia affinis</i> Ahl, 1939
<i>Arrivoletia poecilochotus</i> Peters, 1863
<i>Arrivoletia stenocephala</i> Pfeffer, 1893
<i>Arrivoletia sylvaticus</i> (Laurent, 1954)
<i>Arrivoletia variabilis</i> Matschie, 1893
<i>Arrivoletia venodicyclodes</i> Hewitt, 1933
<i>Ascapthus montanus</i> Mittleman and Myers, 1949
<i>Ascapthus truesi</i> Stejneger, 1899
<i>Asus dentifrons</i> (Loweislager, 1933)
<i>Ateiolepnathus nitor</i> (Barrio, 1973)
<i>Ateolopus barbotini</i> Lescure, 1981
<i>Ateolopus bomolochus</i> Peters, 1973
<i>Ateolopus certus</i> Barbour, 1923
<i>Ateolopus chiquensis</i> Shreve, 1936
<i>Ateolopus cruciger</i> (Lichtenstein and Mertens, 1856)
<i>Ateolopus elegans</i> (Boulenger, 1882)
<i>Ateolopus exiguus</i> Boettger, 1892
<i>Ateolopus ferci</i> Lynch, 1993
<i>Ateolopus fuscus</i> Duméril and Bibron, 1841
<i>Ateolopus franciscus</i> Lescure, 1974
<i>Ateolopus gyphus</i> Dunn, 1931
<i>Ateolopus hallekii</i> Peters, 1973
<i>Ateolopus hoogmoedi</i> Lescure, 1974
<i>Ateolopus grescens</i> (Cornelis, 1849)
<i>Ateolopus helmsmus</i> Ruiz-Carranza, Ardila-Robayo, and Hernández-Camacho, 1994
<i>Ateolopus limosus</i> Ibañez, Jaramillo, and Solís, 1995
<i>Ateolopus mindoensis</i> Peters, 1973
<i>Ateolopus mucidogeniss</i> (Reyer, 1974)
<i>Ateolopus mulica</i> Rueda-Almonacid and Hoyos, 1992
<i>Ateolopus nanay</i> Coloma, 2002
<i>Ateolopus onorei</i> Coloma, Llofers, Duellman, and Miranda-Leiva, 2007
<i>Ateolopus pulcher</i> Boulenger, 1882
<i>Ateolopus simulatus</i> Ruiz-Carranza and Osorno-Muñoz, 1994
<i>Ateolopus sonori</i> La Marca, 1983
<i>Ateolopus spumarius</i> Cope, 1871
<i>Ateolopus varius</i> (Lichtenstein and Mertens, 1856)
<i>Ateolopus zelandi</i> Dunn, 1933
<i>Aubria subsagittata</i> (Duméril, 1856)
<i>Austrochaperina adelphie</i> (Zweifel, 1965)
<i>Austrochaperina fryi</i> (Zweifel, 1965)
<i>Austrochaperina gracilipes</i> Fry, 1912
<i>Austrochaperina pluvialis</i> (Zweifel, 1965)
<i>Austrochaperina robusta</i> Fry, 1912
<i>Babina edroptera</i> (Boulenger, 1909)
<i>Babina hoisi</i> (Boulenger, 1892)
<i>Babina okavensis</i> (Boettger, 1895)
<i>Barbouria dussumieri</i> Taylor and Noble, 1924
<i>Barycholos fernetzi</i> (Miranda-Ribeiro, 1937)
<i>Batrachyla tantanica</i> Barrio, 1967
<i>Batrachyla tepopus</i> Bell, 1943
<i>Batrachyla taeniata</i> (Girard, 1855)
<i>Bismmeria angolae</i> Andreone, Rosa, Noll, Crotini, Vences, and Raxworthy, 2010
<i>Bismmeria wittii</i> (Gülb, 1974)
<i>Bokermannohyla alvarengi</i> (Bokermann, 1956)
<i>Bokermannohyla circumdata</i> (Cope, 1871)
<i>Bokermannohyla hyla</i> (Reyer, 1985)
<i>Bokermannohyla martinii</i> (Bokermann, 1964)
<i>Bokermannohyla nanuzae</i> (Bokermann and Sazima, 1973)
<i>Bokermannohyla pseudopodiceps</i> (Miranda-Ribeiro, 1937)
<i>Bombina bombina</i> (Linnaeus, 1761)
<i>Bombina maxima</i> (Boulenger, 1905)
<i>Bombina orientalis</i> (Boulenger, 1909)
<i>Bombina pachypus</i> (Bonaparte, 1838)
<i>Bombina variegata</i> (Linnaeus, 1758)
<i>Bophis albibrans</i> (Boulenger, 1886)
<i>Bophis ankaratra</i> Andreone, 1993
<i>Bophis boehmei</i> Glaw and Vences, 1992
<i>Bophis bolinae</i> Vences and Glaw, 2002
<i>Bophis brachyrrh</i> (Boettger, 1882)
<i>Bophis calcaratus</i> Viallan, Vences, and Glaw, 2010
<i>Bophis douflii</i> (Anguill, 1934)
<i>Bophis elvinae</i> Andreone, 1993
<i>Bophis goudoti</i> Tschudi, 1838
<i>Bophis gulbei</i> (McCarthy, 1978)
<i>Bophis ides</i> (Steindachner, 1867)
<i>Bophis lichenoides</i> Viallan, Glaw, Andreone, and Cadle, 1998
<i>Bophis illianae</i> Köhler, Glaw, and Vences, 2008
<i>Bophis luciae</i> Glaw, Köhler, De la Riva, Vences, and Vences, 2010
<i>Bophis luteus</i> (Boulenger, 1882)
<i>Bophis madagascariensis</i> (Peters, 1874)
<i>Bophis marquetensis</i> Glaw and Vences, 1994
<i>Bophis microtypanum</i> (Boettger, 1881)
<i>Bophis occidentalis</i> Glaw and Vences, 1994
<i>Bophis opisthoton</i> (Boulenger, 1886)
<i>Bophis pauliani</i> (Gülb, 1953)
<i>Bophis peregrites</i> Cadle, 1995
<i>Bophis pyrrhus</i> Glaw, Vences, Andreone, and Viallan, 2001
<i>Bophis quasiboehmei</i> Vences, Köhler, Crotini, and Glaw, 2010
<i>Bophis rappodes</i> (Ahl, 1928)
<i>Bophis reliculatus</i> Bornmren-Schlosser, 1979
<i>Bophis septentrionalis</i> Glaw and Vences, 1994
<i>Bophis sibilans</i> Glaw and Thiesmeier, 1993
<i>Bophis tanjoka</i> Köhler, Glaw, and Vences, 2008
<i>Bophis tschircomyx</i> (Duméril, 1853)
<i>Bophis tsilomaro</i> Vences, Andreone, Glaw, and Glaw, 2010
<i>Bophis viridis</i> Bornmren-Schlosser, 1979
<i>Brachycephalus discipulus</i> (Zackisch, 1971)
<i>Brachycephalus ephippium</i> (Spix, 1824)
<i>Brachycephalus rodolphi</i> Miranda-Ribeiro, 1920
<i>Brachycephalus cerniense</i> (Boulenger, 1886)
<i>Brachycephalus fenei</i> (Boulenger, 1887)
<i>Brachycephalus intermedius</i> (Smith, 1921)
<i>Brevicaps aliparus</i> Peters, 1882
<i>Brevicaps fuscus</i> Hewitt, 1925
<i>Brevicaps gibbosus</i> (Linnaeus, 1758)
<i>Brevicaps macrops</i> Boulenger, 1907
<i>Brevicaps montanus</i> Power, 1926
<i>Brevicaps mossambicus</i> Peters, 1854
<i>Brevicaps verrucosus</i> Repp, 1842
<i>Bromelohyla bromelacea</i> (Schmidt, 1933)
<i>Bryophryne caphites</i> (Lynch, 1975)
<i>Bryophryne gymnotis</i> Lehr and Calenazzi, 2009
<i>Bryophryne harasaueri</i> Lehr and Calenazzi, 2009
<i>Bryophryne nubilosa</i> Lehr and Calenazzi, 2008
<i>Bryophryne zonata</i> Lehr and Calenazzi, 2009
<i>Buergera buergeri</i> (Temminck and Schlegel, 1838)
<i>Buergera japonica</i> (Halobwell, 1861)
<i>Buergera robusta</i> (Boulenger, 1909)
<i>Bufo barkerensis</i> Barbour, 1908
<i>Bufo bufo</i> (Linnaeus, 1758)
<i>Bufo eichwaldi</i> Litvinchuk, Borkin, Skorinov, and Rosanov, 2008
<i>Bufo garzanti</i> Cantón, 1942
<i>Bufo japonicus</i> Temminck and Schlegel, 1838
<i>Bufo stejnegeri</i> Schmidt, 1931
<i>Cacosternum boettgeri</i> (Boulenger, 1882)
<i>Cacosternum nanum</i> Boulenger, 1887
<i>Cacosternum platys</i> Rose, 1950
<i>Callisaurus guttata</i> (Blyth, 1866)
<i>Callisaurus kiswamsu</i> de Sá, Loader, and Channing, 2004
<i>Callisaurus krefftii</i> Noden, 1911
<i>Calypsocephale garyi</i> (Duméril and Bibron, 1841)
<i>Cardioglossa leucomystax</i> (Boulenger, 1903)
<i>Cardioglossa pulchra</i> Schiøtz, 1963
<i>Cardioglossa schoetzi</i> Amel, 1983
<i>Centrolene outax</i> (Lynch and Duellman, 1973)
<i>Centrolene buckleyi</i> (Boulenger, 1882)
<i>Centrolene dacotileum</i> (Ruiz-Carranza and Lynch, 1991)
<i>Centrolene heloderma</i> (Duellman, 1981)
<i>Centrolene lynchi</i> (Duellman, 1980)
<i>Centrolene perspicillum</i> (Lynch and Duellman, 1973)
<i>Centrolene purpuratum</i> Ruiz-Carranza and Lynch, 1995
<i>Centrolene savageri</i> (Ruiz-Carranza and Lynch, 1991)
<i>Ceratobatrachus guentheri</i> Boulenger, 1884
<i>Ceratophrys aurita</i> (Raddi, 1823)
<i>Ceratophrys calcarata</i> Boulenger, 1890
<i>Ceratophrys cornuta</i> (Linnaeus, 1758)
<i>Ceratophrys cranwelli</i> Barrio, 1960
<i>Ceratophrys joazeirensis</i> Mercadal de Barrio, 1966
<i>Ceratophrys ornata</i> (Bell, 1943)
<i>Ceratophrys storeri</i> (Steindachner, 1882)
<i>Ceratophrys testudo</i> Andersson, 1946
<i>Chacophrys pierotti</i> (Vellard, 1948)
<i>Chaperina lucas</i> Mocquard, 1892
<i>Charadrahyla allipotens</i> (Duellman, 1968)
<i>Charadrahyla nephila</i> (Mendelson and Campbell, 1999)

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<i>Charadrius taenopus</i> (Günther, 1901)
<i>Chasmocleis albopunctata</i> (Boettger, 1895)
<i>Chasmocleis analapa</i> Walker and Duetiman, 1974
<i>Chasmocleis ewingreanae</i> Petros and Szurao, 2008
<i>Chasmocleis bessleri</i> Dunn, 1949
<i>Chasmocleis leucosticta</i> (Boulenger, 1888)
<i>Chasmocleis panamensis</i> Dunn, Trapido, and Evans, 1948
<i>Chasmocleis shufeldtensis</i> Dunn, 1949
<i>Chasmocleis ventrimaculata</i> (Anderson, 1946)
<i>Chimerella mariaeanae</i> (Cianeros-Heredia and McDiarmid, 2006)
<i>Chiromantis donae</i> (Boulenger, 1903)
<i>Chiromantis hansenae</i> (Cochran, 1927)
<i>Chiromantis nongkhorensis</i> (Cochran, 1927)
<i>Chiromantis petersi</i> Boulenger, 1882
<i>Chiromantis rufescens</i> (Günther, 1889)
<i>Chiromantis vittatus</i> (Boulenger, 1887)
<i>Chiromantis xerampelina</i> Peters, 1854
<i>Chirostus koehleri</i> (Mertens, 1940)
<i>Clinotarsus curtipis</i> (Jerdon, 1854)
<i>Cochranella eukrenos</i> (Savage and Starnett, 1967)
<i>Cochranella grandiosa</i> (Taylor, 1949)
<i>Cochranella mache</i> Guayasamin and Bonaccorso, 2004
<i>Colostethus federici</i> (Svensson, 1971)
<i>Colostethus ngounala</i> (Cope, 1868)
<i>Colostethus panamensis</i> (Dunn, 1933)
<i>Colostethus pratti</i> (Boulenger, 1899)
<i>Colostethus ruficeps</i> Kaptein, 1967
<i>Corraua allenii</i> (Barbour and Loveridge, 1927)
<i>Corraua grahami</i> (Boulenger, 1906)
<i>Corraua robusta</i> Nielsen, 1958
<i>Cophixalus aenigma</i> Hoskin, 2004
<i>Cophixalus bombiens</i> Zweifel, 1985
<i>Cophixalus boeleni</i> Zweifel, 1985
<i>Cophixalus infacelus</i> Zweifel, 1985
<i>Cophixalus mcdonaldi</i> Zweifel, 1985
<i>Cophixalus monticola</i> (Richard, Dennis, Trenerry, and Werren, 1994)
<i>Cophixalus neglectus</i> Zweifel, 1982
<i>Cophixalus ornatus</i> (Fry, 1912)
<i>Cophixalus savatilis</i> Zweifel and Parker, 1977
<i>Cophixalus variabilis</i> Kraus and Allison, 2006
<i>Cophyla barra</i> Vences, Andreone, and Glaw, 2005
<i>Cophyla occulta</i> (Glaw and Vences, 1992)
<i>Cophyla phyllodactyla</i> Boettger, 1880
<i>Corythomantis greeningi</i> Boulenger, 1896
<i>Craugastor andrei</i> (Boulenger, 1898)
<i>Craugastor andrei</i> (Savage, 1974)
<i>Craugastor augusti</i> (Dugès, 1879)
<i>Craugastor brandfordi</i> (Cope, 1868)
<i>Craugastor chiesi</i> (Savage, 1987)
<i>Craugastor crassidigitus</i> (Taylor, 1952)
<i>Craugastor decoratus</i> (Taylor, 1942)
<i>Craugastor emleri</i> (Dunn and Emelen, 1932)
<i>Craugastor fitzingeri</i> (Schmidt, 1857)
<i>Craugastor golimieri</i> (Peters, 1863)
<i>Craugastor levisinus</i> (Werner, 1896)
<i>Craugastor laticeps</i> (Duméril, 1853)
<i>Craugastor loei</i> (Shannon and Wierler, 1955)
<i>Craugastor longirostris</i> (Boulenger, 1898)
<i>Craugastor megacephalus</i> (Cope, 1875)
<i>Craugastor melanosictus</i> (Cope, 1875)
<i>Craugastor mexicanus</i> (Broch, 1877)
<i>Craugastor mimus</i> (Taylor, 1955)
<i>Craugastor nobeli</i> (Barbour and Dunn, 1921)
<i>Craugastor odoefferus</i> (Cope, 1875)
<i>Craugastor polypichus</i> (Cope, 1888)
<i>Craugastor punctatulus</i> (Peters, 1863)
<i>Craugastor raniformis</i> (Boulenger, 1898)
<i>Craugastor rhodops</i> (Cope, 1867)
<i>Craugastor rostralis</i> (Werner, 1896)
<i>Craugastor rugosus</i> (Peters, 1873)
<i>Craugastor rufinus</i> (Campbell and Savage, 2000)
<i>Craugastor sabrinus</i> (Campbell and Savage, 2000)
<i>Craugastor sandersoni</i> (Schmidt, 1941)
<i>Craugastor thymenianus</i> (Cope, 1863)
<i>Craugastor talamancas</i> (Dunn, 1931)
<i>Craugastor taurus</i> (Taylor, 1958)
<i>Craugastor underwoodi</i> (Boulenger, 1896)
<i>Crepidophryne choppe</i> Vaughan and Mendelson, 2007
<i>Crepidophryne epistoca</i> (Cope, 1875)
<i>Crima bingus</i> (Martin, Tyler, and Davies, 1980)
<i>Crima deserticola</i> (Liem and Ingram, 1977)
<i>Crima fimbriata</i> Doughty, Anolis, and Price, 2009
<i>Crima georgiana</i> Tschudi, 1838
<i>Crima glauerti</i> Loveridge, 1933
<i>Crima insignifera</i> Moore, 1954
<i>Crima nimbus</i> (Koussenev, Ziegler, Brown, Davies, and Littlejohn, 1994)
<i>Crima parvinsignifera</i> Main, 1957
<i>Crima pseudinsignifera</i> Main, 1957
<i>Crima remota</i> (Tyler and Parker, 1974)
<i>Crima riparia</i> Littlejohn and Martin, 1965
<i>Crima signifera</i> Girard, 1853
<i>Crima sloceri</i> Littlejohn, 1959
<i>Crima subinsignifera</i> Littlejohn, 1957
<i>Crima tasmaniensis</i> (Günther, 1864)
<i>Crima trinitatis</i> (Slaughter and Main, 1968)
<i>Crossodactylus caramaschi</i> Bastos and Pombal, 1995
<i>Cruziohyla calceifer</i> (Boulenger, 1902)
<i>Cruziohyla crassipodus</i> (Furkholz, 1957)
<i>Cryptobatrachus fahmyi</i> (Peracca, 1914)
<i>Cryptobatrachus remotus</i> Infante-Rivero, Rojas-Runjaic, and Barrio-Amorós, 2009
<i>Cryptohyla greshoffi</i> (Schilthuis, 1899)
<i>Cryptohyla peyeri</i> McCouart, 1904
<i>Dendrobates auratus</i> (Girard, 1855)
<i>Dendrobates eucomelas</i> Steindachner, 1864
<i>Dendrobates incinctus</i> (Lwin, 1797)
<i>Dendrobates truncatus</i> (Cope, 1861)
<i>Dendrophysicus berthaultzae</i> (Bockstein, 1994)
<i>Dendrophysicus brevipollicatus</i> (Jiménez de la Espada, 1870)
<i>Dendrophysicus leucomystax</i> (Bockstein, 1998)
<i>Dendrophysicus minutus</i> (Mein, 1941)
<i>Dendrosophus anceps</i> (Lutz, 1925)
<i>Dendrosophus berthaultzae</i> (Bokermann, 1962)
<i>Dendrosophus bifurcus</i> (Anderson, 1946)
<i>Dendrosophus bryantalis</i> (Sps., 1824)
<i>Dendrosophus bogerti</i> (Cochran and Goin, 1970)
<i>Dendrosophus bokermanni</i> (Goin, 1960)
<i>Dendrosophus branderi</i> (Cochran, 1948)
<i>Dendrosophus brevifrons</i> (Duellman and Crump, 1974)
<i>Dendrosophus carmifex</i> (Duellman, 1969)
<i>Dendrosophus columbianus</i> (Boulenger, 1892)
<i>Dendrosophus cruzi</i> (Pombal and Bastos, 1998)
<i>Dendrosophus decipiens</i> (Lutz, 1925)
<i>Dendrosophus delavayi</i> (Köhler and Lötters, 2001)
<i>Dendrosophus dufrei</i> (Gomes and Pezold, 1985)
<i>Dendrosophus ebraccatus</i> (Cope, 1874)
<i>Dendrosophus elegans</i> (West-Neuweit, 1824)
<i>Dendrosophus eleanae</i> (Napoli and Caramaschi, 2000)
<i>Dendrosophus gaucheri</i> (Lescure and Marty, 2000)
<i>Dendrosophus giesleri</i> (Mertens, 1950)
<i>Dendrosophus grandis</i> (Goin, 1965)
<i>Dendrosophus jimi</i> (Napoli and Caramaschi, 1999)
<i>Dendrosophus jamaicae</i> (Köhler and Lötters, 2001)
<i>Dendrosophus koehleri</i> (Duellman and Trueb, 1989)
<i>Dendrosophus labialis</i> (Peters, 1863)
<i>Dendrosophus leali</i> (Bokermann, 1964)
<i>Dendrosophus leucophyllatus</i> (Leivers, 1833)
<i>Dendrosophus luteocellatus</i> (Roux, 1927)
<i>Dendrosophus marmoratus</i> (Laurent, 1768)
<i>Dendrosophus melanargyreus</i> (Cope, 1887)
<i>Dendrosophus meridenas</i> (Rivero, 1961)
<i>Dendrosophus microcephalus</i> (Cope, 1886)
<i>Dendrosophus microps</i> (Peters, 1872)
<i>Dendrosophus minusculus</i> (Rivero, 1971)
<i>Dendrosophus minutus</i> (Peters, 1872)
<i>Dendrosophus mystax</i> (Vigle and Guberstein-Vigle, 1980)
<i>Dendrosophus nahdereri</i> (Lutz and Bokermann, 1963)
<i>Dendrosophus nanus</i> (Boulenger, 1889)
<i>Dendrosophus parviceps</i> (Boulenger, 1892)
<i>Dendrosophus phaeodactylus</i> (Steindachner, 1866)
<i>Dendrosophus rhodopeus</i> (Günther, 1858)
<i>Dendrosophus rivero</i> (Cochran and Goin, 1970)
<i>Dendrosophus robertmermeri</i> (Taylor, 1937)
<i>Dendrosophus rubicundulus</i> (Reinhardt and Lötters, 1982)
<i>Dendrosophus saili</i> (Jungfer, Reiche, and Plakunek, 2010)
<i>Dendrosophus sanborni</i> (Schmidt, 1944)
<i>Dendrosophus sarayacuensis</i> (Shreve, 1935)
<i>Dendrosophus seniculus</i> (Cope, 1868)

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<i>Dendropsophus soarezi</i> (Caramaschi and Jim, 1963)
<i>Dendropsophus subocularis</i> (Dunn, 1934)
<i>Dendropsophus triangulum</i> (Günther, 1869)
<i>Dendropsophus tilaniatus</i> (Bokermann, 1965)
<i>Dendropsophus wallfordi</i> (Bokermann, 1962)
<i>Dendropsophus werneri</i> (Cochran, 1952)
<i>Dermatorhinus muelleri</i> (Boettger, 1885)
<i>Diplazia spatulata</i> (Günther, 1862)
<i>Diaporus anthrax</i> (Lynch, 2001)
<i>Diaporus diastema</i> (Cope, 1875)
<i>Diaporus gularis</i> (Boulenger, 1898)
<i>Diaporus hyaliformis</i> (Cope, 1875)
<i>Diaporus ventrimaculatus</i> Chavez, García-Rodríguez, Mora, and Leal, 2009
<i>Dichmanopus gotschli</i> Anderson, 1993
<i>Diagoglossus feanese</i> Busack, 1986
<i>Diagoglossus nigriventris</i> Mendelssohn and Steinitz, 1943
<i>Diagoglossus pilus</i> Olf, 1837
<i>Diagoglossus sardus</i> Tschudi, 1837
<i>Diagoglossus scovazzi</i> Camerano, 1878
<i>Duellmanohyla rufocula</i> (Taylor, 1952)
<i>Duellmanohyla salvadori</i> (McCranie and Wilson, 1986)
<i>Duellmanohyla sorata</i> (Wilson and McCranie, 1985)
<i>Duellmanohyla uranochroa</i> (Cope, 1875)
<i>Duellaphrynus arabicus</i> Heyden, 1927
<i>Duellaphrynus eukoraei</i> (Bogert and Senanayake, 1966)
<i>Duellaphrynus circus</i> (Wogan, Win, Thin, Lewin, Shein, Kyi, and Tun, 2003)
<i>Duellaphrynus dhufarensis</i> (Parker, 1931)
<i>Duellaphrynus himalayanus</i> (Günther, 1864)
<i>Duellaphrynus melanostictus</i> (Schneider, 1799)
<i>Duellaphrynus olivaceus</i> (Bleeker, 1874)
<i>Duellaphrynus scaber</i> (Schneider, 1799)
<i>Duellaphrynus stomatichus</i> (Lütken, 1864)
<i>Dyscophus antongali</i> Grandidier, 1877
<i>Dyscophus guinei</i> (Grandidier, 1875)
<i>Dyscophus insularis</i> Grandidier, 1872
<i>Ecnomiophyla fimbriemembra</i> (Taylor, 1943)
<i>Ecnomiophyla miliaria</i> (Cope, 1866)
<i>Ecnomiophyla molympianum</i> (Cope, 1863)
<i>Ecnomiophyla raborum</i> Mendelson, Savage, Griffith, Ross, Kubicki, and Gagliardo, 2008
<i>Eridorhinia perezi</i> Jiménez de la Espada, 1870
<i>Elachistocleis bicolor</i> (Guérin-Meneville, 1838)
<i>Elachistocleis erythrogaster</i> Kiewit and Di Bernardo, 1998
<i>Elachistocleis ovalis</i> (Schneider, 1799)
<i>Elachistocleis surinamensis</i> (Daudin, 1802)
<i>Eleutherodactylus abbotti</i> Cochran, 1952
<i>Eleutherodactylus smaragdus</i> Hedges, Thomas, and Franz, 1987
<i>Eleutherodactylus antillensis</i> (Rehnardt and Lütken, 1863)
<i>Eleutherodactylus atkinsi</i> Dunn, 1925
<i>Eleutherodactylus audeni</i> Cochran, 1934
<i>Eleutherodactylus auriculatus</i> (Cope, 1862)
<i>Eleutherodactylus bakeri</i> Cochran, 1935
<i>Eleutherodactylus bellingeri</i> Lynch, 1965
<i>Eleutherodactylus bartonsmithi</i> Schwartz, 1960
<i>Eleutherodactylus brittoni</i> Schmidt, 1920
<i>Eleutherodactylus cochraniae</i> Grant, 1932
<i>Eleutherodactylus cooki</i> Grant, 1932
<i>Eleutherodactylus coqui</i> Thomas, 1966
<i>Eleutherodactylus cuniculipes</i> Schwartz, 1964
<i>Eleutherodactylus cundalli</i> Dunn, 1926
<i>Eleutherodactylus cystignathoides</i> (Cope, 1877)
<i>Eleutherodactylus electrae</i> Dunn, 1926
<i>Eleutherodactylus flavescens</i> Noble, 1923
<i>Eleutherodactylus gryllus</i> Schmidt, 1920
<i>Eleutherodactylus guttatus</i> (Cope, 1879)
<i>Eleutherodactylus hedricki</i> Rivero, 1963
<i>Eleutherodactylus ibera</i> Estrada and Hedges, 1996
<i>Eleutherodactylus inoptatus</i> (Barbour, 1914)
<i>Eleutherodactylus jasperi</i> Drewry and Jones, 1976
<i>Eleutherodactylus johnstonei</i> Barbour, 1914
<i>Eleutherodactylus juanivernei</i> Ross-López and Thomas, 2007
<i>Eleutherodactylus leleri</i> Schwartz, 1965
<i>Eleutherodactylus leprus</i> (Cope, 1879)
<i>Eleutherodactylus limbellus</i> (Cope, 1862)
<i>Eleutherodactylus bocourti</i> Schmidt, 1920
<i>Eleutherodactylus mariposa</i> Hedges, Estrada, and Thomas, 1992
<i>Eleutherodactylus marnooi</i> (Cope, 1878)
<i>Eleutherodactylus martinicensis</i> (Tschudi, 1838)
<i>Eleutherodactylus minutus</i> Noble, 1923
<i>Eleutherodactylus modestus</i> (Taylor, 1942)
<i>Eleutherodactylus monensis</i> (Meeuwirth, 1921)
<i>Eleutherodactylus nitidus</i> (Peters, 1870)
<i>Eleutherodactylus parapetes</i> Hedges and Thomas, 1987
<i>Eleutherodactylus piplaris</i> (Taylor, 1940)
<i>Eleutherodactylus planirostris</i> (Cope, 1862)
<i>Eleutherodactylus portoricensis</i> Schmidt, 1927
<i>Eleutherodactylus richmondi</i> Stejneger, 1904
<i>Eleutherodactylus rivulanti</i> Diaz, Estrada, and Hedges, 2001
<i>Eleutherodactylus ronaldi</i> Schwartz, 1960
<i>Eleutherodactylus schwarzi</i> Thomas, 1968
<i>Eleutherodactylus thorectes</i> Hedges, 1988
<i>Eleutherodactylus unicolor</i> Stejneger, 1904
<i>Eleutherodactylus velleyi</i> Dunn, 1925
<i>Eleutherodactylus verrucipes</i> (Cope, 1866)
<i>Eleutherodactylus wightmanae</i> Schmidt, 1920
<i>Engystomops coloradum</i> (Cannaleale and Duellman, 1984)
<i>Engystomops petersi</i> Jiménez de la Espada, 1872
<i>Engystomops pustulatus</i> (Shreve, 1941)
<i>Engystomops pustulosus</i> (Cope, 1864)
<i>Engystomops pygmaeus</i> Ron, Toral, Rivera, and Terán-Valdez, 2010
<i>Engystomops randi</i> (Ron, Cannaleale, and Coloma, 2004)
<i>Epideles calumita</i> (Laurenti, 1766)
<i>Epipedobates anthonyi</i> Noble, 1941
<i>Epipedobates boulengeri</i> (Barbour, 1909)
<i>Epipedobates machallae</i> (Coloma, 1995)
<i>Epipedobates tricolor</i> (Boulenger, 1899)
<i>Epispharax andina</i> (Rivero, 1968)
<i>Epispharax callistomma</i> (Guayasamin and Trueb, 2007)
<i>Epispharax prosopeus</i> (Boettger, 1902)
<i>Eupemphix nattereri</i> Steindachner, 1863
<i>Euphyllotis cyanophyllotis</i> (Schneider, 1799)
<i>Euphyllotis hexadactylus</i> (Lesson, 1834)
<i>Eupsophia calcareata</i> (Günther, 1881)
<i>Eupsophia confulmensis</i> Ortiz, Ibarra-Vidal, and Formas, 1989
<i>Eupsophia emilioguglii</i> Formas, 1989
<i>Eupsophia migueli</i> Formas, 1978
<i>Eupsophia nahuelbutensis</i> Ortiz and Ibarra-Vidal, 1992
<i>Eupsophia roseus</i> (Duméril and Bibron, 1841)
<i>Eupsophia vertebrae</i> Grandison, 1961
<i>Excidobates capiviensis</i> (Myers, 1982)
<i>Excidobates mystriensis</i> (Myers, 1982)
<i>Everodonta catraiche</i> (Poyas and Wilson, 1987)
<i>Everodonta chimalapa</i> (Mendelson and Campbell, 1994)
<i>Everodonta melanomma</i> (Taylor, 1940)
<i>Everodonta perkinsi</i> (Campbell and Brodie, 1992)
<i>Everodonta smaragdina</i> (Taylor, 1940)
<i>Everodonta sumichrasti</i> Brodie, 1979
<i>Everodonta zere</i> (Mendelson and Campbell, 1994)
<i>Felhyia palpebralis</i> (Smith, 1924)
<i>Felivarya andamanensis</i> (Stoliczka, 1870)
<i>Felivarya cancinosa</i> (Graham-Smith, 1929)
<i>Felivarya caperata</i> Kuramoto, Joshy, Kurabayashi, and Sumida, 2007
<i>Felivarya greeni</i> (Boulenger, 1905)
<i>Felivarya keralensis</i> (Dabols, 1851)
<i>Felivarya kirtisinghei</i> (Manamendra-Arachchi and Gabadage, 1996)
<i>Felivarya kudremukhensis</i> Kuramoto, Joshy, Kurabayashi, and Sumida, 2007
<i>Felivarya limochensis</i> (Graham-Smith, 1929)
<i>Felivarya mudduraja</i> Kuramoto, Joshy, Kurabayashi, and Sumida, 2007
<i>Felivarya nicobarensis</i> (Stoliczka, 1870)
<i>Felivarya rufescens</i> (Jerdon, 1854)
<i>Felivarya syhadrensis</i> (Annandale, 1919)
<i>Felivarya triora</i> Stuart, Chayankem, Chan-ard, and Inger, 2006
<i>Plectonotus fissus</i> (Miranda-Ribeiro, 1920)
<i>Plectonotus fitzingeri</i> (Parker, 1934)
<i>Plectonotus pygmaeus</i> (Boettger, 1893)
<i>Gastrophryne carolinensis</i> (Holbrook, 1835)
<i>Gastrophryne elegans</i> (Boulenger, 1882)
<i>Gastrophryne olivacea</i> (Hallowell, 1856)
<i>Gastrophryne pictiventris</i> (Cope, 1866)
<i>Gastrophryne uita</i> (Cope, 1866)
<i>Gastrotheca antiochochae</i> (De la Riva and Chaparro, 2005)
<i>Gastrotheca cornuta</i> (Boulenger, 1899)
<i>Gastrotheca excubitor</i> Duellman and Fritts, 1972
<i>Gastrotheca guerini</i> (Boulenger, 1882)
<i>Gastrotheca helensis</i> Dunn, 1944
<i>Gastrotheca itoneoda</i> Duellman and Trueb, 1988
<i>Gastrotheca laurizcae</i> De la Riva, 1992
<i>Gastrotheca itoneida</i> Duellman and Hillis, 1987

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<i>Gastrotheca marsupata</i> (Duméril and Bérton, 1841)
<i>Gastrotheca monticola</i> Barbour and Noble, 1920
<i>Gastrotheca ophiylax</i> Duellman and Pyles, 1960
<i>Gastrotheca ovifera</i> (Lichtenstein and Westland, 1854)
<i>Gastrotheca peniana</i> (Boulenger, 1920)
<i>Gastrotheca plumbea</i> (Boulenger, 1882)
<i>Gastrotheca pseudata</i> Duellman and Hillis, 1987
<i>Gastrotheca rosbamiiae</i> Fowler, 1913
<i>Gastrotheca ruzi</i> Duellman and Burrowes, 1986
<i>Geobatrachus walker</i> Ruthven, 1915
<i>Geocornia alba</i> Vardoli-Johnson and Roberts, 1989
<i>Geocornia baevia</i> (Günther, 1864)
<i>Geocornia leai</i> (Fletcher, 1895)
<i>Geocornia lutea</i> (Main, 1853)
<i>Geocornia rosea</i> (Harrison, 1927)
<i>Geocornia victoriana</i> (Boulenger, 1888)
<i>Geocornia vitellina</i> Vardoli-Johnson and Roberts, 1989
<i>Gephyromantis asper</i> (Boulenger, 1882)
<i>Gephyromantis boulengeri</i> Methuen, 1920
<i>Gephyromantis corvus</i> (Glaw and Vences, 1994)
<i>Gephyromantis granulatus</i> (Boulenger, 1881)
<i>Gephyromantis leucocephalus</i> Angel, 1930
<i>Gephyromantis juleza</i> (Methuen and Hewitt, 1913)
<i>Gephyromantis moesi</i> (Glaw and Vences, 2002)
<i>Gephyromantis redimitus</i> (Boulenger, 1889)
<i>Gephyromantis sculpturatus</i> (Ahl, 1929)
<i>Gephyromantis silvius</i> (Vences, Glaw, and Andreone, 1997)
<i>Ghatophryne ornata</i> (Günther, 1876)
<i>Glandirana rugosa</i> (Temminck and Schlegel, 1838)
<i>Glyphoplossus nodosus</i> Günther, 1869
<i>Guibemantis bicalcaratus</i> (Boettger, 1913)
<i>Guibemantis flavobrunneus</i> (Blommers-Schlösser, 1979)
<i>Guibemantis fiber</i> (Peracca, 1933)
<i>Guibemantis pulcher</i> (Boulenger, 1882)
<i>Guibemantis tonieri</i> (Ahl, 1928)
<i>Hadrasus kneriatus</i> (Spix, 1824)
<i>Hadromophryne natalensis</i> (Hewitt, 1913)
<i>Hemiphrachyne boliviana</i> (Parker, 1927)
<i>Heleoporus albopunctatus</i> Gray, 1841
<i>Heleoporus australiacus</i> (Shaw and Nodder, 1996)
<i>Heleoporus barycragus</i> Lee, 1967
<i>Heleoporus eyrei</i> (Günth, 1845)
<i>Heleoporus mortuus</i> Lee and Main, 1954
<i>Heleoporus psammophilus</i> Lee and Main, 1954
<i>Heleophryne hewitt</i> Boycott, 1906
<i>Heleophryne rosei</i> Hewitt, 1910
<i>Heleophryne rosei</i> Hewitt, 1925
<i>Hemiphractus fasciatus</i> Peters, 1892
<i>Hemiphractus phisaroni</i> (Noble, 1917)
<i>Hemiphractus proboscideus</i> (Jiménez de la Espada, 1870)
<i>Hemiphractus scutellus</i> (Spix, 1824)
<i>Hemius guttatus</i> (Rüpp, 1842)
<i>Hemius marmoratus</i> (Peters, 1854)
<i>Heterixalus aboguttatus</i> (Boulenger, 1882)
<i>Heterixalus boettgeri</i> (Mocquard, 1902)
<i>Heterixalus futeodriatus</i> (Anderson, 1910)
<i>Heterixalus madagascariensis</i> (Duméril and Bérton, 1841)
<i>Heterixalus punctatus</i> (Glaw and Vences, 1994)
<i>Heterixalus rutenbergi</i> (Boettger, 1881)
<i>Heterixalus tricolor</i> (Boettger, 1881)
<i>Hobombantis ornata</i> (Peters, 1878)
<i>Hoplobatrachus crassus</i> (Jerdon, 1854)
<i>Hoplobatrachus occipitalis</i> (Günther, 1858)
<i>Hoplobatrachus rugulosus</i> (Wiegmann, 1834)
<i>Hoplobatrachus tigrinus</i> (Baudin, 1802)
<i>Hoplophryne rogersi</i> Barbour and Loveridge, 1928
<i>Hyla masoni</i> (Boulenger, 1884)
<i>Hyla sumatrana</i> Yang, 1991
<i>Humerana mopus</i> (Boulenger, 1918)
<i>Hyalinobatrachium aureoguttatum</i> (Blarera-Rodríguez and Ruiz-Carranza, 1989)
<i>Hyalinobatrachium begeri</i> (Cannatella, 1980)
<i>Hyalinobatrachium carlesvili</i> Castroviejo-Fisher, Padial, Chaparro, Aguiayo, and De la Riva, 2009
<i>Hyalinobatrachium chiriqui</i> (Taylor, 1958)
<i>Hyalinobatrachium colymbiophilum</i> (Taylor, 1949)
<i>Hyalinobatrachium crurifasciatum</i> Myers and Donnelly, 1997
<i>Hyalinobatrachium esmeraldi</i> Ruiz-Carranza and Lynch, 1998
<i>Hyalinobatrachium feischmanni</i> (Boettger, 1893)
<i>Hyalinobatrachium fragile</i> (Rivers, 1895)
<i>Hyalinobatrachium hesperidenae</i> (Ayscaghenia, 1992)
<i>Hyalinobatrachium ibama</i> Ruiz-Carranza and Lynch, 1998
<i>Hyalinobatrachium ignicolum</i> Noonan and Bonett, 2003
<i>Hyalinobatrachium orientale</i> (Rivers, 1905)
<i>Hyalinobatrachium pellicudum</i> (Lynch and Duellman, 1973)
<i>Hyalinobatrachium ruedai</i> Ruiz-Carranza and Lynch, 1998
<i>Hyalinobatrachium salmianae</i> (Taylor, 1952)
<i>Hyalinobatrachium taylori</i> (Goin, 1968)
<i>Hyalinobatrachium valentini</i> (Dunn, 1931)
<i>Hyalinobatrachium vireofractum</i> (Starnett and Savage, 1973)
<i>Hyla andersoni</i> Baird, 1854
<i>Hyla annectans</i> (Jerdon, 1870)
<i>Hyla ansonae</i> (Linnaeus, 1758)
<i>Hyla arenicolor</i> Cope, 1866
<i>Hyla avivoca</i> Viosca, 1928
<i>Hyla bocourti</i> Mocquard, 1899
<i>Hyla chinensis</i> Günther, 1858
<i>Hyla chrysocelis</i> Cope, 1880
<i>Hyla cinerea</i> (Schneider, 1799)
<i>Hyla euphorbiacea</i> Günther, 1858
<i>Hyla eximia</i> Baird, 1854
<i>Hyla felixarabica</i> Gvozdev, Kotlik and Moravec, 2010
<i>Hyla femoralis</i> Kots, 1860
<i>Hyla gralliosa</i> LeConte, 1856
<i>Hyla hallowelli</i> Thompson, 1912
<i>Hyla immaculata</i> Boettger, 1888
<i>Hyla intermedia</i> Boulenger, 1882
<i>Hyla japonica</i> Günther, 1859
<i>Hyla mendonensis</i> Boettger, 1874
<i>Hyla molleni</i> Bedriaga, 1890
<i>Hyla orientalis</i> Bedriaga, 1890
<i>Hyla pictata</i> Broch, 1877
<i>Hyla sarda</i> (De Betta, 1853)
<i>Hyla savignyi</i> Audouin, 1827
<i>Hyla simplex</i> Boettger, 1901
<i>Hyla squirella</i> Bosc, 1800
<i>Hyla suweonensis</i> Kuramoto, 1980
<i>Hyla taittingensis</i> Liu and Hu, 1966
<i>Hyla versicolor</i> LeConte, 1865
<i>Hyla walkeri</i> Stuart, 1954
<i>Hyla wrightorum</i> Taylor, 1939
<i>Hyilarana aurantilaca</i> (Boulenger, 1904)
<i>Hyilarana aurata</i> (Günther, 2003)
<i>Hyilarana barjerana</i> (Leong and Lim, 2003)
<i>Hyilarana baramica</i> (Boettger, 1900)
<i>Hyilarana chatconota</i> (Schlegel, 1837)
<i>Hyilarana daemeli</i> (Steindachner, 1868)
<i>Hyilarana erythraea</i> (Schlegel, 1837)
<i>Hyilarana glandulosa</i> (Boulenger, 1882)
<i>Hyilarana grisea</i> (Van Kampen, 1913)
<i>Hyilarana guntheri</i> (Boulenger, 1882)
<i>Hyilarana labialis</i> (Boulenger, 1887)
<i>Hyilarana laterimaculata</i> (Barbour and Noble, 1916)
<i>Hyilarana latouchi</i> (Boulenger, 1899)
<i>Hyilarana luctuosa</i> (Peters, 1871)
<i>Hyilarana luzonensis</i> (Boulenger, 1896)
<i>Hyilarana malabarica</i> (Fisch, 1838)
<i>Hyilarana nicobariensis</i> (Stoliczka, 1870)
<i>Hyilarana nigrovittata</i> (Blyth, 1856)
<i>Hyilarana picturata</i> (Boulenger, 1920)
<i>Hyilarana raniceps</i> (Peters, 1871)
<i>Hyilarana signata</i> (Günther, 1872)
<i>Hyilarana talpehenas</i> (Van Denburgh, 1909)
<i>Hyilarana temporalis</i> (Günther, 1864)
<i>Hyilarana volkneri</i> (Günther, 2003)
<i>Hyilodes asper</i> (Müller, 1824)
<i>Hyilodes freyi</i> Canedo and Pombal, 2007
<i>Hyilodes heyeri</i> Haddad, Pombal, and Basalos, 1996
<i>Hyilodes ornatus</i> (Bokermann, 1967)
<i>Hyilodes phyllodes</i> Heyer and Cocroft, 1986
<i>Hyilodes pipilans</i> Canedo and Pombal, 2007
<i>Hyigophorus prokezi</i> Kraus and Allison, 2009
<i>Hyloscirtus sylvestris</i> Lill, 1843
<i>Hyloscirtus altyolax</i> (Duellman, 1972)
<i>Hyloscirtus armatus</i> (Boulenger, 1902)
<i>Hyloscirtus charaxii</i> (Vielzeuf, 1910)
<i>Hyloscirtus colymba</i> (Dunn, 1931)
<i>Hyloscirtus denticulatus</i> (Duellman, 1972)

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<i>Hyloscirtus laniopygion</i> (Duellman, 1973)
<i>Hyloscirtus jasonius</i> (Rivero, 1970)
<i>Hyloscirtus lindae</i> (Duellman and Altig, 1978)
<i>Hyloscirtus palmeri</i> (Boulenger, 1906)
<i>Hyloscirtus pambolichus</i> (Duellman and Berger, 1982)
<i>Hyloscirtus phyllonathus</i> (Melin, 1941)
<i>Hyloscirtus simmonsii</i> (Duellman, 1989)
<i>Hyloscirtus tapachaleca</i> (Kotzein, Coloma, and Paredes-Recalde, 2003)
<i>Hyloscirtus tigrinus</i> Mueses-Canteras and Angaroy-Crolio, 2008
<i>Hyloxaalus awa</i> (Coloma, 1995)
<i>Hyloxaalus aureiventris</i> (Kneller and Henle, 1985)
<i>Hyloxaalus bocagei</i> Jiménez de la Espada, 1870
<i>Hyloxaalus cevallosi</i> (Rivero, 1991)
<i>Hyloxaalus echyridiscus</i> (Edwards, 1971)
<i>Hyloxaalus infraguttatus</i> (Boulenger, 1996)
<i>Hyloxaalus insulatus</i> (Duellman, 2004)
<i>Hyloxaalus rexipus</i> (Ross, 1986)
<i>Hyloxaalus sauli</i> (Edwards, 1974)
<i>Hyloxaalus vertebralis</i> (Boulenger, 1899)
<i>Hymenochirus boettgeri</i> (Toniér, 1896)
<i>Hymenochirus curtipis</i> Noble, 1924
<i>Hyperolius argus</i> Peters, 1854
<i>Hyperolius boerneriasi</i> Schiøtz, 1967
<i>Hyperolius boettgeriae</i> Mertens, 1938
<i>Hyperolius camerunensis</i> Amiet, 2004
<i>Hyperolius castaneus</i> Ahl, 1931
<i>Hyperolius cinnamomeiventris</i> Bocage, 1866
<i>Hyperolius concolor</i> (Hallowell, 1844)
<i>Hyperolius dirlheymanni</i> Lötters and Schmitz, 2004
<i>Hyperolius discodiscus</i> Ahl, 1931
<i>Hyperolius frontalis</i> Laurent, 1950
<i>Hyperolius fusciventris</i> Peters, 1876
<i>Hyperolius glaucicolor</i> Peters, 1876
<i>Hyperolius guttulatus</i> Günther, 1858
<i>Hyperolius horsfieldi</i> (Schlegel, 1837)
<i>Hyperolius kribiaensis</i> Schiøtz and Westergaard, 1999
<i>Hyperolius kivuensis</i> Ahl, 1931
<i>Hyperolius langi</i> Noble, 1924
<i>Hyperolius lateralis</i> Laurent, 1948
<i>Hyperolius marginatus</i> Peters, 1854
<i>Hyperolius mariae</i> Barbour and Loveridge, 1928
<i>Hyperolius marmoratus</i> Kapp, 1942
<i>Hyperolius mitchelli</i> Loveridge, 1953
<i>Hyperolius molieri</i> (Bednaga, 1892)
<i>Hyperolius montianus</i> (Ager, 1924)
<i>Hyperolius nasutus</i> Günther, 1865
<i>Hyperolius nimbae</i> Laurent, 1956
<i>Hyperolius nitidulus</i> Peters, 1875
<i>Hyperolius ocellatus</i> Günther, 1858
<i>Hyperolius pardalis</i> Laurent, 1948
<i>Hyperolius parkeri</i> Loveridge, 1933
<i>Hyperolius phantasticus</i> (Boulenger, 1899)
<i>Hyperolius pickersgilli</i> Raw, 1962
<i>Hyperolius poweri</i> Loveridge, 1938
<i>Hyperolius punctulatus</i> (Pfeiffer, 1893)
<i>Hyperolius pusillus</i> (Cope, 1862)
<i>Hyperolius curruveitatus</i> Bocage, 1866
<i>Hyperolius riggenbachi</i> (Nieden, 1910)
<i>Hyperolius rubrovermiculatus</i> Schiøtz, 1975
<i>Hyperolius semiflavus</i> Hewitt, 1927
<i>Hyperolius spingularis</i> Stevens, 1971
<i>Hyperolius stenodiscus</i> Ahl, 1931
<i>Hyperolius stylatus</i> Schiøtz, 1967
<i>Hyperolius thomasi</i> Bocage, 1886
<i>Hyperolius torrentis</i> Schiøtz, 1967
<i>Hyperolius tuberculatus</i> (Mocquard, 1897)
<i>Hyperolius tuberculatus</i> Smith, 1849
<i>Hyperolius viridiflavus</i> (Duméril and Bibron, 1841)
<i>Hypodactylus nigrovittatus</i> (Anderson, 1946)
<i>Hyposachus variolosus</i> (Cope, 1866)
<i>Hyposbaes alboargyratus</i> (Spix, 1824)
<i>Hyposbaes albingeri</i> (Nielsen, 1923)
<i>Hyposbaes albopunctatus</i> (Spix, 1824)
<i>Hyposbaes balzani</i> (Boulenger, 1896)
<i>Hyposbaes beckeri</i> (Caramaschi and Cruz, 2004)
<i>Hyposbaes bischoffi</i> (Boulenger, 1897)
<i>Hyposbaes boana</i> (Linnaeus, 1758)
<i>Hyposbaes burii</i> (Caramaschi and Cruz, 1999)
<i>Hyposbaes canguis</i> (Centron, 1901)
<i>Hyposbaes calcareatus</i> (Troschel, 1848)
<i>Hyposbaes calipheura</i> (Boulenger, 1902)
<i>Hyposbaes cherearensis</i> (Spix, 1824)
<i>Hyposbaes cipoensis</i> (Lutz, 1966)
<i>Hyposbaes cordobae</i> (Barrio, 1965)
<i>Hyposbaes crepitans</i> (Wied-Neuwied, 1834)
<i>Hyposbaes curupi</i> García, Favovich, and Haddad, 2007
<i>Hyposbaes dentei</i> (Bokermann, 1967)
<i>Hyposbaes ericae</i> (Caramaschi and Cruz, 2006)
<i>Hyposbaes faber</i> (Wied-Neuwied, 1821)
<i>Hyposbaes fasciatus</i> (Günther, 1858)
<i>Hyposbaes geographica</i> (Spix, 1824)
<i>Hyposbaes gladiator</i> Köhler, Kosciński, Padial, Chaparro, Handford, Lougheed, and De la Riva, 2010
<i>Hyposbaes golianus</i> (Lutz, 1968)
<i>Hyposbaes guentheri</i> (Boulenger, 1886)
<i>Hyposbaes helprinii</i> (Noble, 1923)
<i>Hyposbaes jimenaei</i> Señaris and Ayarzagüena, 2006
<i>Hyposbaes piquini</i> (Lutz, 1966)
<i>Hyposbaes lanciformis</i> Cope, 1871
<i>Hyposbaes lateralis</i> (Caramaschi and Cruz, 2004)
<i>Hyposbaes kydymedus</i> (Braun and Braun, 1977)
<i>Hyposbaes lundii</i> (Burmeister, 1856)
<i>Hyposbaes marginatus</i> (Boulenger, 1887)
<i>Hyposbaes marianae</i> (Carrizo, 1922)
<i>Hyposbaes multifasciatus</i> (Günther, 1859)
<i>Hyposbaes nympha</i> Favovich, Moravec, Cisneros-Heredia, and Köhler, 2006
<i>Hyposbaes palestensis</i> (Duellman, De la Riva, and Wild, 1997)
<i>Hyposbaes paranalis</i> Carvalho and Giaretta, 2010
<i>Hyposbaes pardalis</i> (Spix, 1824)
<i>Hyposbaes pellucens</i> (Weimer, 1901)
<i>Hyposbaes phaeoleura</i> (Caramaschi and Cruz, 2000)
<i>Hyposbaes picturatus</i> (Boulenger, 1899)
<i>Hyposbaes popu</i> García, Picot, and Haddad, 2008
<i>Hyposbaes polytaenius</i> (Cope, 1870)
<i>Hyposbaes prasinus</i> (Burmeister, 1856)
<i>Hyposbaes pugna</i> (Schmidt, 1857)
<i>Hyposbaes puchtelii</i> (Duméril and Bibron, 1841)
<i>Hyposbaes punctatus</i> (Schneider, 1799)
<i>Hyposbaes raniceps</i> Cope, 1862
<i>Hyposbaes rogersi</i> (Kosciowski, 1895)
<i>Hyposbaes rosenbergi</i> (Boulenger, 1898)
<i>Hyposbaes rufellus</i> (Fouquette, 1961)
<i>Hyposbaes semipudatus</i> (Lutz, 1925)
<i>Hyposbaes semilineatus</i> (Spix, 1824)
<i>Hyposbaes stilesi</i> (Rivero, 1972)
<i>Hyposbaes stilesi</i> Knoll, 2006
<i>Hyposbaes wavvini</i> (Parker, 1936)
<i>Incilius alvarius</i> (Girard, 1859)
<i>Incilius euomoei</i> (O'Neill and Mendelson, 2004)
<i>Incilius bocourti</i> (Brocchi, 1877)
<i>Incilius canaliellus</i> (Cope, 1877)
<i>Incilius cockeri</i> (Cope, 1866)
<i>Incilius confusus</i> (Cope, 1862)
<i>Incilius holdridgei</i> (Taylor, 1962)
<i>Incilius luteiventris</i> (Boulenger, 1891)
<i>Incilius majordomus</i> Savage, Ugarte, and Donnelly, 2013
<i>Incilius marmoratus</i> (Wiegmann, 1833)
<i>Incilius mazatlanensis</i> (Taylor, 1945)
<i>Incilius melanochlorus</i> (Cope, 1877)
<i>Incilius nebulifer</i> (Girard, 1854)
<i>Incilius perferens</i> (Savage, 1967)
<i>Incilius velleros</i> (Wiegmann, 1833)
<i>Indirana beddomi</i> (Günther, 1876)
<i>Indirana brachyterus</i> (Günther, 1876)
<i>Indirana semipalmata</i> (Boulenger, 1882)
<i>Indirana baluensis</i> (Boulenger, 1896)
<i>Indirana lui</i> (Yang, 1963)
<i>Indirana tsalanzei</i> (Smith, 1921)
<i>Ingerophrynus bipoecatus</i> (Gravenhorst, 1829)
<i>Ingerophrynus celebensis</i> (Günther, 1859)
<i>Ingerophrynus evergens</i> (Peters, 1871)
<i>Ingerophrynus galeatus</i> (Günther, 1864)
<i>Ingerophrynus parvus</i> (Boulenger, 1887)
<i>Ingerophrynus philippinus</i> (Boulenger, 1887)
<i>Ingerophrynus quadriporcatus</i> (Boulenger, 1887)
<i>Insuetophrynus acaripicus</i> Barrio, 1970

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<i>Ichthyocnema guentheri</i> (Steindachner, 1864)
<i>Ichthyocnema holt</i> (Cochran, 1948)
<i>Ichthyocnema manezinho</i> (Garcia, 1996)
<i>Ichthyocnema parva</i> (Grard, 1853)
<i>Ichthyocnema vittata</i> (Bakermann, 1976)
<i>Isthmohyla debilis</i> (Taylor, 1952)
<i>Isthmohyla infucata</i> (Duellman, 1968)
<i>Isthmohyla lanceaster</i> (Barbour, 1928)
<i>Isthmohyla melacaena</i> (McCrane and Castañeda, 2006)
<i>Isthmohyla picador</i> (Dunn, 1937)
<i>Isthmohyla pseudoguma</i> (Günther, 1901)
<i>Isthmohyla rivularis</i> (Taylor, 1952)
<i>Isthmohyla sica</i> (Starrett, 1966)
<i>Isthmohyla variltoctota</i> (Duellman, 1968)
<i>Isthmohyla zetek</i> (Saige, 1929)
<i>Itapothyla langadoffi</i> (Duméril and Bibrón, 1841)
<i>Kalophrynus baileyi</i> Kiew, 1967
<i>Kalophrynus heterochrus</i> Boulenger, 1900
<i>Kalophrynus interlineatus</i> (Blyth, 1855)
<i>Kalophrynus palmatus</i> Kiew, 1964
<i>Kalophrynus pleurostigma</i> Tschudi, 1838
<i>Kaloula aureata</i> Nuphard, 1989
<i>Kaloula bealei</i> (Müller, 1836)
<i>Kaloula borealis</i> (Barbour, 1908)
<i>Kaloula picta</i> (Duméril and Bibrón, 1841)
<i>Kaloula pulchra</i> Gray, 1831
<i>Kaloula rigida</i> Taylor, 1952
<i>Kaloula rugifera</i> Stejneger, 1924
<i>Kaloula saprobatica</i> Parker, 1934
<i>Kaloula verrucosa</i> Boulenger, 1904
<i>Kassina maculata</i> (Duméril, 1853)
<i>Kassina maculosa</i> (Stemfeld, 1917)
<i>Kassina schiøtzii</i> Rödel, Gräfe, Buxholt and Ernst, 2002
<i>Kassina senegalensis</i> (Duméril and Bibrón, 1841)
<i>Kassina somalica</i> Scortecchi, 1932
<i>Kurixalus blascowii</i> (Taylor, 1962)
<i>Kurixalus effingeri</i> (Boettger, 1895)
<i>Kurixalus ibidocatus</i> (Kuramola and Wang, 1987)
<i>Kurixalus odontosternus</i> Ye and Fei, 1993
<i>Kurixalus verrucosus</i> (Boulenger, 1893)
<i>Laliostoma labrosus</i> (Cope, 1868)
<i>Laniarctes conigulus</i> (Peters, 1863)
<i>Lechniodus fletcheri</i> (Boulenger, 1890)
<i>Leiopelma archeyi</i> Turbott, 1942
<i>Leiopelma hamiltoni</i> McCulloch, 1919
<i>Leiopelma hochstetteri</i> Fitzinger, 1861
<i>Leiopelma pakeka</i> Bell, Daugherty, and Hay, 1996
<i>Leptodactylus asper</i> Boulgelli, 1899
<i>Leptodactylus laevis</i> Boulgelli, 1899
<i>Leptodactylus llanensis</i> Reig and Cei, 1963
<i>Leptodactylus njyoberg</i> Smith, 1925
<i>Leptodermum abbotti</i> (Cochran, 1926)
<i>Leptobranchium chapaense</i> (Bourret, 1937)
<i>Leptobranchium gunungense</i> Malmkus, 1996
<i>Leptobranchium hasseltii</i> (Tschudi, 1838)
<i>Leptobranchium hendricksoni</i> Taylor, 1962
<i>Leptobranchium montanum</i> Fischer, 1865
<i>Leptobranchium nigropus</i> Berry and Hendrickson, 1963
<i>Leptobranchium smithi</i> Matsui, Nabhitabhata, and Panha, 1999
<i>Leptodactylus andreae</i> Müller, 1923
<i>Leptodactylus bolivianus</i> Boulenger, 1898
<i>Leptodactylus bufonius</i> Boulenger, 1894
<i>Leptodactylus chaquensis</i> Cei, 1960
<i>Leptodactylus cotombiensis</i> Heyer, 1994
<i>Leptodactylus didymus</i> Heyer, García-López, and Cardoso, 1996
<i>Leptodactylus elenae</i> Heyer, 1978
<i>Leptodactylus felix</i> Müller, 1928
<i>Leptodactylus fragilis</i> (Brooch, 1877)
<i>Leptodactylus furcatus</i> Sazima and Bokermann, 1978
<i>Leptodactylus fuscus</i> (Schneider, 1799)
<i>Leptodactylus gracilis</i> (Duméril and Bibrón, 1840)
<i>Leptodactylus hylaedactylus</i> (Cope, 1868)
<i>Leptodactylus kneri</i> Heyer, 1972
<i>Leptodactylus labrosus</i> Jiménez de la Espada, 1875
<i>Leptodactylus labyrinthicus</i> (Spix, 1824)
<i>Leptodactylus laticeps</i> Boulenger, 1918
<i>Leptodactylus latissus</i> Jiménez de la Espada, 1875
<i>Leptodactylus latrans</i> (Steffen, 1815)
<i>Leptodactylus leptodactylodes</i> (Andersson, 1946)
<i>Leptodactylus lineatus</i> (Schneider, 1799)
<i>Leptodactylus macrosternum</i> Miranda-Ribeiro, 1926
<i>Leptodactylus marmoratus</i> (Steindachner, 1867)
<i>Leptodactylus melanonotus</i> (Hallowell, 1861)
<i>Leptodactylus mystaceus</i> (Spix, 1824)
<i>Leptodactylus mystacinus</i> (Burmester, 1861)
<i>Leptodactylus natanensis</i> Lutz, 1935
<i>Leptodactylus notoaktites</i> Heyer, 1978
<i>Leptodactylus pentadactylus</i> (Laurenti, 1768)
<i>Leptodactylus petersi</i> (Steindachner, 1864)
<i>Leptodactylus piaumanni</i> Ahl, 1936
<i>Leptodactylus podicipinus</i> (Cope, 1862)
<i>Leptodactylus poecilochilus</i> (Cope, 1862)
<i>Leptodactylus pustulatus</i> (Peters, 1870)
<i>Leptodactylus rhodomystax</i> Boulenger, 1884
<i>Leptodactylus rhodonotus</i> (Günther, 1859)
<i>Leptodactylus rufopus</i> Noble, 1953
<i>Leptodactylus savagei</i> Heyer, 2005
<i>Leptodactylus syphax</i> Bokermann, 1969
<i>Leptodactylus troglodytes</i> Lutz, 1928
<i>Leptodactylus validus</i> Garman, 1888
<i>Leptodactylus vastus</i> Lutz, 1930
<i>Leptodactylus wagneri</i> (Peters, 1862)
<i>Leptolax applebyi</i> Rowley and Cao, 2009
<i>Leptolax arayai</i> Matsui, 1997
<i>Leptolax Jourd' hui</i> Dubois, 1983
<i>Leptolax gracilis</i> (Günther, 1872)
<i>Leptolax lui</i> Fei and Ye, 1990
<i>Leptolax pabohtrotes</i> (Boulenger, 1893)
<i>Leptolax pictus</i> Malmkus, 1992
<i>Leptolax ventripunctatus</i> Fei, Ye, and Li, 1990
<i>Leptopelis argenteus</i> (Pfeffer, 1883)
<i>Leptopelis aubryi</i> (Duméril, 1856)
<i>Leptopelis barbouri</i> Ahl, 1929
<i>Leptopelis brevirostris</i> (Wiener, 1898)
<i>Leptopelis christyi</i> (Boulenger, 1912)
<i>Leptopelis concolor</i> Ahl, 1929
<i>Leptopelis flavonaculatus</i> (Günther, 1864)
<i>Leptopelis lividus</i> Ahl, 1929
<i>Leptopelis misoni</i> (Boulenger, 1895)
<i>Leptopelis mossambicus</i> Hovinton, 1985
<i>Leptopelis natanensis</i> (Smith, 1949)
<i>Leptopelis occidentalis</i> Schietz, 1967
<i>Leptopelis parker</i> Barbour and Lovendge, 1928
<i>Leptopelis regazzi</i> (Boulenger, 1896)
<i>Leptopelis rufus</i> Reichenow, 1874
<i>Leptopelis spirituscaecis</i> Rödel, 2007
<i>Leptopelis susanne</i> Largen, 1977
<i>Leptopelis uluguruensis</i> Barbour and Loveridge, 1928
<i>Leptopelis vermiculatus</i> (Boulenger, 1909)
<i>Leptopelis viridis</i> (Günther, 1859)
<i>Leptopelis yaldeni</i> Largen, 1977
<i>Leptophryne borbonica</i> (Tschudi, 1838)
<i>Limnodynastes convexicaulus</i> (Macleay, 1878)
<i>Limnodynastes depressus</i> Tyler, 1976
<i>Limnodynastes dorsalis</i> (Gray, 1841)
<i>Limnodynastes dumerilii</i> (Peters, 1863)
<i>Limnodynastes fletcheri</i> Boulenger, 1888
<i>Limnodynastes interioris</i> Fry, 1913
<i>Limnodynastes signatus</i> (Tyler, Martin, and Davies, 1978)
<i>Limnodynastes peronii</i> (Duméril and Bibrón, 1841)
<i>Limnodynastes salmini</i> Steindachner, 1867
<i>Limnodynastes tasmanensis</i> (Günther, 1858)
<i>Limnodynastes terraesanae</i> Fry, 1915
<i>Limnodynastes macroglossa</i> (Duméril and Bibrón, 1841)
<i>Limnometes bythii</i> (Boulenger, 1907)
<i>Limnometes debanus</i> (Smith, 1922)
<i>Limnometes finchi</i> (Inger, 1966)
<i>Limnometes fragilis</i> (Lu and Hu, 1973)
<i>Limnometes fulviventris</i> Ye and Fei, 1994
<i>Limnometes gyldenstolpei</i> (Andersson, 1916)
<i>Limnometes ingeni</i> (Kiew, 1978)
<i>Limnometes kochi</i> (Smith, 1922)
<i>Limnometes kuhlii</i> (Tschudi, 1838)
<i>Limnometes laticeps</i> (Boulenger, 1882)
<i>Limnometes leporinus</i> (Andersson, 1923)
<i>Limnometes leytenis</i> (Boettger, 1893)
<i>Limnometes macrocephalus</i> (Inger, 1954)

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<i>Limnometes macrodon</i> (Duméril and Bérton, 1841)
<i>Limnometes melanurus</i> (Kiew, 1984)
<i>Limnometes modestus</i> (Boulenger, 1882)
<i>Limnometes namyeri</i> (Stejneger, 1901)
<i>Limnometes palawanensis</i> (Boulenger, 1894)
<i>Limnometes paramacrodon</i> (Inger, 1966)
<i>Limnometes picalekua</i> (Stoliczka, 1873)
<i>Lophyne schlaginhaufeni</i> (Wandolock, 1911)
<i>Lithobates areolatus</i> (Baird and Girard, 1852)
<i>Lithobates berlandieri</i> (Baird, 1859)
<i>Lithobates laurii</i> (Mecham, Littlejohn, Oldham, Brown, and Brown, 1973)
<i>Lithobates brownorum</i> (Sanders, 1973)
<i>Lithobates capito</i> (LeConte, 1855)
<i>Lithobates catesbeianus</i> (Shaw, 1802)
<i>Lithobates chiricahuensis</i> (Platz and Mecham, 1979)
<i>Lithobates clamitans</i> (Latreille, 1801)
<i>Lithobates forsteri</i> (Boulenger, 1883)
<i>Lithobates gryllus</i> (Stejneger, 1901)
<i>Lithobates heckscherti</i> (Wright, 1924)
<i>Lithobates maculatus</i> (Brooks, 1877)
<i>Lithobates magraculatus</i> (Frost and Bagnara, 1974)
<i>Lithobates montezumae</i> (Baird, 1854)
<i>Lithobates neovolcanicus</i> (Hillis and Frost, 1985)
<i>Lithobates oaxacoae</i> (Meyer, 1985)
<i>Lithobates onca</i> (Cope, 1875)
<i>Lithobates palmipes</i> (Spix, 1824)
<i>Lithobates palustris</i> (LeConte, 1825)
<i>Lithobates pipiens</i> (Schreber, 1782)
<i>Lithobates palvondra</i> (Webb, 2001)
<i>Lithobates pustulosus</i> (Boulenger, 1883)
<i>Lithobates septentrionalis</i> (Baird, 1854)
<i>Lithobates sphencephalus</i> (Cope, 1886)
<i>Lithobates sylvaticus</i> (LeConte, 1825)
<i>Lithobates tarahumarae</i> (Boulenger, 1917)
<i>Lithobates taylori</i> (Smith, 1959)
<i>Lithobates vallanti</i> (Brooks, 1877)
<i>Lithobates virgatipes</i> (Cope, 1891)
<i>Lithobates warszewitschi</i> (Schmidt, 1857)
<i>Lithobates yavapaiensis</i> (Platz and Frost, 1984)
<i>Litoria adelaidensis</i> (Gray, 1841)
<i>Litoria alboguttata</i> (Günther, 1867)
<i>Litoria ambiguaensis</i> (Horn, 1883)
<i>Litoria andrimalin</i> McDonald, 1997
<i>Litoria arfakiana</i> (Peters and Doria, 1878)
<i>Litoria aurea</i> (Lesser, 1826)
<i>Litoria australis</i> (Gray, 1842)
<i>Litoria barringtonensis</i> (Copland, 1957)
<i>Litoria bicolor</i> (Gray, 1842)
<i>Litoria booroolongensis</i> (Moore, 1961)
<i>Litoria brevipalmata</i> Tyler, Martin, and Watson, 1972
<i>Litoria brevipes</i> (Peters, 1871)
<i>Litoria burrowsi</i> (Scott, 1942)
<i>Litoria caerulea</i> (White, 1790)
<i>Litoria castanea</i> Steindachner, 1867
<i>Litoria cavemolae</i> Tyler and Davies, 1979
<i>Litoria cheesmani</i> (Tyler, 1964)
<i>Litoria chloria</i> (Boulenger, 1892)
<i>Litoria ciliroga</i> (Peron, 1807)
<i>Litoria congensita</i> (Peters and Doria, 1878)
<i>Litoria cooloolensis</i> Liem, 1974
<i>Litoria coplandi</i> (Tyler, 1968)
<i>Litoria corbeni</i> (Wells and Wellington, 1985)
<i>Litoria cryptotis</i> Tyler and Martin, 1977
<i>Litoria cultripes</i> (Parker, 1940)
<i>Litoria cyclorhyncha</i> (Boulenger, 1882)
<i>Litoria dahlii</i> (Boulenger, 1896)
<i>Litoria darlingtoni</i> (Loveridge, 1945)
<i>Litoria daviesae</i> Mahony, Knowles, Foster, and Donnellan, 2001
<i>Litoria dayi</i> (Günther, 1897)
<i>Litoria dentata</i> (Kefauver, 1888)
<i>Litoria dorsalis</i> Macleay, 1875
<i>Litoria electrica</i> Ingram and Corben, 1990
<i>Litoria eucrenia</i> (Lunenburg, 1900)
<i>Litoria ewingi</i> (Duméril and Bérton, 1841)
<i>Litoria fallax</i> (Peters, 1880)
<i>Litoria feschelii</i> Tschudi, 1838
<i>Litoria geminicauda</i> (Horn, 1883)
<i>Litoria gilleni</i> Spencer, 1896
<i>Litoria gracilenta</i> (Peters, 1869)
<i>Litoria laevis</i> Menzies, 1983
<i>Litoria mermis</i> (Peters, 1867)
<i>Litoria infratrifasciata</i> (Günther, 1867)
<i>Litoria puvionensis</i> (Duméril and Bérton, 1841)
<i>Litoria jungguy</i> Donnellan and Mahony, 2004
<i>Litoria kumaes</i> Menzies and Tyler, 2004
<i>Litoria latipalmata</i> (Günther, 1867)
<i>Litoria lesueurii</i> (Duméril and Bérton, 1841)
<i>Litoria littlejohni</i> White, Whitford, and Mahony, 1994
<i>Litoria longipes</i> (Tyler and Martin, 1977)
<i>Litoria longirostris</i> Tyler and Davies, 1977
<i>Litoria lorica</i> Davies and McDonald, 1979
<i>Litoria maculosa</i> (Tyler and Martin, 1977)
<i>Litoria maini</i> (Tyler and Martin, 1977)
<i>Litoria manya</i> (Van Beurden and McDonald, 1980)
<i>Litoria meriana</i> (Tyler, 1969)
<i>Litoria microbelta</i> (Cogger, 1966)
<i>Litoria moorei</i> (Copland, 1957)
<i>Litoria myola</i> Hoskin, 2007
<i>Litoria nanodes</i> (Anderson, 1916)
<i>Litoria nasuta</i> (Gray, 1842)
<i>Litoria nigrofrenata</i> (Günther, 1867)
<i>Litoria novaeollandiae</i> (Steindachner, 1867)
<i>Litoria nudidigita</i> (Copland, 1963)
<i>Litoria nyakalensis</i> Liem, 1974
<i>Litoria oblongumensis</i> Liem and Ingram, 1977
<i>Litoria pallida</i> Davies, Martin, and Watson, 1983
<i>Litoria parawingii</i> Watson, Loftus-Hills, and Littlejohn, 1971
<i>Litoria pearsoniana</i> (Copland, 1981)
<i>Litoria peronii</i> (Tschudi, 1838)
<i>Litoria personata</i> Tyler, Davies, and Martin, 1978
<i>Litoria phyllonchae</i> (Günther, 1883)
<i>Litoria piparata</i> Tyler and Davies, 1985
<i>Litoria platycephala</i> (Günther, 1873)
<i>Litoria pulchra</i> (Wandolock, 1911)
<i>Litoria pygmaea</i> (Meyer, 1875)
<i>Litoria raniformis</i> (Kefauver, 1867)
<i>Litoria reveillei</i> Ingram, Corben, and Hosmer, 1982
<i>Litoria riebolta</i> Liem, 1974
<i>Litoria rothi</i> (De Vis, 1884)
<i>Litoria rubella</i> (Gray, 1842)
<i>Litoria serrata</i> (Anderson, 1916)
<i>Litoria spenceri</i> Dubois, 1964
<i>Litoria splendida</i> Tyler, Davies, and Martin, 1977
<i>Litoria stictota</i> Doughty and Anstis, 2007
<i>Litoria subglaudivosa</i> Tyler and Anstis, 1983
<i>Litoria thesaurensis</i> (Peters, 1877)
<i>Litoria torneri</i> (Nelson, 1923)
<i>Litoria tyleri</i> Martin, Watson, Gartside, Littlejohn, and Loftus-Hills, 1979
<i>Litoria vagluzi</i> (Tyler, Davies, and Martin, 1981)
<i>Litoria verreauxi</i> (Duméril, 1853)
<i>Litoria verrucosa</i> (Tyler and Martin, 1977)
<i>Litoria wallumensis</i> (Copland, 1957)
<i>Litoria wilcoxi</i> (Günther, 1864)
<i>Litoria xanthomera</i> Davies, McDonald, and Adams, 1986
<i>Luavalua romeri</i> (Smith, 1953)
<i>Macrognathopis alipao</i> Carvalho, 1948
<i>Mannophryne collaris</i> (Boulenger, 1912)
<i>Mannophryne herminae</i> (Boettger, 1893)
<i>Mannophryne leonardo</i> Manzanilla, La Marca, Jowers, Sánchez, and García-Paris, 2007
<i>Mannophryne neblina</i> (Test, 1956)
<i>Mannophryne obliterata</i> (Riviero, 1984)
<i>Mannophryne olivacea</i> (Ferd, 1853)
<i>Mannophryne ocellata</i> Barrio-Amorós, Santos, and Molina, 2010
<i>Mannophryne rivero</i> (Donoso-Barrio, 1965)
<i>Mannophryne speen</i> La Marca, 2009
<i>Mannophryne trinitatis</i> (Gaiman, 1888)
<i>Mannophryne venezuelensis</i> Manzanilla, Jowers, La Marca, and García-Paris, 2007
<i>Mannophryne vulcano</i> Barrio-Amorós, Santos, and Molina, 2010
<i>Mannophryne yuzitzi</i> (La Marca, 1999)
<i>Mantella aurantiaca</i> Mocquard, 1900
<i>Mantella baroni</i> Boulenger, 1888
<i>Mantella bernhardi</i> Mercet, Gile, Peyrieras, Böhme, and Busse, 1994
<i>Mantella betaleo</i> (Granddier, 1872)
<i>Mantella cowani</i> Boulenger, 1882
<i>Mantella crocea</i> Portak and Böhme, 1990
<i>Mantella ebenau</i> (Boettger, 1880)
<i>Mantella expectata</i> Busse and Böhme, 1992

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<i>Mantella harenheem</i> Busse, 1961
<i>Mantella brevipes</i> Muthuen and Hewitt, 1913
<i>Mantella madagascariensis</i> (Grandidier, 1872)
<i>Mantella manery</i> Vences, Glaw, and Böhme, 1999
<i>Mantella microsternum</i> Stanislawski, 1996
<i>Mantella nigricans</i> Guibé, 1978
<i>Mantella pulchra</i> Parker, 1925
<i>Mantella viridis</i> Fritsch and Böhme, 1988
<i>Mantidactylus aerumnalis</i> (Peracca, 1893)
<i>Mantidactylus abotrenalis</i> (Müller, 1892)
<i>Mantidactylus asperatus</i> Muthuen, 1920
<i>Mantidactylus belyi</i> Mocquard, 1895
<i>Mantidactylus betsiaricus</i> (Boulenger, 1882)
<i>Mantidactylus charlotteae</i> Vences and Glaw, 2004
<i>Mantidactylus delormei</i> Angel, 1938
<i>Mantidactylus femoralis</i> (Boulenger, 1882)
<i>Mantidactylus grandis</i> Mocquard, 1895
<i>Mantidactylus guttulatus</i> (Boulenger, 1881)
<i>Mantidactylus lugubris</i> (Duméril, 1853)
<i>Mantidactylus madecassus</i> (Milne and Guibé, 1950)
<i>Mantidactylus major</i> Boulenger, 1886
<i>Mantidactylus melanopleura</i> (Mocquard, 1901)
<i>Mantidactylus opiparis</i> (Peracca, 1893)
<i>Megophrya kobayashi</i> Makinus and Matsui, 1997
<i>Megophrya montana</i> Kuhl and Van Hasselt, 1822
<i>Megophrya nasuta</i> (Schlegel, 1858)
<i>Megophrya stenegei</i> Taylor, 1920
<i>Melanophryniscus devincenzi</i> Klappenbach, 1968
<i>Melanophryniscus dorsalis</i> (Mertens, 1932)
<i>Melanophryniscus filirogatus</i> (Mertens, 1937)
<i>Melanophryniscus klappenbachi</i> Prigioni and Langone, 2000
<i>Melanophryniscus macrogranulosus</i> Braun, 1973
<i>Melanophryniscus montevideus</i> (Philippi, 1862)
<i>Melanophryniscus moreirae</i> (Miranda-Ribeiro, 1920)
<i>Melanophryniscus stelnzeri</i> (Weyerbergh, 1875)
<i>Menstogenys amoroletatus</i> (Matsu, 1986)
<i>Menstogenys jerboa</i> (Günther, 1872)
<i>Menstogenys kinabaluensis</i> (Inger, 1966)
<i>Menstogenys orphnozetensis</i> (Matsu, 1986)
<i>Menstogenys poecilus</i> (Inger and Grills, 1983)
<i>Menstogenys whitehead</i> (Boulenger, 1887)
<i>Metanaphryne talana</i> (Peters, 1878)
<i>Metacrinia nicholia</i> (Harrison, 1927)
<i>Metaphrynella pollicaris</i> (Boulenger, 1850)
<i>Metaphrynella sundana</i> (Peters, 1867)
<i>Micrixalus nudus</i> Pilla, 1978
<i>Micrixalus saxicola</i> (Jerdon, 1854)
<i>Microbatrachella capensis</i> (Boulenger, 1910)
<i>Microhyla achalina</i> Tschudi, 1838
<i>Microhyla annamensis</i> Smith, 1923
<i>Microhyla annectens</i> Boulenger, 1900
<i>Microhyla berdmorei</i> (Blyth, 1856)
<i>Microhyla borneensis</i> Parker, 1928
<i>Microhyla butleri</i> Boulenger, 1900
<i>Microhyla erythrogaster</i> Tarkentonius, 1994
<i>Microhyla fissipes</i> Boulenger, 1884
<i>Microhyla heymona</i> Vogt, 1911
<i>Microhyla karunawati</i> Fernando and Sewardhane, 1998
<i>Microhyla nepenthicola</i> Das and Haas, 2010
<i>Microhyla okinawensis</i> Stejneger, 1901
<i>Microhyla ornata</i> (Duméril and Bibron, 1841)
<i>Microhyla palmipes</i> Boulenger, 1897
<i>Microhyla petrigena</i> Inger and Fjogner, 1979
<i>Microhyla pecta</i> Schenkel, 1901
<i>Microhyla pulchra</i> (Halowell, 1861)
<i>Microhyla rubra</i> (Jerdon, 1854)
<i>Microhyla sholigeri</i> Dutta and Ray, 2009
<i>Micryella inornata</i> (Boulenger, 1890)
<i>Micryella stenegei</i> (Boulenger, 1909)
<i>Minnervia satyridis</i> Dubois, Ohler, and Biju, 2001
<i>Minocheilus heymerli</i> (Rivers, 1911)
<i>Mixophyes balbus</i> Straughan, 1968
<i>Mixophyes carbonensis</i> Mahony, Kornfield, Richards, and McDonald, 2006
<i>Mixophyes coggeri</i> Mahony, Kornfield, Richards, and McDonald, 2006
<i>Mixophyes fasciolatus</i> Günther, 1864
<i>Mixophyes leayi</i> Corbett and Ingram, 1987
<i>Mixophyes teratus</i> Straughan, 1968
<i>Mixophyes schweili</i> Loveridge, 1933
<i>Myersiella microgaster</i> (Duméril and Bibron, 1841)
<i>Nyctobatrachus goudoti</i> (Gray, 1841)
<i>Nannophryne variegata</i> Günther, 1870
<i>Nannophrya ceylonensis</i> Günther, 1869
<i>Nannophrya nannorata</i> Kristiansen, 1946
<i>Nanorana aenea</i> (Smith, 1922)
<i>Nanorana pleskei</i> Günther, 1896
<i>Nasikabatrachus sahyadrensis</i> Biju and Bossuyt, 2003
<i>Natalobatrachus bonebergi</i> Hewitt and Muthuen, 1912
<i>Nectophrynoides aspergma</i> Poynton, Howell, Clarke, and Lovett, 1999
<i>Nectophrynoides tomi</i> (Roux, 1906)
<i>Nectophrynoides vestergaardi</i> Menegon, Salvidio, and Loader, 2004
<i>Nectophrynoides viviparus</i> (Tornei, 1905)
<i>Nelsonophryne stenima</i> (Günther, 1901)
<i>Neobatrachus albiges</i> Roberts, Mahony, Kendrick, and Majors, 1991
<i>Neobatrachus centralis</i>
<i>Neobatrachus kunguensis</i> Mahony and Roberts, 1986
<i>Neobatrachus petioleatus</i> (Werner, 1914)
<i>Neobatrachus pictus</i> Peters, 1863
<i>Neobatrachus suteri</i> Lamb, 1911
<i>Neobatrachus sutor</i> Main, 1957
<i>Neobatrachus willmorei</i> (Parker, 1940)
<i>Nimbaphrynoides occidentalis</i> (Angel, 1943)
<i>Nobilia pygmaea</i> Lehr and Collesuzzi, 2009
<i>Notaden bennetti</i> Günther, 1873
<i>Notaden melanocapchus</i> Hosmer, 1962
<i>Notaden nicholia</i> Parker, 1940
<i>Notaden weigeli</i> Shea and Johnston, 1988
<i>Nyctobatrachus alciae</i> Inger, Shaffer, Koshy, and Bakke, 1984
<i>Nyctobatrachus sumayanti</i> Bhadani and Krishnan, 1955
<i>Nyctibatrachus kempholeysensis</i> (Rao, 1937)
<i>Nyctibatrachus major</i> Boulenger, 1882
<i>Nyctibatrachus petreus</i> Das and Kuris, 2005
<i>Nyctimantis rugicauda</i> Boulenger, 1882
<i>Nyctivalus margaritifer</i> Boulenger, 1882
<i>Nyctivalus pictus</i> (Peters, 1871)
<i>Nymphargus griffithsi</i> (Gün, 1961)
<i>Nymphargus rosada</i> (Ruiz-Carranza and Lynch, 1997)
<i>Ociodorops talienensis</i> (Boulenger, 1896)
<i>Ociodorops levis</i> (Günther, 1858)
<i>Ociodorops lima</i> (Gravenhorst, 1829)
<i>Ociodorops maritima</i> (Peters, 1867)
<i>Ociodorops sumatrensis</i> (Peters, 1871)
<i>Odontophrynus achalensis</i> di Iada, Barla, Marlori, and Cei, 1984
<i>Odontophrynus americanus</i> (Duméril and Bibron, 1841)
<i>Odontophrynus cufimpes</i> Reinhardt and Lütken, 1882
<i>Odontophrynus lavilla</i> Cei, 1985
<i>Odontophrynus occidentalis</i> (Berg, 1896)
<i>Odorrana chloronota</i> (Günther, 1876)
<i>Odorrana hosii</i> (Boulenger, 1891)
<i>Odorrana ishikawae</i> (Stejneger, 1901)
<i>Odorrana khaleni</i> (Stuart, Oloff, and Chan-ard, 2005)
<i>Odorrana livida</i> (Blyth, 1856)
<i>Odorrana schmackeri</i> (Boettger, 1892)
<i>Odorrana supranarina</i> (Matsu, 1994)
<i>Odorrana swimhoana</i> (Boulenger, 1903)
<i>Odorrana tormota</i> (Wu, 1977)
<i>Oophaga anobres</i> (Myers, Udd, and Martinez, 1984)
<i>Oophaga granulifera</i> (Taylor, 1958)
<i>Oophaga histrionica</i> (Berthold, 1845)
<i>Oophaga lehmanni</i> (Myers and Daly, 1978)
<i>Oophaga occulator</i> (Myers and Daly, 1976)
<i>Oophaga pumilio</i> (Schmidt, 1857)
<i>Oophaga speciosa</i> (Schmidt, 1857)
<i>Oophaga sylvatica</i> (Furnhouse, 1956)
<i>Oophaga vicentei</i> (Jungfer, Weygoldt, and Juraske, 1996)
<i>Opophryne microstoma</i> Boulenger, 1903
<i>Opophryne pachyproctus</i> Kuhl, 1865
<i>Opisthotylax immaculatus</i> (Boulenger, 1903)
<i>Oreobates bambusa</i> Vaira and Ferrari, 2008
<i>Oreobates guianensis</i> Jiménez de la Espada, 1872
<i>Oreophrynella nigra</i> Señaris, Ayarzagüena, and Gorzula, 1994
<i>Oreophrynella quechua</i> Boulenger, 1895
<i>Oreophryne guacamayo</i> Hoogmoed, 1987
<i>Osteocephalus albiguttatus</i> (Boulenger, 1882)
<i>Osteocephalus buckleyi</i> (Boulenger, 1882)
<i>Osteocephalus cabrerai</i> (Cochran and Goin, 1970)
<i>Osteocephalus castaneicola</i> Moravec, Aparicio, Guerrero-Reinhard, Calderón, Jungfer, and Gvozdík, 2009

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<i>Osteocephalus exophthalmus</i> Smith and Noonan, 2001
<i>Osteocephalus festae</i> (Peracca, 1904)
<i>Osteocephalus kuschflesci</i> Jungler, Ron, Sepp, and Almendáriz, 2000
<i>Osteocephalus lepreuxi</i> (Duméril and Bibron, 1841)
<i>Osteocephalus mimeticus</i> (Melin, 1941)
<i>Osteocephalus mutator</i> Jungler and Hödl, 2002
<i>Osteocephalus ophagus</i> Jungler and Schwan, 1995
<i>Osteocephalus planiceps</i> Cope, 1874
<i>Osteocephalus taurinus</i> Steindachner, 1862
<i>Osteocephalus verrucosus</i> (Werner, 1901)
<i>Osteocephalus yasuni</i> Ron and Pramuk, 1999
<i>Osteopius dominicensis</i> (Tschudi, 1838)
<i>Osteopius crucialis</i> (Harlan, 1826)
<i>Osteopius dominicensis</i> (Tschudi, 1838)
<i>Osteopius marianae</i> (Dunn, 1926)
<i>Osteopius pulchrinatus</i> (Cope, 1870)
<i>Osteopius septentrionalis</i> (Duméril and Bibron, 1841)
<i>Osteopius vastus</i> (Cope, 1871)
<i>Osteopius wilderi</i> (Dunn, 1925)
<i>Paracrinia haswelli</i> (Fletcher, 1964)
<i>Paradactylia palmata</i> (Culicé, 1974)
<i>Paraleimnobotus cardosoi</i> Pomati and Haddad, 1999
<i>Pedostibes jozei</i> (Boulenger, 1892)
<i>Pedostibes rugosus</i> Inger, 1968
<i>Pedostibes tuberculosa</i> Günther, 1876
<i>Pelobates cutripes</i> (Cuvier, 1829)
<i>Pelobates fissus</i> (Lacépède, 1766)
<i>Pelobates synticus</i> Boettger, 1889
<i>Pelobates varadii</i> Pasteur and Bona, 1959
<i>Pelodytes caucasicus</i> Boulenger, 1895
<i>Pelodytes ibericus</i> Sánchez-Herriz, Barbadillo-Escrivá, Machordom, and Sanchiz, 2000
<i>Pelodytes punctatus</i> (Daudin, 1802)
<i>Pelophryne agi</i> Ding, 1983
<i>Pelophryne misera</i> (Mocquard, 1890)
<i>Pelophryne signata</i> (Boulenger, 1895)
<i>Pelophylax bedriagai</i> (Carniero, 1882)
<i>Pelophylax bergeri</i> (Günther, 1866)
<i>Pelophylax carallianus</i> (Arkani, 1958)
<i>Pelophylax crenatus</i> (Bleeri, Hotz, Tunner, Heppich, and Uzzell, 1994)
<i>Pelophylax chosrovici</i> (Okada, 1931)
<i>Pelophylax cretensis</i> (Bleeri, Hotz, Tunner, Heppich, and Uzzell, 1994)
<i>Pelophylax apertus</i> (Schneider, Sotiriadou, and Kyriakopoulou-Sklavounou, 1984)
<i>Pelophylax fukienensis</i> (Pope, 1929)
<i>Pelophylax kurtmulleri</i> (Gayda, 1940)
<i>Pelophylax lessonei</i> (Carniero, 1882)
<i>Pelophylax nigromaculatus</i> (Hallowell, 1861)
<i>Pelophylax perezii</i> (López-Seoane, 1885)
<i>Pelophylax rubicundus</i> (Peters, 1771)
<i>Pelophylax saharicus</i> (Boulenger, 1913)
<i>Pelophylax shqipericus</i> (Hotz, Uzzell, Günther, Tunner, and Heppich, 1987)
<i>Pelophryne catalaunensis</i> (Schwarz, 1959)
<i>Pelophryne emousi</i> Cope, 1862
<i>Pelophryne gurdachi</i> (Rubiak, 1959)
<i>Pelophryne lemuri</i> Cope, 1869
<i>Pelophryne longipinna</i> (Stepenger, 1905)
<i>Pelophryne peltocephala</i> (Tschudi, 1838)
<i>Phasmahyla cochraniae</i> (Bokermann, 1966)
<i>Phasmahyla cruzi</i> Carvalho-e-Silva, Silva, and Carvalho-e-Silva, 2009
<i>Phasmahyla exilis</i> (Cruz, 1980)
<i>Phasmahyla guttata</i> (Lutz, 1924)
<i>Phasmahyla jendasi</i> (Bokermann) and Szalma, 1978
<i>Phasmahyla spectabilis</i> Cruz, Feio, and Nascimento, 2008
<i>Philaetusa acutirostris</i> (Peters, 1867)
<i>Philaetusa anomala</i> Smith, 1931
<i>Philaetusa surffasciatus</i> (Schlegel, 1837)
<i>Philaetusa bunitus</i> Inger, Stuebing, and Tan, 1995
<i>Philaetusa erythrophthalmus</i> Stuebing and Wong, 2000
<i>Philaetusa petersi</i> (Boulenger, 1900)
<i>Philaetusa sanctisilvatus</i> Das and Chanda, 1997
<i>Philaetusa vermiculatus</i> (Boulenger, 1900)
<i>Philaetusa vilgieri</i> (Boulenger, 1897)
<i>Phloria frosti</i> Spencer, 1901
<i>Phloria kundegringeri</i> (Gram and Corben, 1975)
<i>Phloria ovoides</i> Parker, 1940
<i>Phloria pughi</i> Knowles, Mahony, Armstrong, and Donnellan, 2004
<i>Phloria richmondensis</i> Knowles, Mahony, Armstrong, and Donnellan, 2004
<i>Phloria sphagnicola</i> (Moore, 1958)
<i>Phrycinantia boulengeri</i> Perret, 1986
<i>Phrycinantia leonardi</i> (Boulenger, 1906)
<i>Phrycinantia verrucosus</i> (Boulenger, 1912)
<i>Phrynella pulchra</i> Boulenger, 1887
<i>Phrynobatrachus acridoides</i> (Cope, 1867)
<i>Phrynobatrachus africanus</i> (Hallowell, 1858)
<i>Phrynobatrachus allenii</i> Parker, 1936
<i>Phrynobatrachus auritus</i> Boulenger, 1900
<i>Phrynobatrachus batesi</i> (Boulenger, 1906)
<i>Phrynobatrachus calcareatus</i> (Peters, 1863)
<i>Phrynobatrachus circogaster</i> Perret, 1957
<i>Phrynobatrachus graui</i> (Nieden, 1911)
<i>Phrynobatrachus krefftii</i> Boulenger, 1909
<i>Phrynobatrachus latifrons</i> AH, 1924
<i>Phrynobatrachus nelsiana</i> (Smith, 1849)
<i>Phrynobatrachus philophilus</i> Rödel and Ernst, 2002
<i>Phrynomantis aspera</i> (Gravenhorst, 1829)
<i>Phrynomantis xalapensis</i> (Inger, 1964)
<i>Phrynomantis sineschensis</i> Werner, 1910
<i>Phrynomantis bifasciatus</i> (Smith, 1847)
<i>Phrynomantis microps</i> Peters, 1875
<i>Phrynomantis marginata</i> (Izacksohn and Cruz, 1976)
<i>Phyllobates aurotaenia</i> (Boulenger, 1913)
<i>Phyllobates bicolor</i> Duméril and Bibron, 1841
<i>Phyllobates lugubris</i> (Schmidt, 1871)
<i>Phyllobates terribilis</i> Myers, Day, and Malkin, 1978
<i>Phyllobates vittatus</i> (Cope, 1863)
<i>Phyllodytes luteolus</i> (Wied-Neuwied, 1824)
<i>Phyllomedusa atelopoides</i> Duellman, Cadle, and Cannatella, 1988
<i>Phyllomedusa ayeaye</i> (Lutz, 1966)
<i>Phyllomedusa azurea</i> Cope, 1862
<i>Phyllomedusa bahiana</i> Lutz, 1925
<i>Phyllomedusa bicolor</i> (Boddaert, 1772)
<i>Phyllomedusa boliviana</i> Boulenger, 1902
<i>Phyllomedusa burmeisteri</i> Boulenger, 1882
<i>Phyllomedusa camba</i> De la Riva, 1999
<i>Phyllomedusa disticta</i> Lutz, 1950
<i>Phyllomedusa hypochondrialis</i> (Daudin, 1800)
<i>Phyllomedusa ihernigi</i> Boulenger, 1885
<i>Phyllomedusa nelsii</i> Barrio-Amorós, 2006
<i>Phyllomedusa noredestina</i> Caramasch, 2006
<i>Phyllomedusa onades</i> Brandão, 2002
<i>Phyllomedusa palliata</i> Peters, 1873
<i>Phyllomedusa roderi</i> Mertens, 1926
<i>Phyllomedusa sauvagii</i> Boulenger, 1882
<i>Phyllomedusa tarsus</i> (Cope, 1868)
<i>Phyllomedusa torpedoides</i> Pomati and Haddad, 1992
<i>Phyllomedusa tomatoperna</i> (Cope, 1868)
<i>Phyllomedusa trimaculata</i> Mertens, 1926
<i>Phyllomedusa vaillanti</i> Boulenger, 1882
<i>Phyllomedusa venusta</i> Duellman and Trueb, 1967
<i>Physalaemus abronotatus</i> (Steindachner, 1864)
<i>Physalaemus altaviani</i> Haddad and Szalma, 2004
<i>Physalaemus biligonigerus</i> (Cope, 1861)
<i>Physalaemus centralis</i> Bokermann, 1962
<i>Physalaemus cinctus</i> Bokermann, 1966
<i>Physalaemus cuqui</i> Lobo, 1993
<i>Physalaemus cuvieri</i> Fitzinger, 1826
<i>Physalaemus ephippiger</i> (Steindachner, 1864)
<i>Physalaemus fischeri</i> (Boulenger, 1890)
<i>Physalaemus gracilis</i> (Boulenger, 1883)
<i>Physalaemus kneri</i> (Kleinand and Lötken, 1862)
<i>Physalaemus lateralis</i> (Steindachner, 1864)
<i>Physalaemus laei</i> Braun and Braun, 1977
<i>Physalaemus maculiventris</i> (Lutz, 1925)
<i>Physalaemus narmorensis</i> (Reinhardt and Lötken, 1862)
<i>Physalaemus morevae</i> (Miranda-Ribeiro, 1937)
<i>Physalaemus nanus</i> (Boulenger, 1886)
<i>Physalaemus offera</i> (Lichtenstein and Mariens, 1856)
<i>Physalaemus orophius</i> Cassini, Cruz, and Caramasch, 2010
<i>Physalaemus rograndensis</i> Milsáed, 1960
<i>Physalaemus signifer</i> (Günther, 1853)
<i>Physalaemus souzei</i> Izacksohn, 1965
<i>Physalaemus springeri</i> (Miranda-Ribeiro, 1926)
<i>Physotretes australis</i> (Boulenger, 1917)
<i>Pipa arrabali</i> Izacksohn, 1976
<i>Pipa aspera</i> Müller, 1924
<i>Pipa carvalhoi</i> (Miranda-Ribeiro, 1937)
<i>Pipa parva</i> Ruhlén and Gaige, 1923
<i>Pipa pipa</i> (Linnaeus, 1758)

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<i>Platymeria corrugatus</i> (Duméril, 1853)
<i>Platymeria donalis</i> (Duméril, 1853)
<i>Platymeria guentheri</i> (Boulenger, 1882)
<i>Platymeria puppyi</i> (Boulenger, 1884)
<i>Platymeria hazelae</i> Taylor, 1920
<i>Platymeria insulatus</i> Brown and Alcalá, 1970
<i>Platymeria luzonensis</i> Brown, Alcalá, Diesmos, and Alcalá, 1997
<i>Platymeria panayensis</i> Brown, Brown, and Alcalá, 1997
<i>Platymeria poillensis</i> (Taylor, 1922)
<i>Platymeria vitianus</i> (Duméril, 1853)
<i>Platymeria vitianus</i> (Günard, 1853)
<i>Platypleis grandis</i> (Boulenger, 1889)
<i>Platypleis tetra</i> Andreone, Fenolio, and Walvoord, 2003
<i>Platypleis tuberosa</i> (Methuen, 1920)
<i>Platyplectrum ornatum</i> (Gray, 1942)
<i>Platyplectrum spenceri</i> (Parker, 1940)
<i>Plectrohyla distincta</i> (Cope, 1877)
<i>Plectrohyla chrysopleura</i> Wilson, McCranie, and Cruz-Díaz, 1994
<i>Plectrohyla dasypus</i> McCranie and Wilson, 1981
<i>Plectrohyla ephrenera</i> (Mella, Carrasco-Márquez, Smith, and Campbell, 2005)
<i>Plectrohyla gutierrezensis</i> Brocchi, 1977
<i>Plectrohyla hartwegi</i> Duellman, 1968
<i>Plectrohyla</i> cf. Stuart, 1942
<i>Plectrohyla matudae</i> Hartweg, 1941
<i>Plectrohyla penitenter</i> (Adler, 1965)
<i>Plectrohyla sagorum</i> Hartweg, 1941
<i>Pseudodactylopsa guentheri</i> Glaw and Venes, 2007
<i>Pseudodactylopsa inguinalis</i> Boulenger, 1882
<i>Pseudodactylopsa mihanika</i> Venes, Klarwirth, Nussbaum, and Glaw, 2003
<i>Pseudodactylopsa notosticta</i> (Günther, 1877)
<i>Pseudodema bibroni</i> Tschudi, 1838
<i>Pseudodema brachyops</i> (Cope, 1869)
<i>Pseudodema fuscomaculatum</i> (Steindachner, 1864)
<i>Pseudodema marmoratum</i> (Duméril and Bibron, 1840)
<i>Pseudodema thau</i> (Lesson, 1826)
<i>Polypedates collettii</i> (Boulenger, 1900)
<i>Polypedates cruciger</i> Blyth, 1852
<i>Polypedates eques</i> Günther, 1858
<i>Polypedates fatigo</i> Manamendra-Arachchi and Pethiyagoda, 2001
<i>Polypedates leucomystax</i> (Gravenhorst, 1829)
<i>Polypedates longinasus</i> (Alt, 1927)
<i>Polypedates macrois</i> (Boulenger, 1891)
<i>Polypedates maculatus</i> (Gray, 1830)
<i>Polypedates megacephalus</i> Hallowell, 1861
<i>Polypedates occidentalis</i> Das and Datta, 2006
<i>Polypedates olophus</i> (Boulenger, 1893)
<i>Polypedates pseudocruigeri</i> Das and Ravichandran, 1998
<i>Poyntonie palauicola</i> Channing and Boycott, 1989
<i>Poyntonophrynus domensis</i> (Bocage, 1865)
<i>Poyntonophrynus lupherensis</i> (Loveridge, 1932)
<i>Pristimantis acelinus</i> (Boulenger, 1896)
<i>Pristimantis actives</i> (Lynch, 1979)
<i>Pristimantis acuminatus</i> (Shreve, 1935)
<i>Pristimantis albertus</i> Duellman and Hedges, 2007
<i>Pristimantis allenae</i> sonius (Barbour and Dunn, 1921)
<i>Pristimantis appendiculatus</i> (Werner, 1894)
<i>Pristimantis bicornutus</i> (Peters, 1863)
<i>Pristimantis bogotensis</i> (Peters, 1863)
<i>Pristimantis boulengeri</i> (Lynch, 1981)
<i>Pristimantis brevifrons</i> (Lynch, 1981)
<i>Pristimantis bromelaeae</i> (Lynch, 1979)
<i>Pristimantis calcaratus</i> (Lynch, 1976)
<i>Pristimantis caryophylleus</i> (Barbour, 1928)
<i>Pristimantis cerasinus</i> (Cope, 1875)
<i>Pristimantis charlotewillensis</i> (Kaiser, Dwyer, Feichtinger, and Schmid, 1995)
<i>Pristimantis consipillatus</i> (Günther, 1858)
<i>Pristimantis crucifer</i> (Boulenger, 1899)
<i>Pristimantis cruentus</i> (Peters, 1873)
<i>Pristimantis delius</i> (Duellman and Mendelson, 1995)
<i>Pristimantis diadematus</i> (Jiménez de la Espada, 1875)
<i>Pristimantis doropisica</i> (Rivers and Serna, 1988)
<i>Pristimantis erythroleura</i> (Boulenger, 1896)
<i>Pristimantis eugeniae</i> (Lynch and Duellman, 1997)
<i>Pristimantis fenestratus</i> (Steindachner, 1864)
<i>Pristimantis festae</i> (Peracca, 1904)
<i>Pristimantis gaigei</i> (Dunn, 1931)
<i>Pristimantis gaudi</i> Jiménez de la Espada, 1870
<i>Pristimantis kichwarum</i> Elmer and Cannatella, 2008
<i>Pristimantis lanthanites</i> (Lynch, 1975)
<i>Pristimantis latidivus</i> (Lynch and Burrows, 1990)
<i>Pristimantis leoni</i> (Lynch, 1976)
<i>Pristimantis leucopus</i> (Lynch, 1976)
<i>Pristimantis litidorsalis</i> (Lynch, 1976)
<i>Pristimantis lymani</i> (Barbour and Noble, 1920)
<i>Pristimantis makini</i> (Lynch, 1980)
<i>Pristimantis mariae</i> (Lynch, 1974)
<i>Pristimantis moro</i> (Savage, 1965)
<i>Pristimantis museosa</i> (BaÑez, Jaramillo, and Arosemena, 1994)
<i>Pristimantis nevius</i> (Lynch, 1994)
<i>Pristimantis ockendeni</i> (Boulenger, 1912)
<i>Pristimantis parva</i> (Lynch and Arlilla-Robayo, 1999)
<i>Pristimantis palmeri</i> (Boulenger, 1912)
<i>Pristimantis paramerus</i> (Rivers, 1984)
<i>Pristimantis pardalis</i> (Barbour, 1928)
<i>Pristimantis peruvius</i> (Lynch, 1976)
<i>Pristimantis peruvianus</i> (Lynch, Ruiz-Caranza, and Arlilla-Robayo, 1994)
<i>Pristimantis peruvianus</i> (Melin, 1941)
<i>Pristimantis platydictylus</i> (Boulenger, 1903)
<i>Pristimantis quiquevarius</i> (Lynch, 1974)
<i>Pristimantis renjilum</i> (Lynch, 2000)
<i>Pristimantis rogersi</i> (Cope, 1866)
<i>Pristimantis riveti</i> (Desape, 1911)
<i>Pristimantis taenatus</i> (Boulenger, 1912)
<i>Pristimantis terraeboliviana</i> (Rivers, 1961)
<i>Pristimantis unistigatus</i> (Günther, 1859)
<i>Pristimantis uranobates</i> (Lynch, 1991)
<i>Pristimantis urchi</i> (Boettger, 1894)
<i>Pristimantis variabilis</i> (Lynch, 1988)
<i>Pristimantis w-nigrum</i> (Boettger, 1892)
<i>Pristimantis walkeri</i> (Lynch, 1974)
<i>Pristimantis zimmermanae</i> (Fryer and Hardy, 1991)
<i>Probreviceps macrodictylus</i> (Nieden, 1928)
<i>Proceratophrys appendiculata</i> (Günther, 1873)
<i>Proceratophrys digitibosa</i> (Peters, 1872)
<i>Proceratophrys boie</i> (Wied-Neuwied, 1824)
<i>Proceratophrys concavdympianum</i> Giacetta, Bernarde, and Kokubum, 2000
<i>Proceratophrys crataepus</i> (Müller, 1863)
<i>Proceratophrys feliposa</i> Jacobson and Pexidlo, 1981
<i>Proceratophrys morato</i> (Jim and Caramaschi, 1980)
<i>Proceratophrys subguttata</i> Jacobson, Cruz, and Pexidlo, 1999
<i>Psithyrus dum</i> (Kiers, 1961)
<i>Pseudacris brachyphona</i> (Cope, 1889)
<i>Pseudacris brimley</i> Brandt and Walker, 1933
<i>Pseudacris calaverina</i> (Cope, 1866)
<i>Pseudacris clarki</i> (Baird, 1854)
<i>Pseudacris crucifer</i> (Wied-Neuwied, 1838)
<i>Pseudacris ferarum</i> (Baird, 1854)
<i>Pseudacris fouquettei</i> Lemmon, Lemmon, Collins, and Cannatella, 2008
<i>Pseudacris hypochondrifica</i> (Hallowell, 1854)
<i>Pseudacris illinoensis</i> Smith, 1961
<i>Pseudacris kalmi</i> Harper, 1955
<i>Pseudacris maculata</i> (Agassiz, 1850)
<i>Pseudacris nigrita</i> (LeConte, 1825)
<i>Pseudacris oculans</i> (Bosc and Daudin, 1801)
<i>Pseudacris ornata</i> (Holbrook, 1836)
<i>Pseudacris regilla</i> (Baird and Girard, 1853)
<i>Pseudacris sierrae</i> (Jameson, Mackey, and Richmond, 1966)
<i>Pseudacris streckeri</i> Wright and Wright, 1933
<i>Pseudacris triarata</i> (Wied-Neuwied, 1838)
<i>Pseudopodaiaea boulengeri</i> (Lataste, 1879)
<i>Pseudopodaiaea brongersmai</i> (Hoogmoed, 1972)
<i>Pseudopodaiaea sedleyi</i> (Strauch, 1879)
<i>Pseudopodaiaea acule</i> (Stock, Scola, Belfiore, Buckley, Lo Valvo, and Arculeo, 2008)
<i>Pseudopodaiaea aurata</i> (Boulenger, 1891)
<i>Pseudopodaiaea variabilis</i> (Pallas, 1769)
<i>Pseudopodaiaea viridis</i> (Laurenti, 1768)
<i>Pseudis bolodactyla</i> Lutz, 1925
<i>Pseudis carolinensis</i> Kner, 2000
<i>Pseudis minuta</i> Günther, 1858
<i>Pseudis paradoxa</i> (Linnaeus, 1758)
<i>Pseudis tocantins</i> Caramaschi and Cruz, 1998
<i>Pseudobufo subasper</i> Tschudi, 1838
<i>Pseudopaludicola boliviana</i> Parker, 1927
<i>Pseudopaludicola felipes</i> (Hensel, 1867)
<i>Pseudopaludicola mystacalis</i> (Cope, 1887)
<i>Pseudopaludicola saltica</i> (Cope, 1887)

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<i>Pseudopaludicola ferretzi</i> Miranda-Ribeiro, 1937
<i>Pseudophilautus embol</i> (Biju and Bossuyt, 2009)
<i>Pseudophilautus cavirostris</i> (Günther, 1869)
<i>Pseudophilautus cuspi</i> (Manamendra-Arachchi and Pethiyagoda, 2005)
<i>Pseudophilautus decori</i> (Manamendra-Arachchi and Pethiyagoda, 2005)
<i>Pseudophilautus femoralis</i> (Günther, 1864)
<i>Pseudophilautus foliola</i> (Manamendra-Arachchi and Pethiyagoda, 2005)
<i>Pseudophilautus hofmanni</i> (Meegaskumbura and Manamendra-Arachchi, 2005)
<i>Pseudophilautus hopotloi</i> (Manamendra-Arachchi and Pethiyagoda, 2005)
<i>Pseudophilautus kari</i> (Biju and Bossuyt, 2009)
<i>Pseudophilautus microgasterum</i> (Günther, 1864)
<i>Pseudophilautus ocularis</i> (Manamendra-Arachchi and Pethiyagoda, 2005)
<i>Pseudophilautus popillae</i> (Meegaskumbura and Manamendra-Arachchi, 2005)
<i>Pseudophilautus popularis</i> (Manamendra-Arachchi and Pethiyagoda, 2005)
<i>Pseudophilautus procax</i> (Manamendra-Arachchi and Pethiyagoda, 2005)
<i>Pseudophilautus reticulatus</i> (Günther, 1864)
<i>Pseudophilautus ros</i> (Manamendra-Arachchi and Pethiyagoda, 2005)
<i>Pseudophilautus silus</i> (Manamendra-Arachchi and Pethiyagoda, 2005)
<i>Pseudophilautus singu</i> (Meegaskumbura, Manamendra-Arachchi, and Pethiyagoda, 2009)
<i>Pseudophilautus sordius</i> (Manamendra-Arachchi and Pethiyagoda, 2005)
<i>Pseudophilautus stiereri</i> (Meegaskumbura and Manamendra-Arachchi, 2005)
<i>Pseudophilautus stictomerus</i> (Günther, 1876)
<i>Pseudophilautus zoro</i> (Manamendra-Arachchi and Pethiyagoda, 2005)
<i>Pseudophryne australis</i> Gray, 1831
<i>Pseudophryne bibroni</i> Günther, 1859
<i>Pseudophryne coriacea</i> Kelenstein, 1988
<i>Pseudophryne corrodore</i> Moore, 1953
<i>Pseudophryne covacevichae</i> Ingram and Corben, 1994
<i>Pseudophryne dendyi</i> Lucas, 1892
<i>Pseudophryne guentheri</i> Boulenger, 1882
<i>Pseudophryne major</i> Parker, 1940
<i>Pseudophryne occidentalis</i> Parker, 1940
<i>Pseudophryne peggileyi</i> Wells and Wellington, 1985
<i>Pseudophryne raveni</i> Ingram and Corben, 1994
<i>Pseudophryne semimarmorata</i> Lucas, 1892
<i>Pyrrhadenia archetiae</i> (Bocour, 1865)
<i>Pyrrhadenia maseanensis</i> (Duméril and Bibron, 1841)
<i>Pyrrhadenia oxyrhynchus</i> (Smith, 1849)
<i>Pyrrhadenia schillakerum</i> (Werner, 1938)
<i>Pyrrhophyla hypomykter</i> McCrane and Wilson, 1993
<i>Pyrrhophyla legleri</i> (Taylor, 1958)
<i>Pyrrhophyla leonardschultzei</i> (Al, 1934)
<i>Pyrrhophyla salvadorensis</i> (Mertens, 1952)
<i>Pyrrhophyla spinipollex</i> (Schmidt, 1936)
<i>Pyrrhophyla adspersa</i> Tschudi, 1838
<i>Pyrrhophyla edulis</i> Peters, 1854
<i>Quasipaa exilispinosa</i> (Lu and Hu, 1975)
<i>Quasipaa fasciculipinna</i> (Inger, 1970)
<i>Quasipaa gonosa</i> (David, 1975)
<i>Ramanella anamalaensis</i> Rao, 1937
<i>Ramanella montana</i> (Jerdon, 1864)
<i>Ramanella obscura</i> (Günther, 1864)
<i>Ramanella triangularis</i> (Günther, 1876)
<i>Ramanella variegata</i> (Stoliczka, 1872)
<i>Rana "kainwani" XXX</i> , 2008
<i>Rana amurenensis</i> Boulenger, 1886
<i>Rana arvalis</i> Nilsson, 1842
<i>Rana aurora</i> Baird and Girard, 1852
<i>Rana boylii</i> Baird, 1854
<i>Rana cascadae</i> Slater, 1939
<i>Rana chensiensis</i> David, 1975
<i>Rana coreana</i> Okada, 1928
<i>Rana dalmatina</i> Fitzinger, 1839
<i>Rana dryidoni</i> Baird and Girard, 1852
<i>Rana dybowskii</i> Günther, 1876
<i>Rana graeca</i> Boulenger, 1891
<i>Rana huarenensis</i> Fei, Ye, and Huang, 1990
<i>Rana berica</i> Boulenger, 1879
<i>Rana italica</i> Dubois, 1987
<i>Rana japonica</i> Boulenger, 1879
<i>Rana pinna</i> Smith, 1921
<i>Rana kunyensis</i> Lu and Li, 2002
<i>Rana laietana</i> Boulenger, 1879
<i>Rana longicauda</i> Stejneger, 1898
<i>Rana luteiventris</i> Thompson, 1913
<i>Rana macrocheilus</i> Boulenger, 1885
<i>Rana muscosa</i> Cope, 1917
<i>Rana ornativentris</i> Werner, 1903
<i>Rana pricei</i> Matsui, 1991
<i>Rana pretiosa</i> Baird and Girard, 1853
<i>Rana pyrenaica</i> Serra-Cobo, 1993
<i>Rana sakurai</i> Matsui and Matsui, 1990
<i>Rana stutzeri</i> Boulenger, 1909
<i>Rana sierrae</i> Cope, 1917
<i>Rana sylvatica</i>
<i>Rana tagoi</i> Okada, 1928
<i>Rana tavassensis</i> Baran and Altun, 1986
<i>Rana temporaria</i> Linnaeus, 1758
<i>Rana tashimensis</i> Stejneger, 1907
<i>Ranitomeya amazonica</i> (Schulte, 1999)
<i>Ranitomeya benedicta</i> Brown, Twomey, Pepper, and Sanchez-Rodriguez, 2008
<i>Ranitomeya bolei</i> (Moraes, 1992)
<i>Ranitomeya bomboles</i> (Myers and Daly, 1980)
<i>Ranitomeya claudiae</i> (Jungfer, Lötters, and Jörgens, 2000)
<i>Ranitomeya cyathostoma</i> Pérez-Felipe, Chávez, Twomey, and Brown, 2010
<i>Ranitomeya defleri</i> Twomey and Brown, 2009
<i>Ranitomeya dorcaswansonae</i> (Rueda-Almonacid, Rada, Sánchez-Pacheco, Velásquez-Avárez, and Quevedo-Gil, 2006)
<i>Ranitomeya duellmani</i> (Schulte, 1999)
<i>Ranitomeya fanfastica</i> (Boulenger, 1884)
<i>Ranitomeya flaviventris</i> (Schulte, 1999)
<i>Ranitomeya fupurilla</i> (Silverstone, 1975)
<i>Ranitomeya imitator</i> (Schulte, 1986)
<i>Ranitomeya intermedia</i> (Schulte, 1999)
<i>Ranitomeya lamasi</i> (Moraes, 1992)
<i>Ranitomeya minuta</i> (Shreve, 1935)
<i>Ranitomeya opisthomelas</i> (Boulenger, 1899)
<i>Ranitomeya reticulata</i> (Boulenger, 1884)
<i>Ranitomeya sirensis</i> (Aichinger, 1991)
<i>Ranitomeya sumneri</i> Brown, Twomey, Pepper, and Sanchez-Rodriguez, 2008
<i>Ranitomeya tolimensis</i> Berna-Bauduis, Lora-Mora, Gallego, and Quevedo-Gil, 2007
<i>Ranitomeya uskari</i> (Brown, Schulte, and Summers, 2006)
<i>Ranitomeya vanzolini</i> (Myers, 1982)
<i>Ranitomeya variegata</i> (Zimmermann and Zimmermann, 1988)
<i>Ranitomeya ventrimaculata</i> (Shreve, 1935)
<i>Ranitomeya virolinensis</i> (Ruiz-Carranza and Ramirez-Pinilla, 1992)
<i>Rhacophorus akrapallagi</i> (Biju and Bossuyt, 2009)
<i>Rhacophorus bombaensis</i> (Amadasi, 1979)
<i>Rhacophorus charius</i> (Rao, 1937)
<i>Rhacophorus glandulosus</i> (Jerdon, 1864)
<i>Rhacophorus jayakari</i> (Biju and Bossuyt, 2009)
<i>Rhacophorus kakati</i> (Biju and Bossuyt, 2009)
<i>Rhacophorus luteolus</i> (Kuremoto and Joshy, 2003)
<i>Rhacophorus neotragus</i> (Biju and Bossuyt, 2009)
<i>Rhacophorus ponmudi</i> (Biju and Bossuyt, 2005)
<i>Rhacophorus tuberculatus</i> (Kuremoto and Joshy, 2003)
<i>Reithrodontomys</i> (Ruthe, 1914)
<i>Rhacophorus angulirostris</i> Al, 1927
<i>Rhacophorus annamensis</i> Smith, 1924
<i>Rhacophorus apicifidulatus</i> (Günther, 1858)
<i>Rhacophorus arboreus</i> (Okada and Kawano, 1924)
<i>Rhacophorus arvalis</i> Lue, Lai, and Chen, 1995
<i>Rhacophorus auriventris</i> Lue, Lai, and Chen, 1994
<i>Rhacophorus belalongensis</i> Dehling and Grafe, 2008
<i>Rhacophorus bipunctatus</i> Al, 1927
<i>Rhacophorus bumasus</i> (Kriekorian, 1938)
<i>Rhacophorus calcadenas</i> Al, 1927
<i>Rhacophorus calcaneus</i> Smith, 1924
<i>Rhacophorus calidius</i> Harvey, Penherton, and Smith, 2002
<i>Rhacophorus cyanopunctatus</i> Marthey and Steid, 1998
<i>Rhacophorus dennysi</i> Blanford, 1881
<i>Rhacophorus dultensis</i> Boulenger, 1892
<i>Rhacophorus everetti</i> Boulenger, 1884
<i>Rhacophorus ewechopygus</i> Inger, Onov, and Darevsky, 1999
<i>Rhacophorus fasciatus</i> Boulenger, 1895
<i>Rhacophorus fazei</i> Boulenger, 1893
<i>Rhacophorus gedingsensis</i> Das and Haas, 2005
<i>Rhacophorus gauri</i> (Inger, 1966)
<i>Rhacophorus georgii</i> Roux, 1904
<i>Rhacophorus harrisoni</i> Inger and Hale, 1959
<i>Rhacophorus hungfensis</i> Liu and Hu, 1961
<i>Rhacophorus kaku</i> Ding, 1983
<i>Rhacophorus klo Ohler and Delorme, 2006</i>
<i>Rhacophorus lateralis</i> Boulenger, 1883
<i>Rhacophorus melanoticus</i> (Jerdon, 1870)
<i>Rhacophorus margaritifer</i> (Schlegel, 1837)
<i>Rhacophorus maximus</i> Günther, 1858

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<i>Rhacophorus mollerecti</i> Boulenger, 1908
<i>Rhacophorus morticola</i> Boulenger, 1898
<i>Rhacophorus nigropalmatus</i> Boulenger, 1895
<i>Rhacophorus orori</i> Ziegler and Köhler, 2001
<i>Rhacophorus oswolti</i> (Stejneger, 1907)
<i>Rhacophorus pardalis</i> Günther, 1858
<i>Rhacophorus prasinatus</i> Moç, Nisch, and Lue, 1983
<i>Rhacophorus prominens</i> Smith, 1924
<i>Rhacophorus pseudomalabaricus</i> Vasudevan and Dutta, 2000
<i>Rhacophorus reinwardti</i> (Schlegel, 1840)
<i>Rhacophorus riodus</i> Li and Hu, 1995
<i>Rhacophorus rufipes</i> Inger, 1998
<i>Rhacophorus schlegelii</i> (Günther, 1858)
<i>Rhacophorus tapieus</i> Liang and Wang, 1978
<i>Rhacophorus viridis</i> (Hallowell, 1861)
<i>Rhacophorus yaoshanensis</i> Liu and Hu, 1962
<i>Rhaebo blombergi</i> (Meyers and Furkhouser, 1951)
<i>Rhaebo glaberrimus</i> (Günther, 1869)
<i>Rhaebo pufillatus</i> (Schneider, 1799)
<i>Rhaebo haemalticus</i> Cope, 1862
<i>Rhaebo hypomelas</i> (Boulenger, 1913)
<i>Rheobates palmatus</i> (Werner, 1899)
<i>Rheobatrachus silus</i> Liem, 1973
<i>Rhodabatrachus vitellinus</i> Mahony, Tyler, and Davies, 1984
<i>Rhinella abei</i> (Baldassera, Caramaschi, and Haddad, 2004)
<i>Rhinella alata</i> (Thomind, 1884)
<i>Rhinella arenarum</i> (Hensel, 1907)
<i>Rhinella arunco</i> (Molina, 1782)
<i>Rhinella azarae</i> (Gallardo, 1965)
<i>Rhinella castaneodactylus</i> (Caldwell, 1951)
<i>Rhinella ceradensis</i> Maciel, Brandão, Campos, and Sebben, 2007
<i>Rhinella crucifer</i> (Wied-Neuwied, 1821)
<i>Rhinella dipalis</i> (Myers and Carvalho, 1945)
<i>Rhinella dorbignyi</i> (Duméril and Bibron, 1841)
<i>Rhinella fernandezae</i> (Gallardo, 1957)
<i>Rhinella granulosa</i> (Spix, 1824)
<i>Rhinella hoogmoedi</i> Caramaschi and Pombal, 2006
<i>Rhinella humboldti</i> (Gallardo, 1965)
<i>Rhinella icterica</i> (Spix, 1824)
<i>Rhinella jimi</i> (Stévoux, 2002)
<i>Rhinella major</i> (Müller and Hellmich, 1936)
<i>Rhinella maritima</i> Chaparro, Pram, and Gluesenkamp, 2007
<i>Rhinella margaritifera</i> (Laurenti, 1768)
<i>Rhinella marina</i> (Linnaeus, 1758)
<i>Rhinella marly</i> Fouquet, Gascho, Blight, and Vélez-Rodriguez, 2007
<i>Rhinella mirandaberoi</i> (Gallardo, 1965)
<i>Rhinella ocellata</i> (Günther, 1858)
<i>Rhinella ornata</i> (Spix, 1824)
<i>Rhinella paraguayensis</i> Avila, Panonzo, and Struassmann, 2010
<i>Rhinella pombali</i> (Baldassera, Caramaschi, and Haddad, 2004)
<i>Rhinella proboscidea</i> (Spix, 1824)
<i>Rhinella quechua</i> (Gallardo, 1961)
<i>Rhinella rubescens</i> (Lutz, 1925)
<i>Rhinella rubropunctata</i> (Guichenot, 1848)
<i>Rhinella schneideri</i> (Werner, 1898)
<i>Rhinella spinulosa</i> (Wiegmann, 1834)
<i>Rhinella stenocephala</i> (Günther, 1858)
<i>Rhinella verreauxi</i> (Brandão, Maciel, and Sebben, 2007)
<i>Rhinoderma darwini</i> Duméril and Bibron, 1841
<i>Rhinophrynus dorsalis</i> Duméril and Bibron, 1841
<i>Rhinophrynus coronatus</i> (Venoso and Glaw, 2003)
<i>Rhinophrynus matavyi</i> D'Cruze, Köhler, Venoso, and Glaw, 2010
<i>Ruiyana modamdi</i> (Cárdena-Heredia, Venegas, Rada, and Schulte, 2008)
<i>Ruiyana sustermani</i> (Ruiz-Correa and Lynch, 1995)
<i>Sachatamia albomaculata</i> (Taylor, 1949)
<i>Sachatamia flex</i> (Savage, 1967)
<i>Sachatamia punctulata</i> (Ruiz-Correa and Lynch, 1995)
<i>Scaphiophrynus brevis</i> (Boulenger, 1896)
<i>Scaphiophrynus calcarata</i> (Mocquard, 1895)
<i>Scaphiophrynus gottliebii</i> Buisson and Böhm, 1992
<i>Scaphiophrynus madagascariensis</i> (Boulenger, 1882)
<i>Scaphiophrynus marmorata</i> Boulenger, 1882
<i>Scaphiophrynus spinosa</i> (Steindachner, 1862)
<i>Scaphiopus couchii</i> Baird, 1854
<i>Scaphiopus holbrooki</i> (Harian, 1835)
<i>Scaphiopus huerfani</i> Strecker, 1910
<i>Scartylia goniurum</i> (Bokermann, 1962)
<i>Scartylia vigilans</i> (Solano, 1971)
<i>Schismaderma carens</i> (Smith, 1848)
<i>Schnax acuminatus</i> (Cope, 1862)
<i>Schnax aliae</i> (Dunn, 1933)
<i>Schnax alter</i> (Lutz, 1973)
<i>Schnax argyrenatus</i> (Miranda-Ribeiro, 1926)
<i>Schnax arnoldi</i> (Favovich, 2005)
<i>Schnax berthae</i> (Barrio, 1962)
<i>Schnax boesemani</i> (Gün, 1966)
<i>Schnax boulengeri</i> (Cope, 1887)
<i>Schnax catharinae</i> (Boulenger, 1888)
<i>Schnax centralis</i> Pombal and Bassi, 1998
<i>Schnax chiquitanus</i> (De la Riva, 1990)
<i>Schnax crosopodopsila</i> (Lutz, 1925)
<i>Schnax cuentomimus</i> (Duellman, 1972)
<i>Schnax curvica</i> (Pugliese, Pombal, and Sazima, 2004)
<i>Schnax duartei</i> (Lutz, 1951)
<i>Schnax electrophorus</i> (Cope, 1875)
<i>Schnax eurylopes</i> (Bokermann, 1968)
<i>Schnax funereus</i> (Cope, 1874)
<i>Schnax fuscomarginatus</i> (Lutz, 1925)
<i>Schnax fuscomaculatus</i> (Lutz, 1925)
<i>Schnax garbei</i> (Miranda-Ribeiro, 1926)
<i>Schnax granulatus</i> (Peters, 1871)
<i>Schnax hayi</i> (Barbour, 1900)
<i>Schnax humilis</i> (A. Lutz and B. Lutz, 1964)
<i>Schnax ictericus</i> (Duellman and Wiens, 1993)
<i>Schnax littoralis</i> (Pombal and Corio, 1991)
<i>Schnax litoreus</i> (Peixoto, 1988)
<i>Schnax longimembris</i> (Lutz, 1968)
<i>Schnax mairiqui</i> (Barrio-Amorós, Orellana, and Chacón-Ortiz, 2004)
<i>Schnax nasica</i> (Cope, 1862)
<i>Schnax nebulosus</i> (Spix, 1824)
<i>Schnax oblongatus</i> (Lutz, 1973)
<i>Schnax oreites</i> (Duellman and Wiens, 1993)
<i>Schnax pachycrus</i> (Miranda-Ribeiro, 1937)
<i>Schnax perereca</i> (Pombal, Haddad, and Kashiwara, 1995)
<i>Schnax persuasiva</i> (Lutz and Lutz, 1939)
<i>Schnax proboscidea</i> (Brongersma, 1933)
<i>Schnax quinquelascotus</i> (Fowler, 1913)
<i>Schnax ranki</i> (Andrade and Cardoso, 1989)
<i>Schnax rizzolii</i> (Bokermann, 1964)
<i>Schnax rogersi</i> (Pugliese, Balle, and Pombal, 2009)
<i>Schnax costatus</i> (Peters, 1863)
<i>Schnax ruber</i> (Laurenti, 1768)
<i>Schnax similis</i> (Cochran, 1953)
<i>Schnax squamirostris</i> (Lutz, 1925)
<i>Schnax staufferi</i> (Cope, 1865)
<i>Schnax sugliatus</i> (Duellman, 1973)
<i>Schnax tapicheiro</i> (A. Lutz and B. Lutz, 1954)
<i>Schnax trilineatus</i> (Hoogmoed and Gorzula, 1979)
<i>Schnax uruguayus</i> (Schmidt, 1944)
<i>Schnax vanisae</i> (Pyburn and Fouquette, 1971)
<i>Schnax x-signatus</i> (Spix, 1824)
<i>Scythophrys sawayae</i> (Cochran, 1953)
<i>Semmodactylus weali</i> (Boulenger, 1882)
<i>Silurana epifitocalis</i> (Fischberg, Colombelli, and Picard, 1982)
<i>Silurana tropicalis</i> Gray, 1864
<i>Silverstoneia ficiator</i> (Dunn, 1931)
<i>Silverstoneia rubicola</i> (Dunn, 1924)
<i>Smilisca baudini</i> (Duméril and Bibron, 1841)
<i>Smilisca cyanopectus</i> (Smith, 1953)
<i>Smilisca dentata</i> (Smith, 1957)
<i>Smilisca fodiens</i> (Boulenger, 1882)
<i>Smilisca phaeota</i> (Cope, 1862)
<i>Smilisca zuma</i> (Cope, 1865)
<i>Smilisca alia</i> (Duellman and Trueb, 1966)
<i>Smilisca sordida</i> (Peters, 1863)
<i>Scopeloglossus sechelenensis</i> (Boettger, 1896)
<i>Spea bomblifrons</i> (Cope, 1863)
<i>Spea hammondi</i> (Baird, 1859)
<i>Spea intermontana</i> (Cope, 1863)
<i>Spea multiplicata</i> (Cope, 1863)
<i>Sphaenorhynchus caraniaschi</i> Toledo, Garcia, Lingnau, and Haddad, 2007
<i>Sphaenorhynchus doriae</i> (Gün, 1957)
<i>Sphaenorhynchus lacteus</i> (Baudin, 1800)
<i>Sphaenorhynchus orophilus</i> (Lutz and Lutz, 1938)
<i>Sphaenorhynchus palustris</i> (Bokermann, 1966)
<i>Sphaenorhynchus planicola</i> (Lutz and Lutz, 1938)
<i>Sphaenorhynchus surdus</i> (Cochran, 1953)

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<i>Sphaerostetha brevicaps</i> (Schneider, 1799)
<i>Sphaerostetha rolandei</i> (Dubois, 1953)
<i>Sphenophryne comula</i> Peters and Doria, 1878
<i>Spicospina harnmocerae</i> lela Roberts, Horwitz, Wardell-Johnson, Maxson, and Mahony, 1997
<i>Spinomantis aglavei</i> (Methuen and Hewitt, 1913)
<i>Spinomantis fimbriatus</i> (Glaw and Vences, 1994)
<i>Spinomantis phantasticus</i> (Glaw and Vences, 1997)
<i>Stauriois pulchatus</i> (Günther, 1858)
<i>Stauriois latopalmatus</i> (Boulenger, 1887)
<i>Stauriois natorator</i> (Günther, 1858)
<i>Stauriois parvus</i> Inger and Halls, 1959
<i>Stauriois tuberlinguis</i> Boulenger, 1916
<i>Stefania ayangananae</i> MacCulloch and Lallrop, 2002
<i>Stefania ginesi</i> Rivero, 1968
<i>Stereocyclops incompressus</i> Cope, 1870
<i>Strabomantis biporcatus</i> Peters, 1863
<i>Strabomantis bufoniformis</i> (Boulenger, 1896)
<i>Strongylopus boneaspesi</i> (Dubois, 1981)
<i>Strongylopus fasciatus</i> (Smith, 1849)
<i>Strongylopus grayi</i> (Smith, 1849)
<i>Stumpffia pygmaea</i> Vences and Glaw, 1991
<i>Stumpffia tetradactyla</i> Vences and Glaw, 1991
<i>Tachycineta seycheliensis</i> (Duméril and Bibrón, 1841)
<i>Taudactylus acolinaria</i> (Anderson, 1916)
<i>Taudactylus diurnus</i> Straughan and Lee, 1966
<i>Taudactylus eungellensis</i> Liem and Hoerner, 1973
<i>Taudactylus liem</i> Ingram, 1980
<i>Taudactylus pleione</i> Czechura, 1986
<i>Taudactylus rheophilus</i> Liem and Hoerner, 1973
<i>Telmatobius culeus</i> (Caimán, 1876)
<i>Telmatobius hintoni</i> Parker, 1940
<i>Telmatobius huayra</i> Lavilla and Egueta-Sandoval, 1995
<i>Telmatobius pelti</i> (Peters, 1873)
<i>Telmatobius marmoratus</i> (Duméril and Bibrón, 1841)
<i>Telmatobufo bukocki</i> Schmidt, 1952
<i>Telmatobufo venustus</i> (Philipp, 1899)
<i>Tepuihyia edelcae</i> (Ayarzagüena, Sefaris, and Gorzula, 1993)
<i>Teratohyla amele</i> (Cisneros-Heredia and Meza-Ramos, 2007)
<i>Teratohyla mida</i> (Lynch and Doolman, 1973)
<i>Teratohyla pulverata</i> (Peters, 1873)
<i>Teratohyla spinosa</i> (Taylor, 1949)
<i>Theloderma asperum</i> (Boulenger, 1886)
<i>Theloderma bicolor</i> (Bourret, 1937)
<i>Theloderma corticale</i> (Boulenger, 1903)
<i>Theloderma gordoni</i> Taylor, 1962
<i>Theloderma horndum</i> (Boulenger, 1903)
<i>Theloderma leporosum</i> Tschudi, 1838
<i>Theloderma lion</i> McLeod and Ahmed, 2007
<i>Theloderma statutum</i> Taylor, 1962
<i>Thoropa megalyparum</i> Caramaschi and Sazima, 1984
<i>Thoropa milneis</i> (Spix, 1824)
<i>Thoropa tephora</i> (Miranda-Ribeiro, 1923)
<i>Tiacoctylia godmani</i> (Günther, 1901)
<i>Tiacoctylia loquax</i> (Günther and Stuart, 1934)
<i>Tiacoctylia picta</i> (Günther, 1901)
<i>Tiacoctylia smithi</i> (Boulenger, 1902)
<i>Tomopterna cryptotis</i> (Boulenger, 1907)
<i>Tomopterna delalandi</i> (Tschudi, 1838)
<i>Tomopterna natalensis</i> (Smith, 1849)
<i>Tomopterna landyi</i> Channing and Bogart, 1996
<i>Trachycephalus coriaceus</i> (Peters, 1867)
<i>Trachycephalus hadroceps</i> (Duellman and Hoogmoed, 1992)
<i>Trachycephalus imitatrix</i> (Miranda-Ribeiro, 1926)
<i>Trachycephalus jordanii</i> (Stejneger and Tietz, 1981)
<i>Trachycephalus mesophaeus</i> (Hensel, 1867)
<i>Trachycephalus nigromaculatus</i> Tschudi, 1838
<i>Trachycephalus reinfinitrix</i> (Coker, 1907)
<i>Trachycephalus typhlus</i> (Linnaeus, 1758)
<i>Trachycephalus venulosus</i>
<i>Trichobatrachus robustus</i> Boulenger, 1900
<i>Triprion petasatus</i> (Cope, 1865)
<i>Uperodon globulosus</i> (Günther, 1864)
<i>Uperodon systoma</i> (Schneider, 1799)
<i>Uperoleia alisana</i> Davies, Villson, McDonald, Tenenry, and Werren, 1993
<i>Uperoleia aspera</i> Tyler, Davies, and Martin, 1981
<i>Uperoleia capitulata</i> Davies, McDonald, and Corben, 1986
<i>Uperoleia fides</i> Davies, McDonald, and Corben, 1986
<i>Uperoleia glandulosa</i> Davies, Mahony, and Roberts, 1985
<i>Uperoleia inundata</i> Tyler, Davies, and Martin, 1981
<i>Uperoleia levisgate</i> Kefauver, 1987
<i>Uperoleia lithomoda</i> Tyler, Davies, and Martin, 1981
<i>Uperoleia lithophni</i> Davies, McDonald, and Corben, 1986
<i>Uperoleia martin</i> Davies and Littlejohn, 1988
<i>Uperoleia mimula</i> Davies, McDonald, and Corben, 1986
<i>Uperoleia mobergii</i> (Anderson, 1913)
<i>Uperoleia rugosa</i> (Anderson, 1916)
<i>Uperoleia russelli</i> (Loveridge, 1933)
<i>Uperoleia talpa</i> Tyler, Davies, and Martin, 1981
<i>Uperoleia tyleri</i> Davies and Littlejohn, 1988
<i>Vandikophrynus gariepensis</i> (Smith, 1848)
<i>Vandikophrynus robinsoni</i> (Branch and Braack, 1996)
<i>Vitreorana antisthenesi</i> (Gün, 1983)
<i>Vitreorana eurygnatha</i> (Lutz, 1925)
<i>Vitreorana oyampiensis</i> (Lescour, 1975)
<i>Vitreorana uranoscopa</i> (Müller, 1824)
<i>Xanthophryne konyavensis</i> (Soman, 1963)
<i>Xanthophryne tigrina</i> Biju, Van Boclaere, Gili, Loader, and Bosuut, 2009
<i>Xenophrys truncata</i> (Jackson, 1959)
<i>Xenophrys aceras</i> (Boulenger, 1903)
<i>Xenophrys brachykola</i> (Inger and Römer, 1961)
<i>Xenophrys longipes</i> (Boulenger, 1886)
<i>Xenophrys major</i> (Boulenger, 1903)
<i>Xenophrys minor</i> Stejneger, 1926
<i>Xenopus amieti</i> Kobel, du Papequier, Fischberg, and Gloor, 1980
<i>Xenopus frazeri</i> Boulenger, 1905
<i>Xenopus gilli</i> Rose and Hewitt, 1927
<i>Xenopus laevis</i> (Daudin, 1802)
<i>Xenopus longipes</i> Loomont and Kobel, 1991
<i>Xenopus muelleri</i> (Peters, 1844)
<i>Xenopus willei</i> Tinsley, Kobel, and Fischberg, 1979

Family	Species	Male Baseline	Male Display	Female Baseline	Female Display	Evidence	Reference
Bufonidae	<i>Amietophrynus gutturalis</i>	Brown, mottled.	Yellow-brown, mottled.	Brown, mottled.	N/A	P	Harper et al., 2010
Bufonidae	<i>Amietophrynus kisoensis</i>	Olive/brown, mottled.	Yellow.	Olive/brown, mottled.	N/A	D/P	Channing & Howell, 2006
Bufonidae	<i>Amietophrynus lemairei</i>	Yellow-green, mottled.	Yellow.	Red-orange/yellow-green, mottled.	N/A	D/P	Bittencourt-Silva, 2014
Bufonidae	<i>Amietophrynus maculatus</i>	Grey-brown/tan. Patches.	Yellow-brown. Patches.	Grey-brown/tan. Patches.	N/A	D/P	Rödel, 2000
Bufonidae	<i>Amietophrynus togoensis</i>	Brown.	Yellow.	Brown.	N/A	D/P	Rödel unpubl. Data
Bufonidae	<i>Bufo bufo</i>	Brown/grey.	Brown/grey.	Brown/grey, spots.	N/A	D	Kuzmin, 1999
Bufonidae	<i>Bufo japonicus</i>	Brown/grey.	Yellow/brown/grey.	Brown/grey.	N/A	D	Maeda & Matsui, 1999
Bufonidae	<i>Duttaphrynus crocus</i>	Tan.	Yellow.	Tan/brown.	N/A	D/P	Wogan et al, 2003
Bufonidae	<i>Duttaphrynus dhufarensis</i>	Grey/light brown, mottled.	Yellow.	Grey/light brown, mottled.	N/A	P	Stock et al, 2004
Bufonidae	<i>Duttaphrynus melanostictus</i>	Brown, mottled.	Yellow/yellow-brown, mottled.	Brown, mottled.	N/A	D	Daniels, 2005
Bufonidae	<i>Inciilius bocourti</i>	Brown, mottled.	Yellow.	Brown, mottled.	N/A	P	Santos-Barrera, G., et al. (2004)
Bufonidae	<i>Inciilius canaliferus</i>	Brown, mottled.	Yellow.	Brown, mottled.	N/A	P	Santos-Barrera, G., et al. (2010)
Bufonidae	<i>Inciilius luetkenii</i>	Brown.	Yellow.	Brown.	N/A	D/P	Doucet & Mennill, 2010
Bufonidae	<i>Pseudepipalea viridis</i>	Grey/brown, green spots.	Light green-grey, yellow-green spots.	Grey/brown, green spots.	N/A	D	Arnold, 2002
Bufonidae	<i>Rhinella cerradensis</i>	Brown.	Yellow-brown. Yellow flanks.	Brown/tan. White flanks.	N/A	D/P	Santana et al, 2010
Bufonidae	<i>Rhinella granulosa</i>	Brown/grey, mottled.	Yellow. Yellow throat.	Brown/grey, darker patches. White throat.	N/A	P	Lima et al, 2005
Bufonidae	<i>Schismaderma carens</i>	Rusty red-brown.	Yellow-brown.	Rusty red-brown.	N/A	P	Channing & Howell, 2006
Bufonidae	<i>Xanthophryne koyayensis</i>	Light brownish-grey. Cream flank.	Yellow. Brown flank.	Light brownish-grey.	N/A	D/P	Biju et al, 2009
Bufonidae	<i>Xanthophryne tigrinus</i>	Light brownish-grey. Cream flank.	Yellow. Black flank.	Light brownish-grey	Yellow	D/P	Biju et al, 2009
Craugastoridae	<i>Ischnocnema vinhai</i>	Brown/tan.	Yellow-brown.	Tan/red-brown.	N/A	D/P	Camurugi et al, 2010
Craugastoridae	<i>Pristimantis boulengeri</i>	Tan/brown, patches.	Yellow. Yellow throat.	Tan/brown, patches. Cream throat.	N/A	P	Castro, F., et al. (2004)
Craugastoridae	<i>Pristimantis ridens</i>	Brown/grey/tan.	Yellow/yellow-brown.	Brown/grey/tan.	N/A	P	Savage, 2002
Craugastoridae	<i>Pristimantis zimmermanae</i>	Light grey.	Yellow.	Light grey.	N/A	P	Lima et al, 2005
Dendrobatidae	<i>Mannophryne trinitatis</i>	Light brown/tan. Grey throat.	Black/dark brown. Black throat.	Light brown/tan. Yellow/orange throat.	N/A	D/P	Barrio-Amoros et al., 2010; Wells, 1980.
Dendrobatidae	<i>Rheobates palmatus</i>	Light brown/tan. Grey throat.	Black/dark brown. Black throat.	Light brown/tan. Yellow/orange throat.	N/A	D/P	Luddecke 1999
Dicroglossidae-Dicroglossinae	<i>Hoplobatrachus occipitalis</i>	Olive/brown.	Yellow. Dark blue throat.	Olive/brown. White throat.	N/A	P	Rödel, 2000
Dicroglossidae-Dicroglossinae	<i>Hoplobatrachus tigrinus</i>	Brown/green mottled.	Yellow. Dark blue throat.	Brown/green mottled. White throat.	N/A	D/P	Glaw & Vences, 2007
Eleutherodactylidae-Eleutherodactylinae	<i>Eleutherodactylus juanariveroi</i>	Brown/tan.	Yellow/yellow-tan.	Brown/tan.	N/A	P	Rios-Lopez & Thomaz, 2007
Hylidae-Hylinae	<i>Bokermannohyla circumdata</i>	Brown, transverse barring. Cream flank.	Brown-yellow, transverse barring. Yellow flank.	Brown, transverse barring. Cream flank.	N/A	P	Garcia & Segalla, 2010
Hylidae-Hylinae	<i>Charadrahyla taeniopus</i>	Brown.	Yellow-brown. Yellow flanks.	Red-brown, yellow tinge.	N/A	P	Santos-Barrera, G., Canseco-Márquez L. (2004)
Hylidae-Hylinae	<i>Dendropsophus bifurcus</i>	Grey-brown. White or yellow patches.	Yellow-brown. Yellow patches.	Grey-brown. White or yellow patches.	N/A	P	Rodríguez & Duellman, 1994
Hylidae-Hylinae	<i>Dendropsophus carnifex</i>	Red-brown.	Yellow.	Red-brown.	N/A	P	Bolivar, W. and J. M. Renjifo (2004)
Hylidae-Hylinae	<i>Dendropsophus columbianus</i>	Brown.	Yellow.	Brown.	N/A	P	Gomez-Hoyos et al., 2012
Hylidae-Hylinae	<i>Dendropsophus decipiens</i>	Light brown.	Yellow.	Light brown.	N/A	P	Abrunhosa et al, 2006
Hylidae-Hylinae	<i>Dendropsophus ebraccatus</i>	Tan/grey-brown.	Yellow/tan-brown.	Tan/brown.	N/A	P	Duellman, 2001; Savage, 2002
Hylidae-Hylinae	<i>Dendropsophus elegans</i>	Light brown, broad darker patch.	Yellow, broad darker patch.	Light brown, broad darker patch.	N/A	P	Abrunhosa et al, 2006
Hylidae-Hylinae	<i>Dendropsophus gaucheri</i>	Grey/brown, mottled.	Yellow.	Grey/brown, mottled.	N/A	P	Fouquet et al., 2011
Hylidae-Hylinae	<i>Dendropsophus giesleri</i>	Grey.	Yellow.	Grey.	N/A	P	Potsch de Carvalho-e-Silva, S. Telles, A. M. (2004)
Hylidae-Hylinae	<i>Dendropsophus joannae</i>	Grey.	Yellow.	Grey.	N/A	P	Reichle, S., et al. (2004)
Hylidae-Hylinae	<i>Dendropsophus luteocellatus</i>	Brown/grey.	Yellow.	Brown/grey.	N/A	P	Mijares, A., et al. (2004)
Hylidae-Hylinae	<i>Dendropsophus melanargyreus</i>	Grey/brown, mottled.	Yellow-brown, mottled.	Brown/grey/red-brown, mottled.	N/A	P	da Silva et al, 2010
Hylidae-Hylinae	<i>Dendropsophus microcephalus</i>	Brown.	Yellow/yellow-brown. Yellow throat.	Brown.	N/A	P	Duellman, 2001; Schwartz & Wells, 1985; Savage, 2002
Hylidae-Hylinae	<i>Dendropsophus microps</i>	Brown/grey.	Yellow. Yellow throat.	Brown/grey. White throat.	N/A	P	Potsch de Carvalho-e-Silva, S., et al. (2010)
Hylidae-Hylinae	<i>Dendropsophus minutus</i>	Brown.	Yellow. Yellow throat.	Brown. White throat.	N/A	D/P	Haddad, 1991; Bell & Zamudio 2012
Hylidae-Hylinae	<i>Dendropsophus nanus</i>	Tan/light brown.	Yellow/tan.	Tan/light brown.	N/A	P	Reichle et al, 2004. In: IUCN 2013.
Hylidae-Hylinae	<i>Dendropsophus phlebodes</i>	Brown/tan/grey.	Yellow/yellow-brown. Yellow throat.	Brown/tan/grey. White throat.	N/A	P	Duellman, 2001; Savage, 2002
Hylidae-Hylinae	<i>Dendropsophus robertmertinsi</i>	Grey/tan.	Yellow.	Grey/tan.	N/A	P	Santos-Barrera et al, 2010.
Hylidae-Hylinae	<i>Dendropsophus sallii</i>	Grey/brown.	Yellow-brown.	Grey/brown.	N/A	P	Jungfer et al, 2010
Hylidae-Hylinae	<i>Dendropsophus sarayacuensis</i>	Dark grey. Cream or yellow patches.	Yellow-brown. Yellow patches.	Dark grey. Cream or yellow patches.	N/A	P	La Marca, E., et al. (2004)
Hylidae-Hylinae	<i>Diaglena spatulata</i>	Brown, mottled.	Yellow/yellow-brown (particularly on flanks), mottled.	Brown, mottled.	N/A	P	Duellman, 2001
Hylidae-Hylinae	<i>Exerodonta smaragdina</i>	Light brown/tan.	Yellow/yellow-brown.	Light brown or tan.	N/A	P	Santos-Barrera, G., Canseco-Márquez, L. (2004)
Hylidae-Hylinae	<i>Exerodonta sumichrasti</i>	Light brown/tan.	Yellow/yellow-brown.	Light brown or tan.	N/A	P	Duellman, 2001
Hylidae-Hylinae	<i>Exerodonta xera</i>	Light brown/tan.	Yellow.	Light brown/tan.	N/A	P	Santos-Barrera, G., Canseco-Márquez L. (2010)
Hylidae-Hylinae	<i>Hypsiboas albopunctatus</i>	Brown/tan.	Yellow/yellow-brown.	Brown/tan.	N/A	D/P	Aquino, L., et al. (2010)
Hylidae-Hylinae	<i>Hypsiboas beckeri</i>	Grey-brown.	Yellow.	Grey-brown.	N/A	P	Stuart, 2006
Hylidae-Hylinae	<i>Hypsiboas bischoffi</i>	Brown/tan.	Yellow/yellow-brown. Yellow throat.	Brown. White throat.	N/A	P	Lutz, 1973
Hylidae-Hylinae	<i>Hypsiboas caingua</i>	Brown.	Yellow-brown.	Brown.	N/A	P	Scott et al., 2004
Hylidae-Hylinae	<i>Hypsiboas calcaratus</i>	Brown, transverse barring.	Yellow.	Brown, transverse barring.	N/A	P	Azevedo-Ramos et al., 2010
Hylidae-Hylinae	<i>Hypsiboas callipleura</i>	Brown.	Yellow/light yellow-brown.	Brown.	N/A	P	Cortez et al., 2004
Hylidae-Hylinae	<i>Hypsiboas cordobae</i>	Grey/brown/green, spotted.	Yellow-brown.	Grey/brown/green, spotted.	N/A	P	Stuart, 2006
Hylidae-Hylinae	<i>Hypsiboas crepitans</i>	Brown/light grey.	Yellow-orange. Yellow throat.	Brown/light grey. White throat.	N/A	P	La Marca et al., 2010
Hylidae-Hylinae	<i>Hypsiboas ericae</i>	Brown/grey. White labial/lateral stripe.	Green/brown. Yellow labial/lateral stripe and throat.	Brown/grey. White labial/lateral stripe.	N/A	P	Garcia et al., 2008
Hylidae-Hylinae	<i>Hypsiboas faber</i>	Brown/tan/grey.	Yellow-brown. Yellow flanks, throat.	Brown/tan/grey. White throat.	N/A	P	Lavilla et al., 2010
Hylidae-Hylinae	<i>Hypsiboas fasciatus</i>	Brown/tan.	Yellow.	Brown/tan.	N/A	D/P	Bell & Zamudio, 2012
Hylidae-Hylinae	<i>Hypsiboas geographicus</i>	Brown/grey/red/orange, patches.	Yellow.	Brown/grey/red/orange, patches.	N/A	P	Duellman, 1973, Bartlett & Bartlett, 2003
Hylidae-Hylinae	<i>Hypsiboas lanciformis</i>	Brown/grey, transverse barring.	Yellow-brown, transverse barring.	Brown/grey, transverse barring.	N/A	P	Lima et al., 2005
Hylidae-Hylinae	<i>Hypsiboas leptolineatus</i>	Dark/light brown/tan.	Yellow.	Dark/light brown/tan.	N/A	P	Garcia et al., 2004
Hylidae-Hylinae	<i>Hypsiboas lundii</i>	Grey/brown, mottled.	Yellow, mottled. Yellow throat.	Grey/brown, mottled.	N/A	P	Caramaschi & Rodrigue, 2004
Hylidae-Hylinae	<i>Hypsiboas marianitae</i>	Tan/brown/grey/green.	Yellow-brown.	Tan/brown/grey/green.	N/A	P	Cortez et al., 2004
Hylidae-Hylinae	<i>Hypsiboas multifasciatus</i>	Tan/brown/grey, transverse bands.	Yellow, transverse bands.	Tan/brown/grey, transverse bands.	N/A	D/P	Bell & Zamudio, 2012
Hylidae-Hylinae	<i>Hypsiboas paranaiba</i>	Brown/tan/orange, patches.	Yellow-brown.	Brown/tan/orange, patches.	N/A	P	De Carvalho et al., 2010
Hylidae-Hylinae	<i>Hypsiboas polytaenius</i>	Brown/grey/tan, striped.	Yellow/yellow-brown.	Brown/grey/tan, striped.	N/A	P	Goncalves da Cruz & Caramaschi, 2010
Hylidae-Hylinae	<i>Hypsiboas raniceps</i>	Brown.	Yellow/yellow-brown.	Brown.	N/A	P	La Marca et al., 2004
Hylidae-Hylinae	<i>Hypsiboas rufitellus</i>	Blue-green, spots.	Yellow-green, spots.	Blue/blue-green, spots.	N/A	P	Savage, 2002
Hylidae-Hylinae	<i>Isthmohyla pseudopuma</i>	Pale cream/reddish-brown. Cream flanks.	Yellow/yellow-brown. Black flanks.	Pale cream/reddish-brown. Cream flanks.	N/A	D/P	Crump & Townsend 1990; Crump 1991; Duellman 1970
Hylidae-Hylinae	<i>Itapohtyia langsdorffii</i>	Green-brown, mottled.	Yellow-green, mottled.	Green-brown, mottled.	N/A	P	Aquino et al, 2004
Hylidae-Hylinae	<i>Osteocephalus lepieurii</i>	Brown/tan.	Yellow.	Brown/tan.	N/A	D/P	Szatecnsy et al., 2010
Hylidae-Hylinae	<i>Osteocephalus yasuni</i>	Brown/tan.	Yellow/yellow brown. Yellow throat.	Brown/tan. White throat.	N/A	P	Ron & Pramuk, 1999
Hylidae-Hylinae	<i>Osteopilus dominicensis</i>	Grey-brown.	Yellow-brown.	Grey-brown.	N/A	P	IUCN SSC Amphibian Specialist Group, 2013
Hylidae-Hylinae	<i>Osteopilus pulchrilineatus</i>	Brown/tan/grey. Cream flank.	Yellow-brown. Yellow flank.	Brown/tan/grey. Cream flank.	N/A	P	IUCN SSC Amphibian Specialist Group, 2013
Hylidae-Hylinae	<i>Osteopilus septentrionalis</i>	Brown/grey, patches. Cream flank.	Yellow. Yellow flank.	Brown/grey, patches. Cream flank.	N/A	P	Hedges et al., 2010
Hylidae-Hylinae	<i>Osteopilus vastus</i>	Brown/green.	Yellow-brown/yellow-green.	Brown/green.	N/A	P	Henderson & Henderson, 2004
Hylidae-Hylinae	<i>Plectrohyla bistincta</i>	Tan/brown/grey. Cream flank.	Yellow-brown/brown. Yellow-brown flank.	Tan/brown/grey. Cream flank.	N/A	P	Duellman, 2001

Family	Species	Male Baseline	Male Display	Female Baseline	Female Display	Evidence	Reference
Hylidae-Hylinae	<i>Plectrohyla matudai</i>	Brown/grey. Grey flank.	Brown. Yellow/yellow-brown flank.	Brown/grey. Grey flank.	N/A	P	Duellman, 2001
Hylidae-Hylinae	<i>Plectrohyla pentheter</i>	Dark brown. Cream flank.	Yellow-brown. Yellow flank.	Dark brown. Cream flank.	N/A	P	Duellman, 2001
Hylidae-Hylinae	<i>Scinax acuminatus</i>	Grey. patches.	Yellow-brown. patches.	Grey. patches.	N/A	P	Aquino et al., 2004
Hylidae-Hylinae	<i>Scinax alter</i>	Cream/light brown.	Yellow.	Cream/light brown.	N/A	P	Potsch de Carvalho-e-Silva & Kwet, 2010
Hylidae-Hylinae	<i>Scinax aromothyella</i>	Grey-brown. patches.	Yellow-brown. patches.	Grey-brown. patches.	N/A	P	Stuart, 2006
Hylidae-Hylinae	<i>Scinax boesemani</i>	Brown.	Yellow-brown.	Brown.	N/A	P	Lima et al., 2005
Hylidae-Hylinae	<i>Scinax catharinae</i>	Brown. patches.	Yellow.	Brown. patches.	N/A	P	Silvano et al., 2004
Hylidae-Hylinae	<i>Scinax chiquitanus</i>	Grey-brown.	Yellow-brown. Yellow flanks.	Grey-brown.	N/A	P	Cortez et al., 2004
Hylidae-Hylinae	<i>Scinax crospeodospilus</i>	Grey-brown. patches.	Yellow-brown. patches.	Grey-brown. patches.	N/A	P	Pavan & Telles, 2004
Hylidae-Hylinae	<i>Scinax curicica</i>	Light grey-brown.	Yellow.	Light grey-brown.	N/A	P	Pugliese et al., 2004
Hylidae-Hylinae	<i>Scinax duartei</i>	Light grey-brown.	Yellow.	Light grey-brown.	N/A	P	Bertoluci & Gonçalves da Cruz, 2010
Hylidae-Hylinae	<i>Scinax elaeochrous</i>	Light grey-brown. Cream flank.	Yellow/yellow-brown. Yellow flank.	Light grey-brown. Cream flank.	N/A	P	Duellman, 2001
Hylidae-Hylinae	<i>Scinax fuscovarius</i>	Olive/brown/grey. patches. Cream flank.	Yellow-brown. patches. Yellow flank.	Olive/brown/grey. patches. Cream flank.	N/A	D/P	Bell & Zamudio, 2012
Hylidae-Hylinae	<i>Scinax granulatus</i>	Light grey-brown. patches.	Yellow-brown. patches.	Light grey-brown. patches.	N/A	P	Scott et al., 2004
Hylidae-Hylinae	<i>Scinax hayii</i>	Light brown/grey.	Yellow.	Light brown/grey.	N/A	D/P	Bell & Zamudio, 2012
Hylidae-Hylinae	<i>Scinax ictericus</i>	Grey-brown. mottled.	Yellow.	Grey-brown. mottled.	N/A	P	Angulo et al., 2004
Hylidae-Hylinae	<i>Scinax manriquei</i>	Grey.	Yellow.	Grey.	N/A	P	Barrio Amorós & La Marca 2010
Hylidae-Hylinae	<i>Scinax nasicus</i>	Light grey-brown. patches.	Yellow.	Grey-brown. mottled.	N/A	P	Kwet et al., 2004
Hylidae-Hylinae	<i>Scinax pachycrus</i>	Tan/brown. White flank.	Yellow-brown. Yellow flank.	Tan-brown. White flank.	N/A	P	Peixoto & Arzabe 2004
Hylidae-Hylinae	<i>Scinax perereca</i>	Grey/tan-brown.	Yellow-brown.	Grey/tan-brown.	N/A	P	Aquino, et al., 2004
Hylidae-Hylinae	<i>Scinax rizzibilis</i>	Grey-brown. Cream flank.	Yellow-brown. Yellow flank.	Grey-brown. Cream flank.	N/A	D/P	Bell & Zamudio, 2012
Hylidae-Hylinae	<i>Scinax similis</i>	Grey/brown. Cream flank, lip.	Yellow-brown. Yellow flank, lip.	Grey/brown. Cream flank, lip.	N/A	P	Rodrigues & Telles 2010
Hylidae-Hylinae	<i>Scinax staufferi</i>	Light grey-brown/cream.	Yellow-brown. Yellow flank.	Light grey-brown/cream.	N/A	P	Savage, 2002
Hylidae-Hylinae	<i>Scinax wandae</i>	Brown/grey.	Yellow-brown. Yellow-orange throat.	Brown/grey. Cream throat.	N/A	P	Rueda et al., 2010
Hylidae-Hylinae	<i>Smilisca baudinii</i>	Green/brown.	Yellow/yellow-brown.	Green/brown	N/A	P	Lannoo, 2005
Hylidae-Hylinae	<i>Smilisca phaeota</i>	Brown/green. patches.	Yellow-brown. patches.	Brown/green. patches.	N/A	P	Savage, 2002
Hylidae-Hylinae	<i>Smilisca puma</i>	Brown/grey/cream. Cream flanks.	Yellow-brown. Yellow flank.	Brown/grey/cream. Cream flanks.	N/A	P	Savage, 2002
Hylidae-Hylinae	<i>Smilisca sila</i>	Brown/grey.	Yellow-brown.	Brown/grey.	N/A	P	Savage, 2002
Hylidae-Hylinae	<i>Smilisca sordida</i>	Brown/grey. patches.	Yellow/yellow-brown.	Brown/grey. patches.	N/A	P	Savage, 2002
Hylidae-Hylinae	<i>Tlalocohyla loquax</i>	Brown/tan.	Yellow. Yellow throat.	Brown/tan.	N/A	P	Savage, 2002
Hylidae-Hylinae	<i>Tlalocohyla smithii</i>	Brown.	Yellow/orange.	Brown.	N/A	P	Duellman, 2001
Hylidae-Hylinae	<i>Trachycephalus coriaceus</i>	Tan/brown.	Yellow-brown.	Tan/brown.	N/A	P	Schiesari & Moreira, 1996
Hylidae-Hylinae	<i>Trachycephalus jordani</i>	Grey/brown.	Yellow.	Grey/brown.	N/A	P	Icochea et al., 2004
Hylidae-Hylinae	<i>Trachycephalus mesophaeus</i>	Light brown. Cream lateral stripe.	Yellow-brown. Yellow lateral stripe.	Light brown. Cream lateral stripe.	N/A	P	Potsch de Carvalho-e-Silva & Garcia, 2004
Hylidae-Hylinae	<i>Trachycephalus typhonius</i>	Grey-brown. Cream labial stripe.	Yellow-brown. Yellow throat, labial stripe.	Grey-brown. Cream throat, labial stripe.	N/A	P	Savage, 2002
Hylidae-Hylinae	<i>Trachycephalus venulosus</i>	Grey-brown. Cream labial stripe.	Yellow-brown. Cream throat, labial stripe.	Grey-brown. Cream throat, labial stripe.	N/A	P	Savage, 2002
Hylidae-Pelodyadinae	<i>Cyclorana brevipes</i>	Brown/grey/cream. patches.	Yellow-brown. patches.	Brown/grey/cream. patches.	N/A	P	Anstis, 2013
Hylidae-Pelodyadinae	<i>Cyclorana longipes</i>	Brown/grey/cream. patches.	Yellow-brown. patches.	Brown/grey/cream. patches.	N/A	P	Anstis, 2013
Hylidae-Pelodyadinae	<i>Cyclorana novaeollandiae</i>	Brown/grey/cream. patches.	Yellow-brown. patches.	Brown/grey/cream. patches.	N/A	P	Hero et al., 2004
Hylidae-Pelodyadinae	<i>Litoria amboinensis</i>	Grey-brown. mottled.	Yellow-brown. mottled.	Grey-brown. mottled.	N/A	P	Menzies, 2006
Hylidae-Pelodyadinae	<i>Litoria bicolor</i>	Green/brown/fawn. White labial stripe.	Green/brown/fawn. Yellow labial stripe/flank.	Green/brown/fawn. White labial stripe.	N/A	P	Anstis, 2013
Hylidae-Pelodyadinae	<i>Litoria booroolongensis</i>	Brown/grey/tan. mottled.	Yellow-brown. mottled.	Brown/grey/tan. mottled.	N/A	O/P	Anstis, 2013
Hylidae-Pelodyadinae	<i>Litoria brevipalmata</i>	Brown. White labial stripe, flank.	Yellow-brown. Yellow labial stripe, flank, throat.	Brown. White labial stripe, flank, throat.	N/A	O/P	Barker et al., 1995
Hylidae-Pelodyadinae	<i>Litoria chloris</i>	Green. White flank.	Green. Yellow flank, throat.	Green. White flank, throat.	N/A	O/P	Barker et al., 1995
Hylidae-Pelodyadinae	<i>Litoria corbeni</i>	Brown/tan. broad patch.	Yellow.	Brown/tan. broad patch.	N/A	O/P	Barker et al., 1995
Hylidae-Pelodyadinae	<i>Litoria dentata</i>	Brown/tan. Cream flank.	Yellow-brown. Yellow flank, throat.	Brown/tan. Cream flank.	N/A	O/P	Barker et al., 1995
Hylidae-Pelodyadinae	<i>Litoria ewingii</i>	Grey/brown/tan	Yellow-brown/brown/grey.	Grey/brown/tan.	N/A	D/O/P	Anstis, 2013
Hylidae-Pelodyadinae	<i>Litoria fallax</i>	Green/brown/fawn. White labial stripe.	Green/brown/fawn. Yellow labial stripe/flank.	Green/brown/fawn. White labial stripe.	N/A	P	Anstis, 2013
Hylidae-Pelodyadinae	<i>Litoria inermis</i>	Fawn/pale brown/grey. White labial stripe/flanks.	Yellow-brown. Yellow flank and labial stripe.	Fawn/pale brown/grey. White labial stripe/flanks.	N/A	O/P	Anstis, 2013
Hylidae-Pelodyadinae	<i>Litoria jervisiensis</i>	Tan.	Faint yellow wash.	Tan.	N/A	D/O/P	Anstis, 2013
Hylidae-Pelodyadinae	<i>Litoria jungguy</i>	Brown/grey/tan.	Yellow/yellow-brown.	Brown/grey/tan.	N/A	D/O/P	Donnelann & Mahoney, 2004
Hylidae-Pelodyadinae	<i>Litoria lesueurii</i>	Brown/grey/tan.	Yellow/yellow-brown.	Brown/grey/tan.	N/A	D/O/P	Donnelann & Mahoney, 2004
Hylidae-Pelodyadinae	<i>Litoria littlejohni</i>	Brown/grey. mottled.	Yellow-brown. mottled.	Brown/grey. mottled.	N/A	P	Anstis, 2013
Hylidae-Pelodyadinae	<i>Litoria nasuta</i>	Brown/tan. striped.	Brown/tan. striped. Yellow wash.	Brown/tan. striped.	N/A	D/O/P	Anstis, 2013
Hylidae-Pelodyadinae	<i>Litoria nigrofrenata</i>	Grey/tan/brown. White/cream flanks.	Tan/yellow-brown. Yellow flanks.	Grey/tan/brown. White/cream flanks.	N/A	D/P	Reynolds, 2010
Hylidae-Pelodyadinae	<i>Litoria pallida</i>	Grey/tan/brown. White/cream flanks.	Tan/yellow-brown. Yellow flanks.	Grey/tan/brown. White/cream flanks.	N/A	D/P	Reynolds, 2010
Hylidae-Pelodyadinae	<i>Litoria paraewingi</i>	Grey/brown/tan	Yellow-brown/brown/grey.	Grey/brown/tan.	N/A	P	Anstis, 2013
Hylidae-Pelodyadinae	<i>Litoria peronii</i>	Brown/grey. mottled.	Yellow-brown. mottled.	Brown/grey. mottled.	N/A	O/P	Barker et al., 1995
Hylidae-Pelodyadinae	<i>Litoria personata</i>	Brown. Cream flank.	Yellow-brown. Yellow flank.	Brown. Cream flank.	N/A	P	Barker et al., 1995
Hylidae-Pelodyadinae	<i>Litoria revelata</i>	Brown/tan. broad patch.	Yellow.	Brown/tan. broad patch.	N/A	O/P	Barker et al., 1995
Hylidae-Pelodyadinae	<i>Litoria rothli</i>	Brown/tan/grey. mottled.	Yellow-brown/tan.	Brown/tan/grey. mottled.	N/A	O/P	Anstis, 2013
Hylidae-Pelodyadinae	<i>Litoria tornieri</i>	Grey/tan/brown. White/cream flanks/labial stripe.	Tan/yellow-brown. Yellow flanks/labial stripe.	Grey/tan/brown. White/cream flanks/labial stripe.	N/A	D/P	Reynolds, 2010
Hylidae-Pelodyadinae	<i>Litoria tyleri</i>	Brown/tan/grey. mottled.	Yellow.	Brown/tan/grey. mottled.	N/A	O/P	Barker et al., 1995
Hylidae-Pelodyadinae	<i>Litoria verreauxii</i>	Brown/tan/grey.	Yellow-brown/brown.	Brown/tan/grey.	N/A	O/P	Anstis, 2013
Hylidae-Pelodyadinae	<i>Litoria watjulumensis</i>	Grey/tan/brown. White/cream flanks.	Tan/yellow-brown. Yellow flanks.	Grey/tan/brown. White/cream flanks.	N/A	D/P	Reynolds, 2010
Hylidae-Pelodyadinae	<i>Litoria wilcoxii</i>	Brown/grey/tan.	Yellow/yellow-brown.	Brown/grey/tan.	N/A	D/O/P	Donnelann & Mahoney, 2004
Leptodactylidae-Leiuperinae	<i>Pleuroderma brachyops</i>	Brown/cream/grey. patches.	Yellow-brown. patches.	Brown/cream/grey. patches.	N/A	P	La Marca, E., et al. 2010
Leptodactylidae-Leiuperinae	<i>Pleuroderma thaul</i>	Brown/cream/grey. patches.	Yellow-brown. Yellow throat.	Brown/cream/grey. patches. White throat.	N/A	P	Duellmann & Veloso, 1977
Mantelliidae-Laliostominae	<i>Aglyptodactylus madagascariensis</i>	Brown/grey/tan.	Yellow.	Brown/grey/tan.	N/A	D/P	Glaw & Vences, 2007
Mantelliidae-Laliostominae	<i>Aglyptodactylus securifer</i>	Brown.	Yellow.	Brown.	N/A	P	Glaw & Vences, 2007
Mantelliidae-Laliostominae	<i>Laliostoma labrosum</i>	Brown/grey. patches. Cream flank.	Yellow-brown. patches. Yellow flank.	Brown/grey. patches. Cream flank.	N/A	P	Glaw & Vences, 2007
Microhylidae-Asterophrynidae	<i>Cophixalus saxatilis</i>	Light tan/grey. patches.	Yellow/yellow-brown. patches.	Tan/grey. Yellow wash.	N/A	O/P	Anstis, 2013
Myobatrachidae-Limnodynastinae	<i>Lechiodus fletcheri</i>	Brown/tan.	Yellow-brown.	Brown/red-brown.	N/A	O/P	Barker et al., 1995
Myobatrachidae-Limnodynastinae	<i>Mixophyes balbus</i>	Brown/tan. Cream flank.	Yellow/yellow-brown. Yellow flank.	Brown/tan. Cream flank.	N/A	O/P	Barker et al., 1995
Myobatrachidae-Limnodynastinae	<i>Mixophyes fleayi</i>	Brown/tan. Cream flank.	Yellow/yellow-brown. Yellow flank.	Brown/tan. Cream flank.	N/A	O/P	Barker et al., 1995
Myobatrachidae-Limnodynastinae	<i>Neobatrachus kunapalari</i>	Brown/grey. patches.	Yellow-brown. patches.	Brown/grey. patches.	N/A	P	Anstis, 2013
Myobatrachidae-Limnodynastinae	<i>Neobatrachus pelobatoides</i>	Brown/grey. patches.	Yellow-brown. patches.	Brown/grey. patches.	N/A	P	Anstis, 2013
Myobatrachidae-Limnodynastinae	<i>Neobatrachus pictus</i>	Brown/grey. patches.	Yellow-brown. patches.	Brown/grey. patches.	N/A	O/P	Barker et al., 1995
Myobatrachidae-Limnodynastinae	<i>Neobatrachus sudelli</i>	Brown/grey. patches.	Yellow-brown. patches.	Brown/grey. patches.	N/A	O/P	Anstis, 2013
Myobatrachidae-Limnodynastinae	<i>Platyplectrum ornatum</i>	Brown/grey/tan. patches/mottling.	Yellow-brown. patches/mottling.	Brown/grey/tan. patches/mottling.	N/A	O/P	Barker et al., 1995; Vanderduys, 2012
Phrynobatrachidae	<i>Phrynobatrachus allenii</i>	Brown.	Yellow.	Brown.	N/A	D	Rödel, 2003
Phrynobatrachidae	<i>Phrynobatrachus latifrons</i>	Brown/grey.	Bright green/yellow. Yellow throat.	Brown/grey. Cream throat.	N/A	D	Rödel, 2000
Phrynobatrachidae	<i>Phrynobatrachus natalensis</i>	Brown/grey.	Green. patches.	Brown/grey.	N/A	D	Rödel, 2000

Family	Species	Male Baseline	Male Display	Female Baseline	Female Display	Evidence	Reference
Ranidae	<i>Clinotarsus curtipes</i>	Orange/yellow	Red-brown.	Red/orange.	N/A	D/P	Bell & Zamudio, 2012
Ranidae	<i>Hylarana aurata</i>	Grey/brown.	Yellow.	Brown.	N/A	D/P	Gunther, 2003
Ranidae	<i>Hylarana grisea</i>	Olive/grey/brown.	Yellow.	Olive/grey/brown.	N/A	D	Bell & Zamudio, 2012
Ranidae	<i>Hylarana volkerjane</i>	Light grey/dark grey.	Yellow.	Tan/red-brown.	N/A	D/P	Gunther, 2003
Ranidae	<i>Odorrana khalam</i>	Brown.	Yellow.	Brown.	N/A	P	Stuart, 2005
Ranidae	<i>Rana arvalis</i>	Brown.	Blue.	Brown.	N/A	D/P	Arnold, 2002. Sztatecsny et al., 2012
Ranidae	<i>Rana graeca</i>	Brown.	Black/dark-brown.	Brown.	N/A	D	Valakos et al., 2008
Ranidae	<i>Rana sylvatica</i>	Brown/tan.	Black/dark-brown.	Brown/tan.	N/A	D/P	Harding, 1997
Ranidae	<i>Rana temporaria</i>	Brown.	Grey-brown.	Brown/tan.	N/A	D	Arnold, 2002
Rhacophoridae-Buergeriinae	<i>Buergeria japonica</i>	Grey-brown.	Yellow.	Grey-brown.	N/A	P	Maeda & Matsui, 1990
Rhacophoridae-Buergeriinae	<i>Buergeria robusta</i>	Grey-brown/tan.	Yellow/yellow-brown.	Grey-brown/tan.	N/A	P	Huang et al., 2001
Rhacophoridae-Rhacophorinae	<i>Polypedates megacephalus</i>	Grey/brown.	Yellow-brown.	Grey/brown.	N/A	P	Huang et al., 2001
Bufonidae	<i>Incilius majordomus</i>	(possibly dichromatic)					
Bufonidae	<i>Incilius nebulifer</i>	(possibly dichromatic)					
Dendrobatidae	<i>Mannophryne collaris</i>	(possibly dichromatic)					
Eleutherodactylidae	<i>Ischnocnema manezinho</i>	(possibly dichromatic)					
Mantellidae	<i>Boophis guibei</i>	(possibly dichromatic)					
Mantellidae	<i>Guibemantis liber</i>	(possibly dichromatic)					

	12s	cytb	Cmyc	Pomc	Rag1	Rhodopsin	Dichromatic	Region	BreedingAggregation
<i>Acris crepitans</i>	148524036	157503194	62001217	302028848	61725332	61725736	no	temperate	yes
<i>Acris gryllus</i>	61697171	157503202	NA	157503332	61725334	61725738	no	temperate	yes
<i>Agalychnis annae</i>	302029228	302028612	143635796	140083466	NA	NA	no	tropical	yes
<i>Agalychnis callidryas</i>	148524011	NA	143635798	140083468	NA	38372031	no	tropical	yes
<i>Agalychnis moreletii</i>	302029234	302028618	143635802	140083472	NA	302028990	no	tropical	yes
<i>Agalychnis saltator</i>	37699585	302028620	NA	140083474	NA	NA	no	tropical	yes
<i>Agalychnis spurrelli</i>	60735493	NA	62001301	62085387	NA	NA	no	tropical	yes
<i>Anotheca spinosa</i>	61697177	112382888	62001219	62085305	61725342	112383020	no	tropical	no
<i>Aparasphenodon bruno</i>	61697178	61697406	NA	NA	61725344	61725752	no	tropical	yes
<i>Aplastodiscus albosignatus</i>	61697207	61697462	NA	NA	NA	61725810	no	tropical	no
<i>Aplastodiscus arildae</i>	61697215	61697478	NA	NA	NA	61725826	no	tropical	no
<i>Aplastodiscus eugenioi</i>	61697280	61697654	NA	NA	NA	61725990	no	tropical	no
<i>Aplastodiscus leucopygius</i>	61697249	61697574	NA	NA	NA	61725914	no	tropical	no
<i>Aplastodiscus perviridis</i>	61697180	61697410	NA	NA	61725348	61725756	no	tropical	no
<i>Argenteohyla siemersi</i>	61697181	61697412	NA	NA	61725350	61725758	no	temperate	yes
<i>Bokermannohyla circumdata</i>	NA	61697522	NA	NA	NA	61725866	dynamic	tropical	yes
<i>Bokermannohyla hylax</i>	87280676	61697558	NA	NA	NA	61725898	no	tropical	yes
<i>Bokermannohyla martinsi</i>	61697252	61697584	NA	NA	NA	61725922	no	tropical	no
<i>Bromeliodhyla bromeliacia</i>	61697223	61697502	NA	NA	NA	61725850	no	tropical	no
<i>Charadrahyla nephila</i>	61697260	61697606	89113761	89113721	NA	61725940	no	tropical	yes
<i>Charadrahyla taeniopus</i>	61697290	61697676	NA	NA	NA	61726012	dynamic	tropical	missing
<i>Chorothomantis greeningi</i>	61697189	61697428	NA	NA	61725364	61725772	ontogenetic	tropical	yes
<i>Cruziohyla calcarifer</i>	302029236	302028622	143635804	302028856	302028936	302028994	no	tropical	no
<i>Cyclorana alboguttata</i>	71148710	NA	NA	NA	NA	NA	no	tropical	yes
<i>Cyclorana australis</i>	61697191	61697432	NA	NA	61725368	61725776	no	tropical	no
<i>Cyclorana brevipes</i>	60735503	NA	NA	NA	NA	NA	dynamic	tropical	yes
<i>Cyclorana cryptotis</i>	284177543	NA	NA	NA	NA	NA	no	tropical	no
<i>Cyclorana cultripes</i>	261600114	NA	NA	NA	NA	NA	no	tropical	yes
<i>Cyclorana longipes</i>	261600115	NA	NA	NA	NA	NA	dynamic	tropical	yes
<i>Cyclorana maculosa</i>	261600116	NA	NA	NA	NA	NA	no	tropical	no
<i>Cyclorana manya</i>	60735489	NA	62001293	62085379	NA	NA	no	tropical	no
<i>Cyclorana novaehollandiae</i>	261600119	NA	NA	NA	NA	NA	dynamic	tropical	yes
<i>Cyclorana platycephala</i>	261600120	NA	NA	NA	NA	NA	no	temperate	yes
<i>Cyclorana vagitus</i>	261600121	NA	NA	NA	NA	NA	no	tropical	no
<i>Cyclorana verrucosa</i>	261600122	NA	NA	NA	NA	NA	no	temperate	no
<i>Dendropsophus anceps</i>	61697208	61697464	NA	NA	NA	61725812	no	tropical	yes
<i>Dendropsophus berthaltutae</i>	61697218	61697490	NA	NA	NA	61725838	no	tropical	no
<i>Dendropsophus bipunctatus</i>	61697219	61697492	NA	NA	NA	61725840	no	tropical	yes
<i>Dendropsophus brevifrons</i>	61697222	61697500	NA	NA	NA	61725848	no	tropical	yes
<i>Dendropsophus carnifex</i>	61697227	61697512	NA	NA	NA	NA	dynamic	tropical	yes
<i>Dendropsophus ebraccatus</i>	61697235	157890887	62001233	62085319	NA	61725878	dynamic	tropical	yes
<i>Dendropsophus elegans</i>	87280673	NA	NA	NA	NA	NA	dynamic	tropical	yes
<i>Dendropsophus giesleri</i>	61697240	61697548	NA	NA	NA	NA	dynamic	tropical	yes
<i>Dendropsophus koechlini</i>	60735461	NA	62001237	62085323	NA	NA	no	tropical	yes
<i>Dendropsophus labialis</i>	61697246	223885925	NA	223885965	NA	61725906	no	tropical	yes
<i>Dendropsophus leali</i>	60735543	NA	NA	NA	NA	NA	no	tropical	yes
<i>Dendropsophus leucophyllatus</i>	87280678	NA	NA	NA	NA	NA	ontogenetic	tropical	no
<i>Dendropsophus marmoratus</i>	61697251	61697582	NA	NA	NA	90183805	no	tropical	no
<i>Dendropsophus microcephalus</i>	148524012	61697588	62001241	62085327	NA	61725926	dynamic	tropical	yes
<i>Dendropsophus minusculus</i>	87280680	NA	NA	NA	NA	NA	no	tropical	yes
<i>Dendropsophus minutus</i>	NA	61697594	NA	NA	NA	90183757	dynamic	tropical	yes
<i>Dendropsophus miyatai</i>	61697258	61697600	NA	NA	NA	61725934	no	tropical	yes
<i>Dendropsophus nanus</i>	60735465	61697604	62001245	62085331	NA	61725938	dynamic	tropical	yes
<i>Dendropsophus parviceps</i>	61697263	61697612	NA	NA	NA	61725946	ontogenetic	tropical	yes
<i>Dendropsophus rhodopeplus</i>	61697269	NA	NA	NA	NA	61725964	no	tropical	yes
<i>Dendropsophus riveroi</i>	87280690	NA	NA	NA	NA	NA	no	tropical	yes
<i>Dendropsophus robertmertensi</i>	60735544	NA	NA	NA	NA	NA	dynamic	tropical	yes
<i>Dendropsophus rubicundulus</i>	61697272	61697636	NA	NA	NA	61725972	no	tropical	yes
<i>Dendropsophus sanborni</i>	61697274	61697640	NA	NA	NA	61725976	no	tropical	yes
<i>Dendropsophus sarayacuensis</i>	61697275	NA	NA	NA	NA	NA	dynamic	tropical	yes
<i>Dendropsophus seniculus</i>	61697277	61697648	NA	NA	NA	61725984	ontogenetic	tropical	yes
<i>Dendropsophus triangulum</i>	61697291	61697680	NA	NA	NA	61726016	no	tropical	yes
<i>Dendropsophus walfordi</i>	61697294	61697686	NA	NA	NA	61726022	no	tropical	yes
<i>Duellmanohyla ruficulis</i>	48093771	48093826	89113747	89113707	61725370	61725782	no	tropical	no
<i>Duellmanohyla soralia</i>	61697195	61697440	62001221	62085307	61725372	61725784	no	tropical	no
<i>Ecnomiohyla miliaria</i>	NA	61697592	NA	NA	NA	61725928	no	tropical	no
<i>Ecnomiohyla myotimpanum</i>	61697256	61697596	62001243	62085329	NA	61725930	no	tropical	no
<i>Exerodonta chimalapa</i>	61697230	61697518	89113753	89113713	NA	61725862	no	tropical	no
<i>Exerodonta melanomma</i>	61697253	61697586	NA	NA	NA	61725924	no	tropical	no
<i>Exerodonta perkinsi</i>	61697264	61697614	89113763	89113723	NA	61725948	no	tropical	no
<i>Exerodonta smaragdina</i>	89113694	NA	89113767	89113729	NA	NA	dynamic	tropical	yes
<i>Exerodonta sumichrasti</i>	60735546	NA	NA	NA	NA	NA	dynamic	tropical	yes
<i>Exerodonta xera</i>	61697297	61697692	89113769	89113731	NA	61726028	dynamic	tropical	no
<i>Hyla andersonii</i>	61697209	60546493	70609747	69208087	NA	61725814	no	temperate	yes
<i>Hyla annectans</i>	61697211	213494118	70609749	69208089	61725392	61725818	no	temperate	yes
<i>Hyla arborea</i>	61697212	213494120	70609751	69208091	61725394	61725820	no	temperate	yes
<i>Hyla arenicolor</i>	148524025	60546495	62001223	62085309	61725398	61725824	no	temperate	yes
<i>Hyla avivoca</i>	61697216	60546489	70609753	NA	61725404	61725832	no	temperate	yes
<i>Hyla chrysoscelis</i>	33943771	60546409	NA	NA	NA	298200507	no	temperate	no
<i>Hyla cinerea</i>	56122896	213494032	62001231	301642270	61725432	38372029	no	temperate	yes
<i>Hyla euphorbiacea</i>	148524028	61697538	70609761	301642276	NA	61725882	no	tropical	yes
<i>Hyla eximia</i>	148524024	61697540	NA	NA	NA	298200491	no	temperate	yes

	12s	cytb	Cmyc	Pomc	Rag1	Rhodopsin	Dichromatic	Region	BreedingAggregation
Hyla_femoralis	148524031	213494034	70609763	69208101	NA	61725888	no	temperate	yes
Hyla_gratiosa	61697241	61697552	256539045	69208103	61725452	61725892	no	temperate	yes
Hyla_immaculata	256539042	NA	256539047	256539055	NA	NA	no	temperate	no
Hyla_intermedia	NA	213494042	NA	NA	NA	NA	no	temperate	yes
Hyla_japonica	148524019	213494126	70609767	69208105	61725456	61725900	no	temperate	yes
Hyla_meridionalis	148524020	213494065	62001239	62085325	NA	295854398	no	temperate	yes
Hyla_molleri	NA	213494114	NA	NA	NA	NA	no	temperate	yes
Hyla_orientalis	NA	213494002	NA	NA	NA	295854396	no	temperate	no
Hyla_plicata	148524029	NA	70609777	301642274	NA	NA	no	tropical	yes
Hyla_sarda	NA	213494083	NA	NA	NA	NA	no	temperate	no
Hyla_savignyi	61697276	213494138	70609783	69208119	NA	61725978	no	temperate	no
Hyla_squirella	61697289	213494154	62001255	62085341	61725540	61726010	no	temperate	no
Hyla_suweonensis	NA	6561850	NA	NA	NA	NA	no	temperate	no
Hyla_tsinlingensis	256539043	NA	256539049	256539057	NA	NA	no	temperate	no
Hyla_versicolor	148524018	60546387	70609791	301642272	61725546	61726020	no	temperate	yes
Hyla_walkerii	148524030	61697688	256539051	256539059	61725548	61726024	no	tropical	yes
Hyla_wrightorum	298200431	NA	62001235	301642278	NA	298200521	no	temperate	yes
Hylomantis_aspera	302029229	NA	NA	NA	NA	302028982	no	tropical	no
Hylomantis_granulosa	302029231	302028614	NA	302028850	61725554	302028986	no	tropical	no
Hylomantis_hulli	302029233	NA	NA	302028852	302028932	302028988	no	tropical	no
Hyloscirtus_armatus	60735515	61697480	NA	NA	NA	61725828	no	tropical	no
Hyloscirtus_charazani	61697229	61697516	NA	NA	NA	61725860	no	tropical	no
Hyloscirtus_colymba	61697231	61697524	NA	62085399	NA	61725868	no	tropical	no
Hyloscirtus_lascinius	87280677	NA	NA	NA	NA	NA	no	tropical	no
Hyloscirtus_lindae	87280679	NA	NA	NA	NA	NA	no	tropical	no
Hyloscirtus_palmeri	61697261	61697608	NA	62085401	NA	61725942	no	tropical	no
Hyloscirtus_pantostictus	37699593	NA	NA	NA	NA	NA	no	tropical	no
Hyloscirtus_phyllognathus	87280687	NA	NA	NA	NA	NA	no	tropical	no
Hyloscirtus_simmonsii	87280694	NA	NA	62085403	NA	NA	no	tropical	no
Hyloscirtus_tapichalaca	NA	61697678	NA	NA	NA	61726014	no	tropical	no
Hypsiboas_albomarginatus	NA	61697458	NA	NA	NA	61725806	no	tropical	yes
Hypsiboas_albopunctatus	NA	61697460	NA	NA	NA	61725808	dynamic	tropical	yes
Hypsiboas_bischoffi	NA	61697494	NA	NA	NA	61725842	dynamic	tropical	yes
Hypsiboas_boans	61697221	61697498	62001227	62085313	NA	61725846	ontogenetic	tropical	no
Hypsiboas_caiangua	NA	61697504	NA	NA	NA	61725852	dynamic	tropical	yes
Hypsiboas_calcaratus	37699597	61697506	NA	NA	NA	NA	dynamic	tropical	no
Hypsiboas_callipleura	NA	304358285	NA	NA	NA	NA	dynamic	tropical	yes
Hypsiboas_crepitans	61697232	61697528	NA	NA	NA	61725872	dynamic	tropical	yes
Hypsiboas_dentei	125661755	NA	NA	NA	NA	NA	no	tropical	no
Hypsiboas_ericae	NA	61697536	NA	NA	NA	61725880	dynamic	tropical	yes
Hypsiboas_faber	87280674	61697542	NA	NA	NA	61725884	dynamic	tropical	yes
Hypsiboas_fasciatus	60735519	61697544	NA	NA	NA	61725886	dynamic	tropical	yes
Hypsiboas_geographicus	61697239	NA	NA	NA	NA	NA	dynamic	tropical	yes
Hypsiboas_guentheri	61697242	61697554	NA	NA	NA	61725894	no	tropical	yes
Hypsiboas_heilprini	61697243	157890889	157890503	157890851	NA	61725896	no	tropical	no
Hypsiboas_joaquini	NA	61697562	NA	NA	NA	61725902	no	tropical	no
Hypsiboas_lanciformis	61697247	61697568	NA	NA	NA	61725908	dynamic	tropical	yes
Hypsiboas_latriatus	NA	61697670	NA	NA	NA	61726006	no	tropical	yes
Hypsiboas_leptolineatus	NA	61697572	NA	NA	NA	61725912	dynamic	temperate	yes
Hypsiboas_lundii	61697250	61697576	NA	NA	NA	61725916	dynamic	tropical	yes
Hypsiboas_marginatus	NA	61697578	NA	NA	NA	61725918	no	tropical	no
Hypsiboas_marianitae	NA	61697580	NA	NA	NA	61725920	dynamic	tropical	yes
Hypsiboas_multifasciatus	61697259	61697602	NA	302028858	NA	61725936	dynamic	tropical	yes
Hypsiboas_palaestes	NA	304358305	NA	NA	NA	NA	no	tropical	missing
Hypsiboas_pardalis	61697262	61697610	NA	NA	NA	61725944	no	tropical	no
Hypsiboas_pellucens	37699599	NA	NA	NA	NA	NA	no	tropical	no
Hypsiboas_picturatus	37699596	NA	NA	NA	NA	NA	no	tropical	yes
Hypsiboas_polytaenius	61697266	61697618	62001247	62085333	NA	61725952	dynamic	tropical	yes
Hypsiboas_prasinus	NA	61697620	NA	NA	NA	61725954	no	tropical	yes
Hypsiboas_pulchellus	NA	61697624	NA	NA	NA	61725958	no	tropical	no
Hypsiboas_punctatus	NA	61697626	NA	NA	NA	61725960	no	tropical	yes
Hypsiboas_raniceps	61697268	61697628	62001249	62085335	NA	61725962	dynamic	tropical	yes
Hypsiboas_rosenbergi	60735530	NA	NA	NA	NA	NA	no	tropical	no
Hypsiboas_rufitellus	61697273	61697638	NA	NA	NA	61725974	dynamic	tropical	yes
Hypsiboas_semiguttatus	NA	61697644	NA	NA	NA	61725980	no	tropical	no
Hypsiboas_semilineatus	NA	61697646	NA	NA	NA	61725982	no	tropical	yes
Hypsiboas_sibleszi	61697278	61697650	NA	NA	NA	61725986	no	tropical	no
Isthmohyla_pseudopuma	61697267	213494079	NA	NA	NA	112383038	dynamic	tropical	yes
Isthmohyla_rivularis	61697270	61697632	NA	NA	NA	61725968	no	tropical	no
Isthmohyla_zeteki	148524035	NA	NA	NA	NA	112383042	no	tropical	no
Itapotihyla_langsdorffii	61697317	61697730	62001257	62085343	NA	61726064	dynamic	tropical	yes
Litoria_adelaidensis	261600123	NA	NA	NA	NA	NA	no	temperate	no
Litoriaamboinensis	284177544	NA	NA	NA	NA	NA	dynamic	tropical	no
Litoria_andiirmalin	261600125	NA	NA	NA	NA	NA	no	tropical	no
Litoria_arfakiana	37699580	NA	NA	NA	NA	NA	no	tropical	no
Litoria_aurea	61697302	61697702	62001295	302028860	NA	61726038	no	temperate	no
Litoria_barringtonensis	261600128	NA	NA	NA	NA	NA	no	temperate	no
Litoria_bicolor	71148699	NA	NA	NA	NA	NA	dynamic	tropical	yes
Litoria_brevipalmata	261600130	NA	NA	NA	NA	NA	dynamic	temperate	yes
Litoria_burrowsi	261600131	NA	NA	NA	NA	NA	no	temperate	no
Litoria_caerulea	37699579	61697704	62001297	62085383	NA	38372033	no	tropical	no

	12s	cytb	Cmyc	Pomc	Rag1	Rhodopsin	Dichromatic	Region	BreedingAggregation
Litoria_cavernicola	261600132	NA	NA	NA	NA	NA	no	tropical	no
Litoria_chloris	71148715	NA	NA	NA	NA	NA	dynamic	tropical	yes
Litoria_congenita	261600133	NA	NA	NA	NA	NA	no	tropical	no
Litoria_coplandi	71148705	NA	NA	NA	NA	NA	no	tropical	no
Litoria_cyclorhyncha	261600135	NA	NA	NA	NA	NA	no	temperate	no
Litoria_dahlui	71148708	NA	NA	NA	NA	NA	no	tropical	no
Litoria_darlingtoni	261600136	NA	NA	NA	NA	NA	no	tropical	no
Litoria_daviesae	261600137	NA	NA	NA	NA	NA	no	temperate	no
Litoria_dentata	261600138	NA	NA	NA	NA	NA	dynamic	temperate	yes
Litoria_dorsalis	261600139	NA	NA	NA	NA	NA	no	tropical	no
Litoria_electrica	261600141	NA	NA	NA	NA	NA	no	tropical	no
Litoria_ewingii	261600145	NA	NA	NA	NA	NA	dynamic	temperate	yes
Litoria_fallax	71148700	NA	NA	NA	NA	NA	dynamic	tropical	yes
Litoria_freyineti	61697304	61697706	NA	NA	61725562	61726042	no	temperate	no
Litoria_genimaculata	90296391	NA	NA	NA	NA	90184039	no	tropical	no
Litoria_gilleni	71148713	NA	NA	NA	NA	NA	no	temperate	no
Litoria_gracilentata	71148717	NA	NA	NA	NA	NA	no	tropical	yes
Litoria_havina	261600151	NA	NA	NA	NA	NA	no	tropical	no
Litoria_inermis	90296380	NA	NA	NA	NA	90184025	dynamic	tropical	no
Litoria_infrafronata	61697305	61697708	NA	NA	61725564	61726044	no	tropical	no
Litoria_jervisiensis	261600156	NA	NA	NA	NA	NA	dynamic	temperate	no
Litoria_jungguy	261600157	NA	NA	NA	NA	NA	dynamic	tropical	yes
Litoria_kumae	261600158	NA	NA	NA	NA	NA	no	tropical	no
Litoria_latopalmata	261600159	NA	NA	NA	NA	NA	no	tropical	no
Litoria_lesueurii	90296373	NA	NA	NA	NA	90184015	dynamic	temperate	yes
Litoria_littlejohni	261600162	NA	NA	NA	NA	NA	dynamic	temperate	no
Litoria_longirostris	261600163	NA	NA	NA	NA	NA	no	tropical	no
Litoria_meiriana	61697306	NA	NA	NA	61725566	61726046	no	tropical	no
Litoria_microbelos	71148695	NA	NA	NA	NA	NA	no	tropical	no
Litoria_moorei	261600170	NA	NA	NA	NA	NA	no	temperate	no
Litoria_nannotis	90296387	NA	NA	NA	NA	90184033	no	tropical	no
Litoria_nasuta	71148702	NA	NA	NA	NA	NA	dynamic	tropical	no
Litoria_nigrofrenata	261600173	NA	NA	NA	NA	NA	dynamic	tropical	no
Litoria_nudidigitus	261600175	NA	NA	NA	NA	NA	no	temperate	no
Litoria_nyakalensis	261600176	NA	NA	NA	NA	NA	no	tropical	no
Litoria_olongburensis	261600177	NA	NA	NA	NA	NA	no	temperate	no
Litoria_pallida	71148704	NA	NA	NA	NA	NA	dynamic	tropical	no
Litoria_paraewingi	261600178	NA	NA	NA	NA	NA	dynamic	temperate	no
Litoria_pearsoniana	261600179	NA	NA	NA	NA	NA	no	temperate	no
Litoria_peronii	60735500	NA	NA	NA	NA	NA	dynamic	temperate	no
Litoria_personata	261600180	NA	NA	NA	NA	NA	dynamic	tropical	no
Litoria_phyllochroa	261600181	NA	NA	NA	NA	NA	no	temperate	no
Litoria_raniformis	261600185	NA	NA	NA	NA	NA	no	temperate	no
Litoria_revelata	261600186	NA	NA	NA	NA	NA	dynamic	tropical	yes
Litoria_rheocola	261600187	NA	NA	NA	NA	NA	no	tropical	no
Litoria_rothii	71148698	NA	NA	NA	NA	NA	dynamic	tropical	no
Litoria_rubella	60735499	NA	NA	NA	NA	NA	no	tropical	yes
Litoria_serrata	261600149	NA	NA	NA	NA	NA	no	tropical	no
Litoria_splendida	71148714	NA	NA	NA	NA	NA	no	tropical	no
Litoria_subglandulosa	284177551	NA	NA	NA	NA	NA	no	temperate	no
Litoria_thesaurensis	284177552	NA	NA	NA	NA	NA	no	tropical	yes
Litoria_tornieri	71148701	NA	NA	NA	NA	NA	dynamic	tropical	no
Litoria_tyleri	261600195	NA	NA	NA	NA	NA	dynamic	temperate	yes
Litoria_verreauxii	261600196	NA	NA	NA	NA	NA	dynamic	temperate	yes
Litoria_watjulumensis	71148706	NA	NA	NA	NA	NA	dynamic	tropical	no
Litoria_wilcoxii	261600197	NA	NA	NA	NA	NA	dynamic	temperate	yes
Litoria_wollastoni	284177553	NA	NA	NA	NA	NA	no	tropical	no
Litoria_xanthomera	71148716	NA	NA	NA	NA	NA	no	tropical	no
Nyctimantis_rugiceps	NA	61697718	NA	NA	NA	NA	no	tropical	no
Nyctimystes_cheesmani	261600199	NA	NA	NA	NA	NA	no	tropical	no
Nyctimystes_dayi	90296389	NA	NA	NA	NA	90184035	no	tropical	no
Nyctimystes_pulcher	61697312	61697720	NA	NA	NA	61726054	no	tropical	no
Osteocephalus_alboguttatus	325559736	NA	NA	NA	NA	NA	no	tropical	no
Osteocephalus_buckleyi	87280696	13095033	157890505	157890855	NA	NA	no	tropical	no
Osteocephalus_cabrerae	61697316	61697728	NA	NA	61725578	61726062	no	tropical	no
Osteocephalus_festae	325559756	NA	NA	NA	NA	NA	no	tropical	no
Osteocephalus_fuscifacies	325559738	NA	NA	NA	NA	NA	no	tropical	no
Osteocephalus_lepieurii	48093817	48093918	NA	NA	61725582	61726066	dynamic	tropical	yes
Osteocephalus_mutabor	87280697	NA	157890507	157890857	NA	NA	no	tropical	no
Osteocephalus_oophagus	61697319	61697734	NA	NA	61725584	61726068	no	tropical	yes
Osteocephalus_planiceps	325559739	NA	157890509	157890859	NA	NA	no	tropical	no
Osteocephalus_taurinus	61697320	61697736	62001259	62085345	61725586	61726070	no	tropical	yes
Osteocephalus_verruciger	325559762	157890897	157890511	157890861	NA	NA	no	tropical	no
Osteocephalus_yasuni	325559743	NA	NA	NA	NA	NA	dynamic	tropical	yes
Osteopilus_crucialis	NA	61697738	NA	157890865	NA	NA	no	tropical	no
Osteopilus_dominicensis	61697322	321228059	NA	157890867	61725588	61726072	dynamic	tropical	yes
Osteopilus_marianae	87280701	321228055	157890515	157890869	NA	NA	no	tropical	no
Osteopilus_pulchilineatus	60735528	157890905	157890517	157890871	NA	NA	dynamic	tropical	yes
Osteopilus_septrionalis	61697323	321227919	157890519	321227519	61725590	NA	dynamic	tropical	yes
Osteopilus_vastus	61697324	321228057	157890521	157890879	NA	NA	dynamic	tropical	no
Osteopilus_wilderi	87280703	NA	157890523	157890881	NA	NA	no	tropical	no

	12s	cytb	Cmyc	Pomc	Rag1	Rhodopsin	Dichromatic	Region	BreedingAggregation
Pachymedusa dactinolor	37699588	61697746	62001303	62085389	61725592	61726074	no	tropical	yes
Phasmahyla cochranæ	61697326	61697748	NA	302028862	302028938	302028996	no	tropical	no
Phasmahyla exilis	302029238	302028626	NA	302028864	302028940	302028998	no	tropical	no
Phasmahyla guttata	61697327	302028628	NA	302028866	61725594	302029000	no	tropical	no
Phasmahyla jandaia	302029240	302028630	NA	302028870	NA	302029002	no	tropical	yes
Phrynomedusa marginata	302029242	302028632	NA	NA	302028942	302029004	no	tropical	no
Phyllodytes auratus	60735475	104304347	62001265	62085351	NA	NA	no	tropical	no
Phyllodytes luteolus	61697332	61697760	NA	302028872	61725604	61726086	no	tropical	no
Phyllomedusa atelopoides	60735505	NA	NA	NA	NA	NA	no	tropical	no
Phyllomedusa ayeaye	302029251	302028656	NA	302028880	302028952	302029026	no	tropical	no
Phyllomedusa azurea	302029255	302028658	NA	NA	302028954	302029034	no	tropical	no
Phyllomedusa bahiana	302029258	302028666	NA	302028886	NA	302029038	no	tropical	yes
Phyllomedusa bicolor	61697334	61697764	NA	NA	61725606	61726090	no	tropical	no
Phyllomedusa boliviana	302029260	302028670	NA	302028888	302028958	302029042	no	tropical	yes
Phyllomedusa burmeisteri	302029262	302028674	NA	302028890	302028960	302029046	no	tropical	yes
Phyllomedusa camba	302029265	NA	NA	302028894	302028962	302029052	no	tropical	yes
Phyllomedusa distincta	302029269	302028686	NA	302028898	NA	302029056	no	tropical	yes
Phyllomedusa hypochondrialis	61697335	61697766	NA	NA	61725608	61726092	no	tropical	yes
Phyllomedusa iheringii	302029271	302028690	NA	302028900	NA	302029058	no	tropical	yes
Phyllomedusa neildi	302029277	302028702	NA	NA	NA	302029070	no	tropical	yes
Phyllomedusa nordestina	302029279	302028708	NA	302028904	302028968	302029074	no	tropical	yes
Phyllomedusa oreades	302029281	302028710	NA	302028910	NA	302029078	no	tropical	no
Phyllomedusa palliata	37699587	302028722	NA	NA	302028970	NA	no	tropical	yes
Phyllomedusa rohdei	302029247	302028640	NA	302028874	302028948	302029012	no	tropical	no
Phyllomedusa sauvagii	302029288	302028724	NA	302028916	302028974	302029094	no	tropical	yes
Phyllomedusa tarsius	61697337	61697770	NA	NA	302028976	61726096	no	tropical	no
Phyllomedusa tetraploidea	302029291	61697772	NA	302028918	302028978	302029098	no	tropical	no
Phyllomedusa tomopterna	37699586	61697774	62001307	302028922	61725610	61726100	no	tropical	no
Phyllomedusa trinitatis	302029294	302028736	NA	NA	302028980	302029102	no	tropical	yes
Phyllomedusa vaillanti	48093819	48093922	NA	NA	61725612	61726102	no	tropical	no
Plectrohyla bistincta	61697220	61697496	NA	NA	NA	61725844	dynamic	tropical	yes
Plectrohyla chrysopleura	60735476	NA	62001267	62085353	NA	NA	no	tropical	no
Plectrohyla guatemalensis	61697342	61697782	70609795	69208131	61725618	61726108	no	tropical	no
Plectrohyla matudai	61697343	61697784	NA	NA	61725620	61726110	dynamic	tropical	no
Plectrohyla pentheter	148524039	NA	NA	NA	NA	NA	dynamic	tropical	yes
Pseudacris brachyphona	33943751	NA	NA	NA	NA	NA	no	temperate	yes
Pseudacris brimleyi	150249259	NA	NA	NA	NA	NA	no	temperate	yes
Pseudacris cadaverina	33943769	242200571	62001229	62085315	NA	61726114	no	temperate	yes
Pseudacris clarkii	150249331	NA	NA	NA	NA	NA	no	temperate	yes
Pseudacris crucifer	33943758	157503250	62001269	62085355	NA	61726116	no	temperate	yes
Pseudacris feriarum	150249399	NA	NA	NA	NA	NA	no	temperate	yes
Pseudacris illinoensis	33943765	NA	NA	NA	NA	NA	no	temperate	yes
Pseudacris kalmi	150249449	NA	NA	NA	NA	NA	no	temperate	yes
Pseudacris maculata	150249348	157503252	NA	157503352	NA	NA	no	temperate	yes
Pseudacris nigrita	150249268	40806858	62001271	62085357	NA	NA	no	temperate	yes
Pseudacris ocularis	33943752	61697792	70609797	69208133	NA	61726118	no	temperate	yes
Pseudacris ornata	33943760	295367245	NA	NA	NA	NA	no	temperate	yes
Pseudacris regilla	33943767	34305541	62001251	62085337	112382286	61726120	no	temperate	yes
Pseudacris streckeri	150249232	40806856	NA	NA	NA	NA	no	temperate	yes
Pseudacris triseriata	150249381	61697796	NA	NA	NA	61726122	no	temperate	yes
Pseudis bolbodactyla	134260669	NA	NA	NA	NA	NA	no	tropical	yes
Pseudis cardosoi	134260660	NA	NA	NA	NA	NA	no	tropical	yes
Pseudis minuta/minutus	61697350	61697798	NA	302028926	61725626	NA	no	tropical	yes
Pseudis paradoxa	37624268	48093924	62001203	62085289	61725628	38372027	no	tropical	no
Pseudis tocantins	134260667	NA	NA	NA	NA	NA	no	tropical	no
Ptychohyla hypomykter	61697356	61697810	70609799	69208135	NA	61726134	no	tropical	yes
Ptychohyla leonhardschultzei	61697357	61697812	NA	NA	61725636	61726136	no	tropical	no
Ptychohyla spinipollex	61697359	61697816	62001275	62085361	61725640	61726140	no	tropical	no
Scarthyla goinorum	61697363	61697822	62001277	62085363	61725644	61726146	no	tropical	no
Scinax acuminatus	61697364	61697824	NA	NA	61725646	61726148	dynamic	tropical	yes
Scinax berthae	61697365	61697826	NA	NA	NA	61726150	no	tropical	no
Scinax boesemani	284177531	NA	NA	NA	NA	NA	dynamic	tropical	yes
Scinax boulengeri	61697366	61697828	NA	NA	61725648	61726152	no	tropical	no
Scinax catharinae	61697367	61697830	62001279	62085365	61725650	61726154	dynamic	temperate	yes
Scinax chiquitanus	284177532	NA	NA	NA	NA	NA	dynamic	tropical	yes
Scinax crosopedospilus	60735483	NA	62001281	62085367	NA	NA	dynamic	tropical	yes
Scinax cruentomma	NA	124518209	NA	NA	NA	NA	no	tropical	yes
Scinax duartei	284177534	NA	NA	NA	NA	NA	dynamic	tropical	yes
Scinax elaeochroa	61697368	61697832	NA	NA	61725652	61726156	dynamic	tropical	yes
Scinax funereus	284177535	NA	NA	NA	NA	NA	no	tropical	yes
Scinax fuscovarius	61697369	61697834	NA	NA	61725654	61726158	dynamic	tropical	yes
Scinax garbei	90296199	NA	NA	NA	NA	90183759	no	tropical	no
Scinax hayii	284177536	NA	NA	NA	NA	NA	dynamic	tropical	yes
Scinax ictericus	284177537	NA	NA	NA	NA	NA	dynamic	tropical	yes
Scinax nasicus	61697370	61697836	NA	NA	61725656	61726160	dynamic	tropical	yes
Scinax nebulosa	NA	33335344	NA	NA	NA	NA	no	tropical	yes
Scinax obtriangulatus	284177538	NA	NA	NA	NA	NA	no	tropical	no
Scinax oreites	284177539	NA	NA	NA	NA	NA	no	tropical	missing
Scinax pedromedinae	284177540	NA	NA	NA	NA	NA	no	tropical	no
Scinax quinquefasciatus	284177541	NA	NA	NA	NA	NA	no	tropical	no
Scinax ruber/rubra	37699575	124507389	NA	NA	61725658	61726162	ontogenetic	tropical	yes

	12s	cytb	Cmyc	Pomc	Rag1	Rhodopsin	Dichromatic	Region	BreedingAggregation
Scinax similis	284177542	NA	NA	NA	NA	NA	dynamic	tropical	yes
Scinax squalirostris	61697371	NA	NA	NA	61725660	61726164	no	tropical	yes
Scinax staufferi	61697372	61697840	NA	302028928	61725662	61726166	dynamic	tropical	yes
Scinax sugillatus	60735484	NA	62001283	62085369	NA	NA	no	tropical	yes
Scinax uruguayus	61697292	61697682	NA	NA	NA	61726018	no	tropical	yes
Scinax x-signatus	NA	124518197	NA	NA	NA	NA	no	tropical	yes
Smilisca baudinii	48093822	112382896	89113773	89113737	NA	112383028	dynamic	tropical	yes
Smilisca cyanosticta	61697374	61697844	62001285	62085371	61725664	112383030	no	tropical	yes
Smilisca fodiens	61697354	112382894	62001273	62085359	NA	112383026	no	temperate	yes
Smilisca phaeota	37699581	112382898	70609803	69208139	NA	61726172	dynamic	tropical	no
Smilisca puma	61697376	112382900	112382843	89113739	61725666	61726174	dynamic	tropical	yes
Smilisca sila	NA	112382902	112382845	89113741	NA	112383036	dynamic	tropical	no
Smilisca sordida	89113701	NA	112382847	89113743	NA	89113703	dynamic	tropical	yes
Sphaenorhynchus dorisae	61697377	61697850	NA	NA	61725668	61726176	no	tropical	yes
Sphaenorhynchus lacteus	48093823	48093930	62001287	62085373	61725670	61726178	no	tropical	yes
Sphaenorhynchus orophilus	87280706	NA	NA	NA	NA	NA	no	tropical	no
Tepuihyla edelcae	61697381	NA	NA	NA	61725676	NA	no	tropical	no
Tlalocohyla loquax	60735523	NA	NA	NA	NA	NA	dynamic	tropical	yes
Tlalocohyla picta	61697265	112382904	NA	89113725	NA	112383044	no	tropical	yes
Tlalocohyla smithii	61697279	112382906	62001253	62085339	NA	61725988	dynamic	tropical	yes
Trachycephalus coriaceus	87280704	157890917	157890525	157890883	NA	NA	dynamic	tropical	yes
Trachycephalus hadrocephs	61697328	61697752	NA	NA	NA	61726078	no	tropical	no
Trachycephalus imitatrix	157890531	NA	NA	NA	NA	NA	no	tropical	yes
Trachycephalus jordani	37699583	157890923	62001289	62085375	61725678	61726186	dynamic	tropical	yes
Trachycephalus mesophaeus	61697329	61697754	NA	NA	NA	61726080	dynamic	tropical	yes
Trachycephalus nigromaculatus	61697383	61697860	NA	NA	NA	61726188	no	tropical	yes
Trachycephalus resinifictrix	61697330	61697756	NA	NA	NA	61726082	no	tropical	no
Trachycephalus venulosus	37699589	157890919	62001263	302028930	NA	61726084	dynamic	tropical	yes
Triprion petasatus	61697385	112382890	62001291	62085377	61725680	112383022	ontogenetic	tropical	yes
Xenohyla truncata	61697386	61697864	NA	NA	NA	NA	no	tropical	yes

Table S4: Substitution models and partition schemes for Hylidae phylogenetic reconstruction.

Locus	Sequences	Length (bp)	Substitution Model
<i>12s</i>	323	464	GTR+I+G
<i>Cytochrome b</i>	208	383	GTR+I+G
<i>Cmyc</i>	86	420	GTR+I+G
<i>Pomc</i>	117	510	GTR+I+G
<i>Rag1</i>	78	428	GTR+I+G
<i>Rhodopsin</i>	193	316	HKY+G

Table S5: Median and upper/lower 95% HPD for correlated dynamic dichromatism and breeding aggregation transition rates from the posterior distribution of BAYESTRAITS analysis.

	q12	q13	q21	q24	q31	q34	q42	q43
<i>Lower 95% HPD</i>	16.338	0	18.498	18.554	18.977	17.259	18.550	12.302
<i>Median</i>	38.239	0	39.015	39.066	39.048	38.853	38.988	37.530
<i>Upper 95% HPD</i>	76.190	0	73.803	73.803	74.550	74.631	74.265	82.946

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